



O*NET-SOC OCCUPATIONAL DEFINITIONS

Sorted by Code

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11-0000 MANAGEMENT OCCUPATIONS

11-1011.00 Chief Executives ¹

Determine and formulate policies and provide the overall direction of companies or private and public sector organizations within the guidelines set up by a board of directors or similar governing body. Plan, direct, or coordinate operational activities at the highest level of management with the help of subordinate executives and staff managers.

11-1011.01 Government Service Executives

Determine and formulate policies and provide overall direction of Federal, State, local, or international government activities. Plan, direct, and coordinate operational activities at the highest level of management with the help of subordinate managers.

11-1011.02 Private Sector Executives

Determine and formulate policies and business strategies and provide overall direction of private sector organizations. Plan, direct, and coordinate operational activities at the highest level of management with the help of subordinate managers.

11-1021.00 General and Operations Managers

Plan, direct, or coordinate the operations of companies or public and private sector organizations. Duties and responsibilities include formulating policies, managing daily operations, and planning the use of materials and human resources, but are too diverse and general in nature to be classified in any one functional area of management or administration, such as personnel, purchasing, or administrative services. Includes owners and managers who head small business establishments whose duties are primarily managerial.

11-1031.00 Legislators

Develop laws and statutes at the Federal, State, or local level.

11-2011.00 Advertising and Promotions Managers

Plan and direct advertising policies and programs or produce collateral materials, such as posters, contests, coupons, or give-aways, to create extra interest in the purchase of a product or service for a department, an entire organization, or on an account basis.

11-2021.00 Marketing Managers

Determine the demand for products and services offered by a firm and its competitors and identify potential customers. Develop pricing strategies with the goal of maximizing the firm's profits or share of the market while ensuring the firm's customers are satisfied. Oversee product development or monitor trends that indicate the need for new products and services.

11-2022.00 Sales Managers

Direct the actual distribution or movement of a product or service to the customer. Coordinate sales distribution by establishing sales territories, quotas, and goals and establish training programs for sales representatives. Analyze sales statistics gathered by staff to determine sales potential and inventory requirements and monitor the preferences of customers.

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² Residual or "All Other" occupation. O*NET data not collected.

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⁴ Military Specific Occupation. O*NET data not collected.

11-2031.00 Public Relations Managers

Plan and direct public relations programs designed to create and maintain a favorable public image for employer or client; or if engaged in fundraising, plan and direct activities to solicit and maintain funds for special projects and nonprofit organizations.

11-3011.00 Administrative Services Managers

Plan, direct, or coordinate supportive services of an organization, such as recordkeeping, mail distribution, telephone operator/receptionist, and other office support services. May oversee facilities planning and maintenance and custodial operations.

11-3021.00 Computer and Information Systems Managers

Plan, direct, or coordinate activities in such fields as electronic data processing, information systems, systems analysis, and computer programming.

11-3031.00 Financial Managers ¹

Plan, direct, and coordinate accounting, investing, banking, insurance, securities, and other financial activities of a branch, office, or department of an establishment.

11-3031.01 Treasurers, Controllers, and Chief Financial Officers

Plan, direct, and coordinate the financial activities of an organization at the highest level of management. Includes financial reserve officers.

11-3031.02 Financial Managers, Branch or Department

Direct and coordinate financial activities of workers in a branch, office, or department of an establishment, such as branch bank, brokerage firm, risk and insurance department, or credit department.

11-3040.00 Human Resources Managers

Plan, direct, and coordinate human resource management activities of an organization to maximize the strategic use of human resources and maintain functions such as employee compensation, recruitment, personnel policies, and regulatory compliance.

11-3041.00 Compensation and Benefits Managers

Plan, direct, or coordinate compensation and benefits activities and staff of an organization.

11-3042.00 Training and Development Managers

Plan, direct, or coordinate the training and development activities and staff of an organization.

11-3049.99 Human Resources Managers, All Other ²

All Human Resources Managers not listed separately.

11-3051.00 Industrial Production Managers

Plan, direct, or coordinate the work activities and resources necessary for manufacturing products in accordance with cost, quality, and quantity specifications.

11-3061.00 Purchasing Managers

Plan, direct, or coordinate the activities of buyers, purchasing officers, and related workers involved in purchasing materials, products, and services.

11-3071.00 Transportation, Storage, and Distribution Managers ¹

Plan, direct, or coordinate transportation, storage, or distribution activities in accordance with governmental policies and regulations.

11-3071.01 Transportation Managers

Plan, direct, and coordinate the transportation operations within an organization or the activities of organizations that provide transportation services.

11-3071.02 Storage and Distribution Managers

Plan, direct, and coordinate the storage and distribution operations within an organization or the activities of organizations that are engaged in storing and distributing materials and products.

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⁴ Military Specific Occupation. O*NET data not collected.

11-9011.00 Farm, Ranch, and Other Agricultural Managers¹

On a paid basis, manage farms, ranches, aquacultural operations, greenhouses, nurseries, timber tracts, cotton gins, packing houses, or other agricultural establishments for employers. Carry out production, financial, and marketing decisions relating to the managed operations following guidelines from the owner. May contract tenant farmers or producers to carry out the day-to-day activities of the managed operation. May supervise planting, cultivating, harvesting, and marketing activities. May prepare cost, production, and other records. May perform physical work and operate machinery.

11-9011.01 Nursery and Greenhouse Managers

Plan, organize, direct, control, and coordinate activities of workers engaged in propagating, cultivating, and harvesting horticultural specialties, such as trees, shrubs, flowers, mushrooms, and other plants.

11-9011.02 Agricultural Crop Farm Managers

Direct and coordinate, through subordinate supervisory personnel, activities of workers engaged in agricultural crop production for corporations, cooperatives, or other owners.

11-9011.03 Fish Hatchery Managers

Direct and coordinate, through subordinate supervisory personnel, activities of workers engaged in fish hatchery production for corporations, cooperatives, or other owners.

11-9012.00 Farmers and Ranchers

On an ownership or rental basis, operate farms, ranches, greenhouses, nurseries, timber tracts, or other agricultural production establishments which produce crops, horticultural specialties, livestock, poultry, finfish, shellfish, or animal specialties. May plant, cultivate, harvest, perform post-harvest activities, and market crops and livestock; may hire, train, and supervise farm workers or supervise a farm labor contractor; may prepare cost, production, and other records. May maintain and operate machinery and perform physical work.

11-9021.00 Construction Managers

Plan, direct, coordinate, or budget, usually through subordinate supervisory personnel, activities concerned with the construction and maintenance of structures, facilities, and systems. Participate in the conceptual development of a construction project and oversee its organization, scheduling, and implementation.

11-9031.00 Education Administrators, Preschool and Child Care Center/Program

Plan, direct, or coordinate the academic and nonacademic activities of preschool and child care centers or programs.

11-9032.00 Education Administrators, Elementary and Secondary School

Plan, direct, or coordinate the academic, clerical, or auxiliary activities of public or private elementary or secondary level schools.

11-9033.00 Education Administrators, Postsecondary

Plan, direct, or coordinate research, instructional, student administration and services, and other educational activities at postsecondary institutions, including universities, colleges, and junior and community colleges.

11-9039.99 Education Administrators, All Other²

All education administrators not listed separately.

11-9041.00 Engineering Managers

Plan, direct, or coordinate activities in such fields as architecture and engineering or research and development in these fields.

11-9051.00 Food Service Managers

Plan, direct, or coordinate activities of an organization or department that serves food and beverages.

11-9061.00 Funeral Directors

Perform various tasks to arrange and direct funeral services, such as coordinating transportation of body to mortuary for embalming, interviewing family or other authorized person to arrange details, selecting pallbearers, procuring official for religious rites, and providing transportation for mourners.

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⁴ Military Specific Occupation. O*NET data not collected.

11-9071.00 Gaming Managers

Plan, organize, direct, control, or coordinate gaming operations in a casino. Formulate gaming policies for their area of responsibility.

11-9081.00 Lodging Managers

Plan, direct, or coordinate activities of an organization or department that provides lodging and other accommodations.

11-9111.00 Medical and Health Services Managers

Plan, direct, or coordinate medicine and health services in hospitals, clinics, managed care organizations, public health agencies, or similar organizations.

11-9121.00 Natural Sciences Managers

Plan, direct, or coordinate activities in such fields as life sciences, physical sciences, mathematics, statistics, and research and development in these fields.

11-9131.00 Postmasters and Mail Superintendents

Direct and coordinate operational, administrative, management, and supportive services of a U.S. post office; or coordinate activities of workers engaged in postal and related work in assigned post office.

11-9141.00 Property, Real Estate, and Community Association Managers

Plan, direct, or coordinate selling, buying, leasing, or governance activities of commercial, industrial, or residential real estate properties.

11-9151.00 Social and Community Service Managers

Plan, organize, or coordinate the activities of a social service program or community outreach organization. Oversee the program or organization's budget and policies regarding participant involvement, program requirements, and benefits. Work may involve directing social workers, counselors, or probation officers.

11-9199.99 Managers, All Other ²

All managers not listed separately.

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13-0000 BUSINESS AND FINANCIAL OPERATIONS MANAGERS

13-1011.00 Agents and Business Managers of Artists, Performers, and Athletes

Represent and promote artists, performers, and athletes to prospective employers. May handle contract negotiation and other business matters for clients.

13-1021.00 Purchasing Agents and Buyers, Farm Products

Purchase farm products either for further processing or resale.

13-1022.00 Wholesale and Retail Buyers, Except Farm Products

Buy merchandise or commodities, other than farm products, for resale to consumers at the wholesale or retail level, including both durable and nondurable goods. Analyze past buying trends, sales records, price, and quality of merchandise to determine value and yield. Select, order, and authorize payment for merchandise according to contractual agreements. May conduct meetings with sales personnel and introduce new products.

13-1023.00 Purchasing Agents, Except Wholesale, Retail, and Farm Products

Purchase machinery, equipment, tools, parts, supplies, or services necessary for the operation of an establishment. Purchase raw or semi-finished materials for manufacturing.

13-1031.00 Claims Adjusters, Examiners, and Investigators ¹

Review settled claims to determine that payments and settlements have been made in accordance with company practices and procedures, ensuring that proper methods have been followed. Report overpayments, underpayments, and other irregularities. Confer with legal counsel on claims requiring litigation.

13-1031.01 Claims Examiners, Property and Casualty Insurance

Review settled insurance claims to determine that payments and settlements have been made in accordance with company practices and procedures. Report overpayments, underpayments, and other irregularities. Confer with legal counsel on claims requiring litigation.

13-1031.02 Insurance Adjusters, Examiners, and Investigators

Investigate, analyze, and determine the extent of insurance company's liability concerning personal, casualty, or property loss or damages, and attempt to effect settlement with claimants. Correspond with or interview medical specialists, agents, witnesses, or claimants to compile information. Calculate benefit payments and approve payment of claims within a certain monetary limit.

13-1032.00 Insurance Appraisers, Auto Damage

Appraise automobile or other vehicle damage to determine cost of repair for insurance claim settlement and seek agreement with automotive repair shop on cost of repair. Prepare insurance forms to indicate repair cost or cost estimates and recommendations.

13-1041.00 Compliance Officers, Except Agriculture, Construction, Health and Safety, and Transportation ¹

Examine, evaluate, and investigate eligibility for or conformity with laws and regulations governing contract compliance of licenses and permits, and other compliance and enforcement inspection activities not classified elsewhere.

13-1041.01 Environmental Compliance Inspectors

Inspect and investigate sources of pollution to protect the public and environment and ensure conformance with Federal, State, and local regulations and ordinances.

13-1041.02 Licensing Examiners and Inspectors

Examine, evaluate, and investigate eligibility for, conformity with, or liability under licenses or permits.

13-1041.03 Equal Opportunity Representatives and Officers

Monitor and evaluate compliance with equal opportunity laws, guidelines, and policies to ensure that employment practices and contracting arrangements give equal opportunity without regard to race, religion, color, national origin, sex, age, or disability.

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13-1041.04 Government Property Inspectors and Investigators

Investigate or inspect government property to ensure compliance with contract agreements and government regulations.

13-1041.05 Pressure Vessel Inspectors

Inspect pressure vessel equipment for conformance with safety laws and standards regulating their design, fabrication, installation, repair, and operation.

13-1041.06 Coroners

Direct activities such as autopsies, pathological and toxicological analyses, and inquests relating to the investigation of deaths occurring within a legal jurisdiction to determine cause of death or to fix responsibility for accidental, violent, or unexplained deaths.

13-1051.00 Cost Estimators

Prepare cost estimates for product manufacturing, construction projects, or services to aid management in bidding on or determining price of product or service. May specialize according to particular service performed or type of product manufactured.

13-1061.00 Emergency Management Specialists

Coordinate disaster response or crisis management activities, provide disaster preparedness training, and prepare emergency plans and procedures for natural (e.g., hurricanes, floods, earthquakes), wartime, or technological (e.g., nuclear power plant emergencies, hazardous materials spills) disasters or hostage situations.

13-1071.00 Employment, Recruitment, and Placement Specialists ¹

Recruit and place workers.

13-1071.01 Employment Interviewers, Private or Public Employment Service

Interview job applicants in employment office and refer them to prospective employers for consideration. Search application files, notify selected applicants of job openings, and refer qualified applicants to prospective employers. Contact employers to verify referral results. Record and evaluate various pertinent data.

13-1071.02 Personnel Recruiters

Seek out, interview, and screen applicants to fill existing and future job openings and promote career opportunities within an organization.

13-1072.00 Compensation, Benefits, and Job Analysis Specialists

Conduct programs of compensation and benefits and job analysis for employer. May specialize in specific areas, such as position classification and pension programs.

13-1073.00 Training and Development Specialists

Conduct training and development programs for employees.

13-1079.99 Human Resources, Training, and Labor Relations Specialists, All Other ²

All human resources, training, and labor relations specialists not listed separately.

13-1081.00 Logisticians

Analyze and coordinate the logistical functions of a firm or organization. Responsible for the entire life cycle of a product, including acquisition, distribution, internal allocation, delivery, and final disposal of resources.

13-1111.00 Management Analysts

Conduct organizational studies and evaluations, design systems and procedures, conduct work simplifications and measurement studies, and prepare operations and procedures manuals to assist management in operating more efficiently and effectively. Includes program analysts and management consultants.

13-1121.00 Meeting and Convention Planners

Coordinate activities of staff and convention personnel to make arrangements for group meetings and conventions.

13-1199.99 Business Operations Specialists, All Other ²

All business operations specialists not listed separately.

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13-2011.00 Accountants and Auditors ¹

Examine, analyze, and interpret accounting records for the purpose of giving advice or preparing statements. Install or advise on systems of recording costs or other financial and budgetary data.

13-2011.01 Accountants

Analyze financial information and prepare financial reports to determine or maintain record of assets, liabilities, profit and loss, tax liability, or other financial activities within an organization.

13-2011.02 Auditors

Examine and analyze accounting records to determine financial status of establishment and prepare financial reports concerning operating procedures.

13-2021.00 Appraisers and Assessors of Real Estate ¹

Appraise real property to determine its fair value. May assess taxes in accordance with prescribed schedules.

13-2021.01 Assessors

Appraise real and personal property to determine its fair value. May assess taxes in accordance with prescribed schedules.

13-2021.02 Appraisers, Real Estate

Appraise real property to determine its value for purchase, sales, investment, mortgage, or loan purposes.

13-2031.00 Budget Analysts

Examine budget estimates for completeness, accuracy, and conformance with procedures and regulations. Analyze budgeting and accounting reports for the purpose of maintaining expenditure controls.

13-2041.00 Credit Analysts

Analyze current credit data and financial statements of individuals or firms to determine the degree of risk involved in extending credit or lending money. Prepare reports with this credit information for use in decision-making.

13-2051.00 Financial Analysts

Conduct quantitative analyses of information affecting investment programs of public or private institutions.

13-2052.00 Personal Financial Advisors

Advise clients on financial plans utilizing knowledge of tax and investment strategies, securities, insurance, pension plans, and real estate. Duties include assessing clients' assets, liabilities, cash flow, insurance coverage, tax status, and financial objectives to establish investment strategies.

13-2053.00 Insurance Underwriters

Review individual applications for insurance to evaluate degree of risk involved and determine acceptance of applications.

13-2061.00 Financial Examiners

Enforce or ensure compliance with laws and regulations governing financial and securities institutions and financial and real estate transactions. May examine, verify correctness of, or establish authenticity of records.

13-2071.00 Loan Counselors

Provide guidance to prospective loan applicants who have problems qualifying for traditional loans. Guidance may include determining the best type of loan and explaining loan requirements or restrictions.

13-2072.00 Loan Officers

Evaluate, authorize, or recommend approval of commercial, real estate, or credit loans. Advise borrowers on financial status and methods of payments. Includes mortgage loan officers and agents, collection analysts, loan servicing officers, and loan underwriters.

13-2081.00 Tax Examiners, Collectors, and Revenue Agents

Determine tax liability or collect taxes from individuals or business firms according to prescribed laws and regulations.

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⁴ Military Specific Occupation. O*NET data not collected.

13-2082.00 Tax Preparers

Prepare tax returns for individuals or small businesses but do not have the background or responsibilities of an accredited or certified public accountant.

13-2099.99 Financial Specialists, All Other ²

All financial specialists not listed separately.

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15-0000 COMPUTER AND MATHEMATICAL SCIENCE OCCUPATIONS

15-1011.00 Computer and Information Scientists, Research

Conduct research into fundamental computer and information science as theorists, designers, or inventors. Solve or develop solutions to problems in the field of computer hardware and software.

15-1021.00 Computer Programmers

Convert project specifications and statements of problems and procedures to detailed logical flow charts for coding into computer language. Develop and write computer programs to store, locate, and retrieve specific documents, data, and information. May program web sites.

15-1031.00 Computer Software Engineers, Applications

Develop, create, and modify general computer applications software or specialized utility programs. Analyze user needs and develop software solutions. Design software or customize software for client use with the aim of optimizing operational efficiency. May analyze and design databases within an application area, working individually or coordinating database development as part of a team.

15-1032.00 Computer Software Engineers, Systems Software

Research, design, develop, and test operating systems-level software, compilers, and network distribution software for medical, industrial, military, communications, aerospace, business, scientific, and general computing applications. Set operational specifications and formulate and analyze software requirements. Apply principles and techniques of computer science, engineering, and mathematical analysis.

15-1041.00 Computer Support Specialists

Provide technical assistance to computer system users. Answer questions or resolve computer problems for clients in person, via telephone or from remote location. May provide assistance concerning the use of computer hardware and software, including printing, installation, word processing, electronic mail, and operating systems.

15-1051.00 Computer Systems Analysts

Analyze science, engineering, business, and all other data processing problems for application to electronic data processing systems. Analyze user requirements, procedures, and problems to automate or improve existing systems and review computer system capabilities, workflow, and scheduling limitations. May analyze or recommend commercially available software. May supervise computer programmers.

15-1061.00 Database Administrators

Coordinate changes to computer databases, test and implement the database applying knowledge of database management systems. May plan, coordinate, and implement security measures to safeguard computer databases.

15-1071.00 Network and Computer Systems Administrators¹

Install, configure, and support an organization's local area network (LAN), wide area network (WAN), and Internet system or a segment of a network system. Maintain network hardware and software. Monitor network to ensure network availability to all system users and perform necessary maintenance to support network availability. May supervise other network support and client server specialists and plan, coordinate, and implement network security measures.

15-1071.01 Computer Security Specialists

Plan, coordinate, and implement security measures for information systems to regulate access to computer data files and prevent unauthorized modification, destruction, or disclosure of information.

15-1081.00 Network Systems and Data Communications Analysts

Analyze, design, test, and evaluate network systems, such as local area networks (LAN), wide area networks (WAN), Internet, intranet, and other data communications systems. Perform network modeling, analysis, and planning. Research and recommend network and data communications hardware and software. Includes telecommunications specialists who deal with the interfacing of computer and communications equipment. May supervise computer programmers.

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⁴ Military Specific Occupation. O*NET data not collected.

15-1099.99 Computer Specialists, All Other ²

All computer specialists not listed separately

15-2011.00 Actuaries

Analyze statistical data, such as mortality, accident, sickness, disability, and retirement rates and construct probability tables to forecast risk and liability for payment of future benefits. May ascertain premium rates required and cash reserves necessary to ensure payment of future benefits.

15-2021.00 Mathematicians

Conduct research in fundamental mathematics or in application of mathematical techniques to science, management, and other fields. Solve or direct solutions to problems in various fields by mathematical methods.

15-2031.00 Operations Research Analysts

Formulate and apply mathematical modeling and other optimizing methods using a computer to develop and interpret information that assists management with decision making, policy formulation, or other managerial functions. May develop related software, service, or products. Frequently concentrates on collecting and analyzing data and developing decision support software. May develop and supply optimal time, cost, or logistics networks for program evaluation, review, or implementation.

15-2041.00 Statisticians

Engage in the development of mathematical theory or apply statistical theory and methods to collect, organize, interpret, and summarize numerical data to provide usable information. May specialize in fields, such as biostatistics, agricultural statistics, business statistics, economic statistics, or other fields.

15-2099.99 Mathematical Science Occupations, All Other ²

All mathematical scientists not listed separately.

15-3011.00 Mathematical Technicians

Apply standardized mathematical formulas, principles, and methodology to technological problems in engineering and physical sciences in relation to specific industrial and research objectives, processes, equipment, and products.

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17-0000 ARCHITECTURE AND ENGINEERING OCCUPATIONS

17-1011.00 Architects, Except Landscape and Naval

Plan and design structures, such as private residences, office buildings, theaters, factories, and other structural property.

17-1012.00 Landscape Architects

Plan and design land areas for such projects as parks and other recreational facilities, airports, highways, hospitals, schools, land subdivisions, and commercial, industrial, and residential sites.

17-1021.00 Cartographers and Photogrammetrists

Collect, analyze, and interpret geographic information provided by geodetic surveys, aerial photographs, and satellite data. Research, study, and prepare maps and other spatial data in digital or graphic form for legal, social, political, educational, and design purposes. May work with Geographic Information Systems (GIS). May design and evaluate algorithms, data structures, and user interfaces for GIS and mapping systems.

17-1022.00 Surveyors

Make exact measurements and determine property boundaries. Provide data relevant to the shape, contour, gravitation, location, elevation, or dimension of land or land features on or near the earth's surface for engineering, mapmaking, mining, land evaluation, construction, and other purposes.

17-2011.00 Aerospace Engineers

Perform a variety of engineering work in designing, constructing, and testing aircraft, missiles, and spacecraft. May conduct basic and applied research to evaluate adaptability of materials and equipment to aircraft design and manufacture. May recommend improvements in testing equipment and techniques.

17-2021.00 Agricultural Engineers

Apply knowledge of engineering technology and biological science to agricultural problems concerned with power and machinery, electrification, structures, soil and water conservation, and processing of agricultural products.

17-2031.00 Biomedical Engineers

Apply knowledge of engineering, biology, and biomechanical principles to the design, development, and evaluation of biological and health systems and products, such as artificial organs, prostheses, instrumentation, medical information systems, and health management and care delivery systems.

17-2041.00 Chemical Engineers

Design chemical plant equipment and devise processes for manufacturing chemicals and products, such as gasoline, synthetic rubber, plastics, detergents, cement, paper, and pulp, by applying principles and technology of chemistry, physics, and engineering.

17-2051.00 Civil Engineers

Perform engineering duties in planning, designing, and overseeing construction and maintenance of building structures, and facilities, such as roads, railroads, airports, bridges, harbors, channels, dams, irrigation projects, pipelines, power plants, water and sewage systems, and waste disposal units. Includes architectural, structural, traffic, ocean, and geo-technical engineers.

17-2061.00 Computer Hardware Engineers

Research, design, develop, and test computer or computer-related equipment for commercial, industrial, military, or scientific use. May supervise the manufacturing and installation of computer or computer-related equipment and components.

17-2071.00 Electrical Engineers

Design, develop, test, or supervise the manufacturing and installation of electrical equipment, components, or systems for commercial, industrial, military, or scientific use.

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17-2072.00 Electronics Engineers, Except Computer

Research, design, develop, and test electronic components and systems for commercial, industrial, military, or scientific use utilizing knowledge of electronic theory and materials properties. Design electronic circuits and components for use in fields such as telecommunications, aerospace guidance and propulsion control, acoustics, or instruments and controls.

17-2081.00 Environmental Engineers

Design, plan, or perform engineering duties in the prevention, control, and remediation of environmental health hazards utilizing various engineering disciplines. Work may include waste treatment, site remediation, or pollution control technology.

17-2111.00 Health and Safety Engineers, Except Mining Safety Engineers and Inspectors¹

Promote worksite or product safety by applying knowledge of industrial processes, mechanics, chemistry, psychology, and industrial health and safety laws.

17-2111.01 Industrial Safety and Health Engineers

Plan, implement, and coordinate safety programs, requiring application of engineering principles and technology, to prevent or correct unsafe environmental working conditions.

17-2111.02 Fire-Prevention and Protection Engineers

Research causes of fires, determine fire protection methods, and design or recommend materials or equipment such as structural components or fire-detection equipment to assist organizations in safeguarding life and property against fire, explosion, and related hazards.

17-2111.03 Product Safety Engineers

Develop and conduct tests to evaluate product safety levels and recommend measures to reduce or eliminate hazards.

17-2112.00 Industrial Engineers

Design, develop, test, and evaluate integrated systems for managing industrial production processes including human work factors, quality control, inventory control, logistics and material flow, cost analysis, and production coordination.

17-2121.00 Marine Engineers and Naval Architects¹

Design, develop, and evaluate the operation of marine vessels, ship machinery, and related equipment, such as power supply and propulsion systems.

17-2121.01 Marine Engineers

Design, develop, and take responsibility for the installation of ship machinery and related equipment including propulsion machines and power supply systems.

17-2121.02 Marine Architects

Design and oversee construction and repair of marine craft and floating structures such as ships, barges, tugs, dredges, submarines, torpedoes, floats, and buoys. May confer with marine engineers.

17-2131.00 Materials Engineers

Evaluate materials and develop machinery and processes to manufacture materials for use in products that must meet specialized design and performance specifications. Develop new uses for known materials. Includes those working with composite materials or specializing in one type of material, such as graphite, metal and metal alloys, ceramics and glass, plastics and polymers, and naturally occurring materials.

17-2141.00 Mechanical Engineers

Perform engineering duties in planning and designing tools, engines, machines, and other mechanically functioning equipment. Oversee installation, operation, maintenance, and repair of such equipment as centralized heat, gas, water, and steam systems.

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17-2151.00 Mining and Geological Engineers, Including Mining Safety Engineers

Determine the location and plan the extraction of coal, metallic ores, nonmetallic minerals, and building materials, such as stone and gravel. Work involves conducting preliminary surveys of deposits or undeveloped mines and planning their development; examining deposits or mines to determine whether they can be worked at a profit; making geological and topographical surveys; evolving methods of mining best suited to character, type, and size of deposits; and supervising mining operations.

17-2161.00 Nuclear Engineers

Conduct research on nuclear engineering problems or apply principles and theory of nuclear science to problems concerned with release, control, and utilization of nuclear energy and nuclear waste disposal.

17-2171.00 Petroleum Engineers

Devise methods to improve oil and gas well production and determine the need for new or modified tool designs. Oversee drilling and offer technical advice to achieve economical and satisfactory progress.

17-2199.99 Engineers, All Other ²

All engineers not listed separately.

17-3011.00 Architectural and Civil Drafters ¹

Prepare detailed drawings of architectural and structural features of buildings or drawings and topographical relief maps used in civil engineering projects, such as highways, bridges, and public works. Utilize knowledge of building materials, engineering practices, and mathematics to complete drawings.

17-3011.01 Architectural Drafters

Prepare detailed drawings of architectural designs and plans for buildings and structures according to specifications provided by architect.

17-3011.02 Civil Drafters

Prepare drawings and topographical and relief maps used in civil engineering projects, such as highways, bridges, pipelines, flood control projects, and water and sewerage control systems.

17-3012.00 Electrical and Electronics Drafters ¹

Prepare wiring diagrams, circuit board assembly diagrams, and layout drawings used for manufacture, installation, and repair of electrical equipment in factories, power plants, and buildings.

17-3012.01 Electronic Drafters

Draw wiring diagrams, circuit board assembly diagrams, schematics, and layout drawings used for manufacture, installation, and repair of electronic equipment.

17-3012.02 Electrical Drafters

Develop specifications and instructions for installation of voltage transformers, overhead or underground cables, and related electrical equipment used to conduct electrical energy from transmission lines or high-voltage distribution lines to consumers.

17-3013.00 Mechanical Drafters

Prepare detailed working diagrams of machinery and mechanical devices, including dimensions, fastening methods, and other engineering information.

17-3019.99 Drafters, All Other ²

All drafters not listed separately.

17-3021.00 Aerospace Engineering and Operations Technicians

Operate, install, calibrate, and maintain integrated computer/communications systems consoles, simulators, and other data acquisition, test, and measurement instruments and equipment to launch, track, position, and evaluate air and space vehicles. May record and interpret test data.

17-3022.00 Civil Engineering Technicians

Apply theory and principles of civil engineering in planning, designing, and overseeing construction and maintenance of structures and facilities under the direction of engineering staff or physical scientists.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

17-3023.00 Electrical and Electronic Engineering Technicians ¹

Apply electrical and electronic theory and related knowledge, usually under the direction of engineering staff, to design, build, repair, calibrate, and modify electrical components, circuitry, controls, and machinery for subsequent evaluation and use by engineering staff in making engineering design decisions.

17-3023.01 Electronics Engineering Technicians

Lay out, build, test, troubleshoot, repair, and modify developmental and production electronic components, parts, equipment, and systems, such as computer equipment, missile control instrumentation, electron tubes, test equipment, and machine tool numerical controls, applying principles and theories of electronics, electrical circuitry, engineering mathematics, electronic and electrical testing, and physics. Usually work under direction of engineering staff.

17-3023.02 Calibration and Instrumentation Technicians

Develop, test, calibrate, operate, and repair electrical, mechanical, electromechanical, electrohydraulic, or electronic measuring and recording instruments, apparatus, and equipment.

17-3023.03 Electrical Engineering Technicians

Apply electrical theory and related knowledge to test and modify developmental or operational electrical machinery and electrical control equipment and circuitry in industrial or commercial plants and laboratories. Usually work under direction of engineering staff.

17-3024.00 Electro-Mechanical Technicians

Operate, test, and maintain unmanned, automated, servo-mechanical, or electromechanical equipment. May operate unmanned submarines, aircraft, or other equipment at worksites, such as oil rigs, deep ocean exploration, or hazardous waste removal. May assist engineers in testing and designing robotics equipment.

17-3025.00 Environmental Engineering Technicians

Apply theory and principles of environmental engineering to modify, test, and operate equipment and devices used in the prevention, control, and remediation of environmental pollution, including waste treatment and site remediation. May assist in the development of environmental pollution remediation devices under direction of engineer.

17-3026.00 Industrial Engineering Technicians

Apply engineering theory and principles to problems of industrial layout or manufacturing production, usually under the direction of engineering staff. May study and record time, motion, method, and speed involved in performance of production, maintenance, clerical, and other worker operations for such purposes as establishing standard production rates or improving efficiency.

17-3027.00 Mechanical Engineering Technicians

Apply theory and principles of mechanical engineering to modify, develop, and test machinery and equipment under direction of engineering staff or physical scientists.

17-3029.99 Engineering Technicians, Except Drafters, All Other ²

All engineering technicians, except drafters, not listed separately.

17-3031.00 Surveying and Mapping Technicians ¹

Perform surveying and mapping duties, usually under the direction of a surveyor, cartographer, or photogrammetrist to obtain data used for construction, mapmaking, boundary location, mining, or other purposes. May calculate mapmaking information and create maps from source data, such as surveying notes, aerial photography, satellite data, or other maps to show topographical features, political boundaries, and other features. May verify accuracy and completeness of topographical maps.

17-3031.01 Surveying Technicians

Adjust and operate surveying instruments, such as the theodolite and electronic distance-measuring equipment, and compile notes, make sketches and enter data into computers.

17-3031.02 Mapping Technicians

Calculate mapmaking information from field notes, and draw and verify accuracy of topographical maps.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-0000 LIFE, PHYSICAL, AND SOCIAL SCIENCE OCCUPATIONS

19-1011.00 Animal Scientists

Conduct research in the genetics, nutrition, reproduction, growth, and development of domestic farm animals.

19-1012.00 Food Scientists and Technologists

Use chemistry, microbiology, engineering, and other sciences to study the principles underlying the processing and deterioration of foods; analyze food content to determine levels of vitamins, fat, sugar, and protein; discover new food sources; research ways to make processed foods safe, palatable, and healthful; and apply food science knowledge to determine best ways to process, package, preserve, store, and distribute food.

19-1013.00 Soil and Plant Scientists¹

Conduct research in breeding, physiology, production, yield, and management of crops and agricultural plants, their growth in soils, and control of pests; or study the chemical, physical, biological, and mineralogical composition of soils as they relate to plant or crop growth. May classify and map soils and investigate effects of alternative practices on soil and crop productivity.

19-1013.01 Plant Scientists

Conduct research in breeding, production, and yield of plants or crops, and control of pests.

19-1013.02 Soil Scientists

Research or study soil characteristics, map soil types, and investigate responses of soils to known management practices to determine use capabilities of soils and effects of alternative practices on soil productivity.

19-1020.01 Biologists

Research or study basic principles of plant and animal life, such as origin, relationship, development, anatomy, and functions.

19-1021.00 Biochemists and Biophysicists¹

Study the chemical composition and physical principles of living cells and organisms, their electrical and mechanical energy, and related phenomena. May conduct research to further understanding of the complex chemical combinations and reactions involved in metabolism, reproduction, growth, and heredity. May determine the effects of foods, drugs, serums, hormones, and other substances on tissues and vital processes of living organisms.

19-1021.01 Biochemists

Research or study chemical composition and processes of living organisms that affect vital processes such as growth and aging to determine chemical actions and effects on organisms such as the action of foods, drugs, or other substances on body functions and tissues.

19-1021.02 Biophysicists

Research or study physical principles of living cells and organisms, their electrical and mechanical energy, and related phenomena.

19-1022.00 Microbiologists

Investigate the growth, structure, development, and other characteristics of microscopic organisms, such as bacteria, algae, or fungi. Includes medical microbiologists who study the relationship between organisms and disease or the effects of antibiotics on microorganisms.

19-1023.00 Zoologists and Wildlife Biologists

Study the origins, behavior, diseases, genetics, and life processes of animals and wildlife. May specialize in wildlife research and management, including the collection and analysis of biological data to determine the environmental effects of present and potential use of land and water areas.

19-1029.99 Biological Scientists, All Other²

All biological scientists not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-1031.00 Conservation Scientists¹

Manage, improve, and protect natural resources to maximize their use without damaging the environment. May conduct soil surveys and develop plans to eliminate soil erosion or to protect rangelands from fire and rodent damage. May instruct farmers, agricultural production managers, or ranchers in best ways to use crop rotation, contour plowing, or terracing to conserve soil and water; in the number and kind of livestock and forage plants best suited to particular ranges and in range; and farm improvements, such as fencing and reservoirs for stock watering.

19-1031.01 Soil Conservationists

Plan and develop coordinated practices for soil erosion control, soil and water conservation, and sound land use.

19-1031.02 Range Managers

Research or study range land management practices to provide sustained production of forage, livestock, and wildlife.

19-1031.03 Park Naturalists

Plan, develop, and conduct programs to inform public of historical, natural, and scientific features of national, state, or local park.

19-1032.00 Foresters

Manage forested lands for economic, recreational, and conservation purposes. May inventory the type, amount, and location of standing timber, appraise the timber's worth, negotiate the purchase, and draw up contracts for procurement. May determine how to conserve wildlife habitats, creek beds, water quality, and soil stability, and how best to comply with environmental regulations. May devise plans for planting and growing new trees, monitor trees for healthy growth, and determine the best time for harvesting. Develop forest management plans for public and privately-owned forested lands.

19-1041.00 Epidemiologists

Investigate and describe the determinants and distribution of disease, disability, and other health outcomes and develop the means for prevention and control.

19-1042.00 Medical Scientists, Except Epidemiologists

Conduct research dealing with the understanding of human diseases and the improvement of human health. Engage in clinical investigation or other research, production, technical writing, or related activities.

19-1099.99 Life Scientists, All Other²

All life scientists not listed separately.

19-2011.00 Astronomers

Observe, research, and interpret celestial and astronomical phenomena to increase basic knowledge and apply such information to practical problems.

19-2012.00 Physicists

Conduct research into the phases of physical phenomena, develop theories and laws on the basis of observation and experiments, and devise methods to apply laws and theories to industry and other fields.

19-2021.00 Atmospheric and Space Scientists

Investigate atmospheric phenomena and interpret meteorological data gathered by surface and air stations, satellites, and radar to prepare reports and forecasts for public and other uses.

19-2031.00 Chemists

Conduct qualitative and quantitative chemical analyses or chemical experiments in laboratories for quality or process control or to develop new products or knowledge.

19-2032.00 Materials Scientists

Research and study the structures and chemical properties of various natural and manmade materials, including metals, alloys, rubber, ceramics, semiconductors, polymers, and glass. Determine ways to strengthen or combine materials or develop new materials with new or specific properties for use in a variety of products and applications.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-2041.00 Environmental Scientists and Specialists, Including Health

Conduct research or perform investigation for the purpose of identifying, abating, or eliminating sources of pollutants or hazards that affect either the environment or the health of the population. Utilizing knowledge of various scientific disciplines may collect, synthesize, study, report, and take action based on data derived from measurements or observations of air, food, soil, water, and other sources.

19-2042.00 Geoscientists, Except Hydrologists and Geographers³

Study the composition, structure, and other physical aspects of the earth. May use geological, physics, and mathematics knowledge in exploration for oil, gas, minerals, or underground water; or in waste disposal, land reclamation, or other environmental problems. May study the earth's internal composition, atmospheres, oceans, and its magnetic, electrical, and gravitational forces. Includes mineralogists, crystallographers, paleontologists, stratigraphers, geodesists, and seismologists.

19-2042.01 Geologists

Study composition, structure, and history of the earth's crust; examine rocks, minerals, and fossil remains to identify and determine the sequence of processes affecting the development of the earth; apply knowledge of chemistry, physics, biology, and mathematics to explain these phenomena and to help locate mineral and petroleum deposits and underground water resources; prepare geologic reports and maps; and interpret research data to recommend further action for study.

19-2043.00 Hydrologists

Research the distribution, circulation, and physical properties of underground and surface waters; study the form and intensity of precipitation, its rate of infiltration into the soil, movement through the earth, and its return to the ocean and atmosphere.

19-2099.99 Physical Scientists, All Other²

All physical scientists not listed separately.

19-3011.00 Economists

Conduct research, prepare reports, or formulate plans to aid in solution of economic problems arising from production and distribution of goods and services. May collect and process economic and statistical data using econometric and sampling techniques.

19-3021.00 Market Research Analysts

Research market conditions in local, regional, or national areas to determine potential sales of a product or service. May gather information on competitors, prices, sales, and methods of marketing and distribution. May use survey results to create a marketing campaign based on regional preferences and buying habits.

19-3022.00 Survey Researchers

Design or conduct surveys. May supervise interviewers who conduct the survey in person or over the telephone. May present survey results to client.

19-3031.00 Clinical, Counseling, and School Psychologists¹

Diagnose and treat mental disorders; learning disabilities; and cognitive, behavioral, and emotional problems using individual, child, family, and group therapies. May design and implement behavior modification programs.

19-3031.01 Educational Psychologists

Investigate processes of learning and teaching and develop psychological principles and techniques applicable to educational problems.

19-3031.02 Clinical Psychologists

Diagnose or evaluate mental and emotional disorders of individuals through observation, interview, and psychological tests, and formulate and administer programs of treatment.

19-3031.03 Counseling Psychologists

Assess and evaluate individuals' problems through the use of case history, interview, and observation and provide individual or group counseling services to assist individuals in achieving more effective personal, social, educational, and vocational development and adjustment.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-3032.00 Industrial-Organizational Psychologists

Apply principles of psychology to personnel, administration, management, sales, and marketing problems. Activities may include policy planning; employee screening, training and development; and organizational development and analysis. May work with management to reorganize the work setting to improve worker productivity.

19-3039.99 Psychologists, All Other ²

All psychologists not listed separately.

19-3041.00 Sociologists

Study human society and social behavior by examining the groups and social institutions that people form, as well as various social, religious, political, and business organizations. May study the behavior and interaction of groups, trace their origin and growth, and analyze the influence of group activities on individual members.

19-3051.00 Urban and Regional Planners

Develop comprehensive plans and programs for use of land and physical facilities of local jurisdictions, such as towns, cities, counties, and metropolitan areas.

19-3091.00 Anthropologists and Archeologists ¹

Study the origin, development, and behavior of humans. May study the way of life, language, or physical characteristics of existing people in various parts of the world. May engage in systematic recovery and examination of material evidence, such as tools or pottery remaining from past human cultures, in order to determine the history, customs, and living habits of earlier civilizations.

19-3091.01 Anthropologists

Research or study the origins and physical, social, and cultural development and behavior of humans and the cultures and organizations they have created.

19-3091.02 Archeologists

Conduct research to reconstruct record of past human life and culture from human remains, artifacts, architectural features, and structures recovered through excavation, underwater recovery, or other means of discovery.

19-3092.00 Geographers

Study nature and use of areas of earth's surface, relating and interpreting interactions of physical and cultural phenomena. Conduct research on physical aspects of a region, including land forms, climates, soils, plants and animals, and conduct research on the spatial implications of human activities within a given area, including social characteristics, economic activities, and political organization, as well as researching interdependence between regions at scales ranging from local to global.

19-3093.00 Historians

Research, analyze, record, and interpret the past as recorded in sources, such as government and institutional records, newspapers and other periodicals, photographs, interviews, films, and unpublished manuscripts, such as personal diaries and letters.

19-3094.00 Political Scientists

Study the origin, development, and operation of political systems. Research a wide range of subjects, such as relations between the United States and foreign countries, the beliefs and institutions of foreign nations, or the politics of small towns or a major metropolis. May study topics, such as public opinion, political decision making, and ideology. May analyze the structure and operation of governments, as well as various political entities. May conduct public opinion surveys, analyze election results, or analyze public documents.

19-3099.99 Social Scientists and Related Workers, All Other ²

All social scientists and related workers not listed separately.

19-4011.00 Agricultural and Food Science Technicians ¹

Work with agricultural scientists in food, fiber, and animal research, production, and processing; assist with animal breeding and nutrition work; under supervision, conduct tests and experiments to improve yield and quality of crops or to increase the resistance of plants and animals to disease or insects. Includes technicians who assist food scientists or food technologists in the research, development, production technology, quality control, packaging, processing, and use of foods.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-4011.01 Agricultural Technicians

Set up and maintain laboratory and collect and record data to assist scientist in biology or related agricultural science experiments.

19-4011.02 Food Science Technicians

Perform standardized qualitative and quantitative tests to determine physical or chemical properties of food or beverage products.

19-4021.00 Biological Technicians

Assist biological and medical scientists in laboratories. Set up, operate, and maintain laboratory instruments and equipment, monitor experiments, make observations, and calculate and record results. May analyze organic substances, such as blood, food, and drugs.

19-4031.00 Chemical Technicians

Conduct chemical and physical laboratory tests to assist scientists in making qualitative and quantitative analyses of solids, liquids, and gaseous materials for purposes, such as research and development of new products or processes, quality control, maintenance of environmental standards, and other work involving experimental, theoretical, or practical application of chemistry and related sciences.

19-4041.00 Geological and Petroleum Technicians ¹

Assist scientists in the use of electrical, sonic, or nuclear measuring instruments in both laboratory and production activities to obtain data indicating potential sources of metallic ore, gas, or petroleum. Analyze mud and drill cuttings. Chart pressure, temperature, and other characteristics of wells or bore holes. Investigate and collect information leading to the possible discovery of new oil fields.

19-4041.01 Geological Data Technicians

Measure, record, and evaluate geological data, using sonic, electronic, electrical, seismic, or gravity-measuring instruments to prospect for oil or gas. May collect and evaluate core samples and cuttings.

19-4041.02 Geological Sample Test Technicians

Test and analyze geological samples, crude oil, or petroleum products to detect presence of petroleum, gas, or mineral deposits indicating potential for exploration and production, or to determine physical and chemical properties to ensure that products meet quality standards.

19-4051.00 Nuclear Technicians ¹

Assist scientists in both laboratory and production activities by performing technical tasks involving nuclear physics, primarily in operation, maintenance, production, and quality control support activities.

19-4051.01 Nuclear Equipment Operation Technicians

Operate equipment used for the release, control, and utilization of nuclear energy to assist scientists in laboratory and production activities.

19-4051.02 Nuclear Monitoring Technicians

Collect and test samples to monitor results of nuclear experiments and contamination of humans, facilities, and environment.

19-4061.00 Social Science Research Assistants ¹

Assist social scientists in laboratory, survey, and other social research. May perform publication activities, laboratory analysis, quality control, or data management. Normally these individuals work under the direct supervision of a social scientist and assist in those activities which are more routine.

19-4061.01 City Planning Aides

Compile data from various sources, such as maps, reports, and field and file investigations, for use by city planner in making planning studies.

19-4091.00 Environmental Science and Protection Technicians, Including Health

Performs laboratory and field tests to monitor the environment and investigate sources of pollution, including those that affect health. Under direction of an environmental scientist or specialist, may collect samples of gases, soil, water, and other materials for testing and take corrective actions as assigned.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

19-4092.00 Forensic Science Technicians

Collect, identify, classify, and analyze physical evidence related to criminal investigations. Perform tests on weapons or substances, such as fiber, hair, and tissue to determine significance to investigation. May testify as expert witnesses on evidence or crime laboratory techniques. May serve as specialists in area of expertise, such as ballistics, fingerprinting, handwriting, or biochemistry.

19-4093.00 Forest and Conservation Technicians

Compile data pertaining to size, content, condition, and other characteristics of forest tracts, under direction of foresters; train and lead forest workers in forest propagation, fire prevention and suppression. May assist conservation scientists in managing, improving, and protecting rangelands and wildlife habitats, and help provide technical assistance regarding the conservation of soil, water, and related natural resources.

19-4099.99 Life, Physical, and Social Science Technicians, All Other ²

All life, physical, and social science technicians not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

21-0000 COMMUNITY AND SOCIAL SERVICES OCCUPATIONS

21-1011.00 Substance Abuse and Behavioral Disorder Counselors

Counsel and advise individuals with alcohol, tobacco, drug, or other problems, such as gambling and eating disorders. May counsel individuals, families, or groups or engage in prevention programs.

21-1012.00 Educational, Vocational, and School Counselors

Counsel individuals and provide group educational and vocational guidance services.

21-1013.00 Marriage and Family Therapists

Diagnose and treat mental and emotional disorders, whether cognitive, affective, or behavioral, within the context of marriage and family systems. Apply psychotherapeutic and family systems theories and techniques in the delivery of professional services to individuals, couples, and families for the purpose of treating such diagnosed nervous and mental disorders.

21-1014.00 Mental Health Counselors

Counsel with emphasis on prevention. Work with individuals and groups to promote optimum mental health. May help individuals deal with addictions and substance abuse; family, parenting, and marital problems; suicide; stress management; problems with self-esteem; and issues associated with aging and mental and emotional health.

21-1015.00 Rehabilitation Counselors

Counsel individuals to maximize the independence and employability of persons coping with personal, social, and vocational difficulties that result from birth defects, illness, disease, accidents, or the stress of daily life. Coordinate activities for residents of care and treatment facilities. Assess client needs and design and implement rehabilitation programs that may include personal and vocational counseling, training, and job placement.

21-1019.99 Counselors, All Other ²

All counselors not listed separately.

21-1021.00 Child, Family, and School Social Workers

Provide social services and assistance to improve the social and psychological functioning of children and their families and to maximize the family well-being and the academic functioning of children. May assist single parents, arrange adoptions, and find foster homes for abandoned or abused children. In schools, they address such problems as teenage pregnancy, misbehavior, and truancy. May also advise teachers on how to deal with problem children.

21-1022.00 Medical and Public Health Social Workers

Provide persons, families, or vulnerable populations with the psychosocial support needed to cope with chronic, acute, or terminal illnesses, such as Alzheimer's, cancer, or AIDS. Services include advising family care givers, providing patient education and counseling, and making necessary referrals for other social services.

21-1023.00 Mental Health and Substance Abuse Social Workers

Assess and treat individuals with mental, emotional, or substance abuse problems, including abuse of alcohol, tobacco, and/or other drugs. Activities may include individual and group therapy, crisis intervention, case management, client advocacy, prevention, and education.

21-1029.99 Social Workers, All Other ²

All social workers not listed separately.

21-1091.00 Health Educators

Promote, maintain, and improve individual and community health by assisting individuals and communities to adopt healthy behaviors. Collect and analyze data to identify community needs prior to planning, implementing, monitoring, and evaluating programs designed to encourage healthy lifestyles, policies and environments. May also serve as a resource to assist individuals, other professionals, or the community, and may administer fiscal resources for health education programs.

21-1092.00 Probation Officers and Correctional Treatment Specialists

Provide social services to assist in rehabilitation of law offenders in custody or on probation or parole. Make recommendations for actions involving formulation of rehabilitation plan and treatment of offender, including conditional release and education and employment stipulations.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

21-1093.00 Social and Human Service Assistants

Assist professionals from a wide variety of fields, such as psychology, rehabilitation, or social work, to provide client services, as well as support for families. May assist clients in identifying available benefits and social and community services and help clients obtain them. May assist social workers with developing, organizing, and conducting programs to prevent and resolve problems relevant to substance abuse, human relationships, rehabilitation, or adult daycare.

21-1099.99 Community and Social Service Specialists, All Other ²

All community and social service specialists not listed separately.

21-2011.00 Clergy

Conduct religious worship and perform other spiritual functions associated with beliefs and practices of religious faith or denomination. Provide spiritual and moral guidance and assistance to members.

21-2021.00 Directors, Religious Activities and Education

Direct and coordinate activities of a denominational group to meet religious needs of students. Plan, direct, or coordinate church school programs designed to promote religious education among church membership. May provide counseling and guidance relative to marital, health, financial, and religious problems.

21-2099.99 Religious Workers, All Other ²

All religious workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

23-0000 LEGAL OCCUPATIONS

23-1011.00 Lawyers

Represent clients in criminal and civil litigation and other legal proceedings, draw up legal documents, and manage or advise clients on legal transactions. May specialize in a single area or may practice broadly in many areas of law.

23-1021.00 Administrative Law Judges, Adjudicators, and Hearing Officers

Conduct hearings to decide or recommend decisions on claims concerning government programs or other government-related matters and prepare decisions. Determine penalties or the existence and the amount of liability, or recommend the acceptance or rejection of claims, or compromise settlements.

23-1022.00 Arbitrators, Mediators, and Conciliators

Facilitate negotiation and conflict resolution through dialogue. Resolve conflicts outside of the court system by mutual consent of parties involved.

23-1023.00 Judges, Magistrate Judges, and Magistrates

Arbitrate, advise, adjudicate, or administer justice in a court of law. May sentence defendant in criminal cases according to government statutes. May determine liability of defendant in civil cases. May issue marriage licenses and perform wedding ceremonies.

23-2011.00 Paralegals and Legal Assistants

Assist lawyers by researching legal precedent, investigating facts, or preparing legal documents. Conduct research to support a legal proceeding, to formulate a defense, or to initiate legal action.

23-2091.00 Court Reporters

Use verbatim methods and equipment to capture, store, retrieve, and transcribe pretrial and trial proceedings or other information. Includes stenocaptioners who operate computerized stenographic captioning equipment to provide captions of live or prerecorded broadcasts for hearing-impaired viewers.

23-2092.00 Law Clerks

Assist lawyers or judges by researching or preparing legal documents. May meet with clients or assist lawyers and judges in court.

23-2093.00 Title Examiners, Abstractors, and Searchers ¹

Search real estate records, examine titles, or summarize pertinent legal or insurance details for a variety of purposes. May compile lists of mortgages, contracts, and other instruments pertaining to titles by searching public and private records for law firms, real estate agencies, or title insurance companies.

23-2093.01 Title Searchers

Compile list of mortgages, deeds, contracts, judgments, and other instruments (chain) pertaining to title by searching public and private records of real estate or title insurance company.

23-2093.02 Title Examiners and Abstractors

Title Examiners: Search public records and examine titles to determine legal condition of property title. Copy or summarize (abstracts) recorded documents which affect condition of title to property (e.g., mortgages, trust deeds, and contracts). May prepare and issue policy that guarantees legality of title. Abstractors: Summarize pertinent legal or insurance details or sections of statutes or case law from reference books for purpose of examination, proof, or ready reference. Search out titles to determine if title deed is correct.

23-2099.99 Legal Support Workers, All Other ²

All legal support workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

25-0000 EDUCATION, TRAINING, AND LIBRARY OCCUPATIONS

25-1011.00 Business Teachers, Postsecondary

Teach courses in business administration and management, such as accounting, finance, human resources, labor relations, marketing, and operations research.

25-1021.00 Computer Science Teachers, Postsecondary

Teach courses in computer science. May specialize in a field of computer science, such as the design and function of computers or operations and research analysis.

25-1022.00 Mathematical Science Teachers, Postsecondary

Teach courses pertaining to mathematical concepts, statistics, and actuarial science and to the application of original and standardized mathematical techniques in solving specific problems and situations.

25-1031.00 Architecture Teachers, Postsecondary

Teach courses in architecture and architectural design, such as architectural environmental design, interior architecture/design, and landscape architecture.

25-1032.00 Engineering Teachers, Postsecondary

Teach courses pertaining to the application of physical laws and principles of engineering for the development of machines, materials, instruments, processes, and services. Includes teachers of subjects, such as chemical, civil, electrical, industrial, mechanical, mineral, and petroleum engineering. Includes both teachers primarily engaged in teaching and those who do a combination of both teaching and research.

25-1041.00 Agricultural Sciences Teachers, Postsecondary

Teach courses in the agricultural sciences. Includes teachers of agronomy, dairy sciences, fisheries management, horticultural sciences, poultry sciences, range management, and agricultural soil conservation.

25-1042.00 Biological Science Teachers, Postsecondary

Teach courses in biological sciences.

25-1043.00 Forestry and Conservation Science Teachers, Postsecondary

Teach courses in environmental and conservation science.

25-1051.00 Atmospheric, Earth, Marine, and Space Sciences Teachers, Postsecondary

Teach courses in the physical sciences, except chemistry and physics.

25-1052.00 Chemistry Teachers, Postsecondary

Teach courses pertaining to the chemical and physical properties and compositional changes of substances. Work may include instruction in the methods of qualitative and quantitative chemical analysis. Includes both teachers primarily engaged in teaching, and those who do a combination of both teaching and research.

25-1053.00 Environmental Science Teachers, Postsecondary

Teach courses in environmental science.

25-1054.00 Physics Teachers, Postsecondary

Teach courses pertaining to the laws of matter and energy. Includes both teachers primarily engaged in teaching and those who do a combination of both teaching and research.

25-1061.00 Anthropology and Archeology Teachers, Postsecondary

Teach courses in anthropology or archeology.

25-1062.00 Area, Ethnic, and Cultural Studies Teachers, Postsecondary

Teach courses pertaining to the culture and development of an area (e.g., Latin America), an ethnic group, or any other group (e.g., women's studies, urban affairs).

25-1063.00 Economics Teachers, Postsecondary

Teach courses in economics.

25-1064.00 Geography Teachers, Postsecondary

Teach courses in geography.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

25-1065.00 Political Science Teachers, Postsecondary

Teach courses in political science, international affairs, and international relations.

25-1066.00 Psychology Teachers, Postsecondary

Teach courses in psychology, such as child, clinical, and developmental psychology, and psychological counseling.

25-1067.00 Sociology Teachers, Postsecondary

Teach courses in sociology.

25-1069.99 Social Sciences Teachers, Postsecondary, All Other ²

All postsecondary social sciences teachers not listed separately.

25-1071.00 Health Specialties Teachers, Postsecondary

Teach courses in health specialties, such as veterinary medicine, dentistry, pharmacy, therapy, laboratory technology, and public health.

25-1072.00 Nursing Instructors and Teachers, Postsecondary

Demonstrate and teach patient care in classroom and clinical units to nursing students. Includes both teachers primarily engaged in teaching and those who do a combination of both teaching and research.

25-1081.00 Education Teachers, Postsecondary

Teach courses pertaining to education, such as counseling, curriculum, guidance, instruction, teacher education, and teaching English as a second language.

25-1082.00 Library Science Teachers, Postsecondary

Teach courses in library science.

25-1111.00 Criminal Justice and Law Enforcement Teachers, Postsecondary

Teach courses in criminal justice, corrections, and law enforcement administration.

25-1112.00 Law Teachers, Postsecondary

Teach courses in law.

25-1113.00 Social Work Teachers, Postsecondary

Teach courses in social work.

25-1121.00 Art, Drama, and Music Teachers, Postsecondary

Teach courses in drama, music, and the arts including fine and applied art, such as painting and sculpture, or design and crafts.

25-1122.00 Communications Teachers, Postsecondary

Teach courses in communications, such as organizational communications, public relations, radio/television broadcasting, and journalism.

25-1123.00 English Language and Literature Teachers, Postsecondary

Teach courses in English language and literature, including linguistics and comparative literature.

25-1124.00 Foreign Language and Literature Teachers, Postsecondary

Teach courses in foreign (i.e., other than English) languages and literature.

25-1125.00 History Teachers, Postsecondary

Teach courses in human history and historiography.

25-1126.00 Philosophy and Religion Teachers, Postsecondary

Teach courses in philosophy, religion, and theology.

25-1191.00 Graduate Teaching Assistants

Assist department chairperson, faculty members, or other professional staff members in college or university by performing teaching or teaching-related duties, such as teaching lower level courses, developing teaching materials, preparing and giving examinations, and grading examinations or papers. Graduate assistants must be enrolled in a graduate school program. Graduate assistants who primarily perform non-teaching duties, such as laboratory research, should be reported in the occupational category related to the work performed.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

25-1192.00 Home Economics Teachers, Postsecondary

Teach courses in child care, family relations, finance, nutrition, and related subjects as pertaining to home management.

25-1193.00 Recreation and Fitness Studies Teachers, Postsecondary

Teach courses pertaining to recreation, leisure, and fitness studies, including exercise physiology and facilities management.

25-1194.00 Vocational Education Teachers Postsecondary

Teach or instruct vocational or occupational subjects at the postsecondary level (but at less than the baccalaureate) to students who have graduated or left high school. Includes correspondence school instructors; industrial, commercial and government training instructors; and adult education teachers and instructors who prepare persons to operate industrial machinery and equipment and transportation and communications equipment. Teaching may take place in public or private schools whose primary business is education or in a school associated with an organization whose primary business is other than education.

25-1199.99 Postsecondary Teachers, All Other ²

All postsecondary teachers not listed separately.

25-2011.00 Preschool Teachers, Except Special Education

Instruct children (normally up to 5 years of age) in activities designed to promote social, physical, and intellectual growth needed for primary school in preschool, day care center, or other child development facility. May be required to hold State certification.

25-2012.00 Kindergarten Teachers, Except Special Education

Teach elemental natural and social science, personal hygiene, music, art, and literature to children from 4 to 6 years old. Promote physical, mental, and social development. May be required to hold State certification.

25-2021.00 Elementary School Teachers, Except Special Education

Teach pupils in public or private schools at the elementary level basic academic, social, and other formative skills.

25-2022.00 Middle School Teachers, Except Special and Vocational Education

Teach students in public or private schools in one or more subjects at the middle, intermediate, or junior high level, which falls between elementary and senior high school as defined by applicable State laws and regulations.

25-2023.00 Vocational Education Teachers, Middle School

Teach or instruct vocational or occupational subjects at the middle school level.

25-2031.00 Secondary School Teachers, Except Special and Vocational Education

Instruct students in secondary public or private schools in one or more subjects at the secondary level, such as English, mathematics, or social studies. May be designated according to subject matter specialty, such as typing instructors, commercial teachers, or English teachers.

25-2032.00 Vocational Education Teachers, Secondary School

Teach or instruct vocational or occupational subjects at the secondary school level.

25-2041.00 Special Education Teachers, Preschool, Kindergarten, and Elementary School

Teach elementary and preschool school subjects to educationally and physically handicapped students. Includes teachers who specialize and work with audibly and visually handicapped students and those who teach basic academic and life processes skills to the mentally impaired.

25-2042.00 Special Education Teachers, Middle School

Teach middle school subjects to educationally and physically handicapped students. Includes teachers who specialize and work with audibly and visually handicapped students and those who teach basic academic and life processes skills to the mentally impaired.

25-2043.00 Special Education Teachers, Secondary School

Teach secondary school subjects to educationally and physically handicapped students. Includes teachers who specialize and work with audibly and visually handicapped students and those who teach basic academic and life processes skills to the mentally impaired.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

25-3011.00 Adult Literacy, Remedial Education, and GED Teachers and Instructors

Teach or instruct out-of-school youths and adults in remedial education classes, preparatory classes for the General Educational Development test, literacy, or English as a Second Language. Teaching may or may not take place in a traditional educational institution.

25-3021.00 Self-Enrichment Education Teachers

Teach or instruct courses other than those that normally lead to an occupational objective or degree. Courses may include self-improvement, nonvocational, and nonacademic subjects. Teaching may or may not take place in a traditional educational institution.

25-3099.99 Teachers and Instructors, All Other ²

All teachers and instructors not listed separately.

25-4011.00 Archivists

Appraise, edit, and direct safekeeping of permanent records and historically valuable documents. Participate in research activities based on archival materials.

25-4012.00 Curators

Administer affairs of museum and conduct research programs. Direct instructional, research, and public service activities of institution.

25-4013.00 Museum Technicians and Conservators

Prepare specimens, such as fossils, skeletal parts, lace, and textiles, for museum collection and exhibits. May restore documents or install, arrange, and exhibit materials.

25-4021.00 Librarians

Administer libraries and perform related library services. Work in a variety of settings, including public libraries, schools, colleges and universities, museums, corporations, government agencies, law firms, non-profit organizations, and healthcare providers. Tasks may include selecting, acquiring, cataloguing, classifying, circulating, and maintaining library materials; and furnishing reference, bibliographical, and readers' advisory services. May perform in-depth, strategic research, and synthesize, analyze, edit, and filter information. May set up or work with databases and information systems to catalogue and access information.

25-4031.00 Library Technicians

Assist librarians by helping readers in the use of library catalogs, databases, and indexes to locate books and other materials; and by answering questions that require only brief consultation of standard reference. Compile records; sort and shelf books; remove or repair damaged books; register patrons; check materials in and out of the circulation process. Replace materials in shelving area (stacks) or files. Includes bookmobile drivers who operate bookmobiles or light trucks that pull trailers to specific locations on a predetermined schedule and assist with providing services in mobile libraries.

25-9011.00 Audio-Visual Collections Specialists

Prepare, plan, and operate audio-visual teaching aids for use in education. May record, catalogue, and file audio-visual materials.

25-9021.00 Farm and Home Management Advisors

Advise, instruct, and assist individuals and families engaged in agriculture, agricultural-related processes, or home economics activities. Demonstrate procedures and apply research findings to solve problems; instruct and train in product development, sales, and the utilization of machinery and equipment to promote general welfare. Includes county agricultural agents, feed and farm management advisers, home economists, and extension service advisers.

25-9031.00 Instructional Coordinators

Develop instructional material, coordinate educational content, and incorporate current technology in specialized fields that provide guidelines to educators and instructors for developing curricula and conducting courses.

25-9041.00 Teacher Assistants

Perform duties that are instructional in nature or deliver direct services to students or parents. Serve in a position for which a teacher or another professional has ultimate responsibility for the design and implementation of educational programs and services.

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² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

25-9099.99 Education, Training, and Library Workers, All Other ²

All education, training, and library workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

27-0000 ARTS, DESIGN, ENTERTAINMENT, SPORTS, AND MEDIA OCCUPATIONS

27-1011.00 Art Directors

Formulate design concepts and presentation approaches, and direct workers engaged in art work, layout design, and copy writing for visual communications media, such as magazines, books, newspapers, and packaging.

27-1012.00 Craft Artists

Create or reproduce hand-made objects for sale and exhibition using a variety of techniques, such as welding, weaving, pottery, and needlecraft.

27-1013.00 Fine Artists, Including Painters, Sculptors, and Illustrators ¹

Create original artwork using any of a wide variety of mediums and techniques, such as painting and sculpture.

27-1013.01 Painters and Illustrators

Paint or draw subject material to produce original artwork or illustrations, using watercolors, oils, acrylics, tempera, or other paint mediums.

27-1013.02 Sketch Artists

Sketch likenesses of subjects according to observation or descriptions either to assist law enforcement agencies in identifying suspects, to depict court room scenes, or for entertainment purposes of patrons, using mediums such as pencil, charcoal, and pastels.

27-1013.03 Cartoonists

Create original artwork using any of a wide variety of mediums and techniques, such as painting and sculpture.

27-1013.04 Sculptors

Design and construct three-dimensional art works, using materials such as stone, wood, plaster, and metal and employing various manual and tool techniques.

27-1014.00 Multi-Media Artists and Animators

Create special effects, animation, or other visual images using film, video, computers, or other electronic tools and media for use in products or creations, such as computer games, movies, music videos, and commercials.

27-1019.99 Artists and Related Workers, All Other ²

All artists and related workers not listed separately.

27-1021.00 Commercial and Industrial Designers

Develop and design manufactured products, such as cars, home appliances, and children's toys. Combine artistic talent with research on product use, marketing, and materials to create the most functional and appealing product design.

27-1022.00 Fashion Designers

Design clothing and accessories. Create original garments or design garments that follow well established fashion trends. May develop the line of color and kinds of materials.

27-1023.00 Floral Designers

Design, cut, and arrange live, dried, or artificial flowers and foliage.

27-1024.00 Graphic Designers

Design or create graphics to meet specific commercial or promotional needs, such as packaging, displays, or logos. May use a variety of mediums to achieve artistic or decorative effects.

27-1025.00 Interior Designers

Plan, design, and furnish interiors of residential, commercial, or industrial buildings. Formulate design which is practical, aesthetic, and conducive to intended purposes, such as raising productivity, selling merchandise, or improving life style. May specialize in a particular field, style, or phase of interior design.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

27-1026.00 Merchandise Displayers and Window Trimmers

Plan and erect commercial displays, such as those in windows and interiors of retail stores and at trade exhibitions.

27-1027.00 Set and Exhibit Designers¹

Design special exhibits and movie, television, and theater sets. May study scripts, confer with directors, and conduct research to determine appropriate architectural styles.

27-1027.01 Set Designers

Design sets for theatrical, motion picture, and television productions.

27-1027.02 Exhibit Designers

Plan, design, and oversee construction and installation of permanent and temporary exhibits and displays.

27-1029.99 Designers, All Other²

All designers not listed separately.

27-2011.00 Actors

Play parts in stage, television, radio, video, or motion picture productions for entertainment, information, or instruction. Interpret serious or comic role by speech, gesture, and body movement to entertain or inform audience. May dance and sing.

27-2012.00 Producers and Directors¹

Produce or direct stage, television, radio, video, or motion picture productions for entertainment, information, or instruction. Responsible for creative decisions, such as interpretation of script, choice of guests, set design, sound, special effects, and choreography.

27-2012.01 Producers

Plan and coordinate various aspects of radio, television, stage, or motion picture production, such as selecting script, coordinating writing, directing and editing, and arranging financing.

27-2012.02 Directors- Stage, Motion Pictures, Television, and Radio

Interpret script, conduct rehearsals, and direct activities of cast and technical crew for stage, motion pictures, television, or radio programs.

27-2012.03 Program Directors

Direct and coordinate activities of personnel engaged in preparation of radio or television station program schedules and programs, such as sports or news.

27-2012.04 Talent Directors

Audition and interview performers to select most appropriate talent for parts in stage, television, radio, or motion picture productions.

27-2012.05 Technical Directors/Managers

Coordinate activities of technical departments, such as taping, editing, engineering, and maintenance, to produce radio or television programs.

27-2021.00 Athletes and Sports Competitors

Compete in athletic events.

27-2022.00 Coaches and Scouts

Instruct or coach groups or individuals in the fundamentals of sports. Demonstrate techniques and methods of participation. May evaluate athletes' strengths and weaknesses as possible recruits or to improve the athletes' technique to prepare them for competition. Those required to hold teaching degrees should be reported in the appropriate teaching category.

27-2023.00 Umpires, Referees, and Other Sports Officials

Officiate at competitive athletic or sporting events. Detect infractions of rules and decide penalties according to established regulations.

27-2031.00 Dancers

Perform dances. May also sing or act.

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² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

27-2032.00 Choreographers

Create and teach dance. May direct and stage presentations.

27-2041.00 Music Directors and Composers ¹

Conduct, direct, plan, and lead instrumental or vocal performances by musical groups, such as orchestras, choirs, and glee clubs. Includes arrangers, composers, choral directors, and orchestrators.

27-2041.01 Music Directors

Direct and conduct instrumental or vocal performances by musical groups, such as orchestras or choirs.

27-2041.02 Music Arrangers and Orchestrators

Write and transcribe musical scores.

27-2041.03 Composers

Compose music for orchestra, choral group, or band.

27-2042.00 Musicians and Singers ¹

Play one or more musical instruments or entertain by singing songs in recital, in accompaniment, or as a member of an orchestra, band, or other musical group. Musical performers may entertain on-stage, radio, TV, film, video, or record in studios.

27-2042.01 Singers

Sing songs on stage, radio, television, or motion pictures.

27-2042.02 Musicians, Instrumental

Play one or more musical instruments in recital, in accompaniment, or as members of an orchestra, band, or other musical group.

27-2099.99 Entertainers and Performers, Sports and Related Workers, All Other ²

All entertainers and performers, sports and related workers not listed separately.

27-3011.00 Radio and Television Announcers

Talk on radio or television. May interview guests, act as master of ceremonies, read news flashes, identify station by giving call letters, or announce song title and artist.

27-3012.00 Public Address System and Other Announcers

Make announcements over loud speaker at sporting or other public events. May act as master of ceremonies or disc jockey at weddings, parties, clubs, or other gathering places.

27-3021.00 Broadcast News Analysts

Analyze, interpret, and broadcast news received from various sources.

27-3022.00 Reporters and Correspondents

Collect and analyze facts about newsworthy events by interview, investigation, or observation. Report and write stories for newspaper, news magazine, radio, or television.

27-3031.00 Public Relations Specialists

Engage in promoting or creating good will for individuals, groups, or organizations by writing or selecting favorable publicity material and releasing it through various communications media. May prepare and arrange displays, and make speeches.

27-3041.00 Editors

Perform variety of editorial duties, such as laying out, indexing, and revising content of written materials, in preparation for final publication.

27-3042.00 Technical Writers

Write technical materials, such as equipment manuals, appendices, or operating and maintenance instructions. May assist in layout work.

27-3043.00 Writers and Authors ¹

Originate and prepare written material, such as scripts, stories, advertisements, and other material.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

27-3043.01 Poets and Lyricists

Write poetry or song lyrics for publication or performance.

27-3043.02 Creative Writers

Create original written works, such as plays or prose, for publication or performance.

27-3043.03 Caption Writers

Write caption phrases of dialogue for hearing-impaired and foreign language-speaking viewers of movie or television productions.

27-3043.04 Copy Writers

Write advertising copy for use by publication or broadcast media to promote sale of goods and services.

27-3091.00 Interpreters and Translators

Translate or interpret written, oral, or sign language text into another language for others.

27-3099.99 Media and Communication Workers, All Other ²

All media and communication workers not listed separately.

27-4011.00 Audio and Video Equipment Technicians

Set up or set up and operate audio and video equipment including microphones, sound speakers, video screens, projectors, video monitors, recording equipment, connecting wires and cables, sound and mixing boards, and related electronic equipment for concerts, sports events, meetings and conventions, presentations, and news conferences. May also set up and operate associated spotlights and other custom lighting systems.

27-4012.00 Broadcast Technicians

Set up, operate, and maintain the electronic equipment used to transmit radio and television programs. Control audio equipment to regulate volume level and quality of sound during radio and television broadcasts. Operate radio transmitter to broadcast radio and television programs.

27-4013.00 Radio Operators

Receive and transmit communications using radiotelegraph or radiotelephone equipment in accordance with government regulations. May repair equipment.

27-4014.00 Sound Engineering Technicians

Operate machines and equipment to record, synchronize, mix, or reproduce music, voices, or sound effects in sporting arenas, theater productions, recording studios, or movie and video productions.

27-4021.00 Photographers ¹

Photograph persons, subjects, merchandise, or other commercial products. May develop negatives and produce finished prints.

27-4021.01 Professional Photographers

Photograph subjects or news worthy events, using still cameras, color or black-and-white film, and variety of photographic accessories.

27-4021.02 Photographers, Scientific

Photograph variety of subject material to illustrate or record scientific/medical data or phenomena, utilizing knowledge of scientific procedures and photographic technology and techniques.

27-4031.00 Camera Operators, Television, Video, and Motion Picture

Operate television, video, or motion picture camera to photograph images or scenes for various purposes, such as TV broadcasts, advertising, video production, or motion pictures.

27-4032.00 Film and Video Editors

Edit motion picture soundtracks, film, and video.

27-4099.99 Media and Communication Equipment Workers, All Other ²

All media and communication equipment workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

29-0000 HEALTHCARE PRACTITIONER AND TECHNICAL OCCUPATIONS

29-1011.00 Chiropractors

Adjust spinal column and other articulations of the body to correct abnormalities of the human body believed to be caused by interference with the nervous system. Examine patient to determine nature and extent of disorder. Manipulate spine or other involved area. May utilize supplementary measures, such as exercise, rest, water, light, heat, and nutritional therapy.

29-1021.00 Dentists, General

Diagnose and treat diseases, injuries, and malformations of teeth and gums and related oral structures. May treat diseases of nerve, pulp, and other dental tissues affecting vitality of teeth.

29-1022.00 Oral and Maxillofacial Surgeons

Perform surgery on mouth, jaws, and related head and neck structure to execute difficult and multiple extractions of teeth, to remove tumors and other abnormal growths, to correct abnormal jaw relations by mandibular or maxillary revision, to prepare mouth for insertion of dental prosthesis, or to treat fractured jaws.

29-1023.00 Orthodontists

Examine, diagnose, and treat dental malocclusions and oral cavity anomalies. Design and fabricate appliances to realign teeth and jaws to produce and maintain normal function and to improve appearance.

29-1024.00 Prosthodontists

Construct oral prostheses to replace missing teeth and other oral structures to correct natural and acquired deformation of mouth and jaws, to restore and maintain oral function, such as chewing and speaking, and to improve appearance.

29-1029.99 Dentists, All Other Specialists ²

All dentists not listed separately.

29-1031.00 Dietitians and Nutritionists

Plan and conduct food service or nutritional programs to assist in the promotion of health and control of disease. May supervise activities of a department providing quantity food services, counsel individuals, or conduct nutritional research.

29-1041.00 Optometrists

Diagnose, manage, and treat conditions and diseases of the human eye and visual system. Examine eyes and visual system, diagnose problems or impairments, prescribe corrective lenses, and provide treatment. May prescribe therapeutic drugs to treat specific eye conditions.

29-1051.00 Pharmacists

Compound and dispense medications following prescriptions issued by physicians, dentists, or other authorized medical practitioners.

29-1061.00 Anesthesiologists

Administer anesthetics during surgery or other medical procedures.

29-1062.00 Family and General Practitioners

Diagnose, treat, and help prevent diseases and injuries that commonly occur in the general population.

29-1063.00 Internists, General

Diagnose and provide non-surgical treatment of diseases and injuries of internal organ systems. Provide care mainly for adults who have a wide range of problems associated with the internal organs.

29-1064.00 Obstetricians and Gynecologists

Diagnose, treat, and help prevent diseases of women, especially those affecting the reproductive system and the process of childbirth.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

29-1065.00 Pediatricians, General

Diagnose, treat, and help prevent children's diseases and injuries.

29-1066.00 Psychiatrists

Diagnose, treat, and help prevent disorders of the mind.

29-1067.00 Surgeons

Treat diseases, injuries, and deformities by invasive methods, such as manual manipulation or by using instruments and appliances.

29-1069.99 Physicians and Surgeons, All Other ²

All physicians and surgeons not listed separately.

29-1071.00 Physician Assistants

Provide healthcare services typically performed by a physician, under the supervision of a physician. Conduct complete physicals, provide treatment, and counsel patients. May, in some cases, prescribe medication. Must graduate from an accredited educational program for physician assistants.

29-1081.00 Podiatrists

Diagnose and treat diseases and deformities of the human foot.

29-1111.00 Registered Nurses

Assess patient health problems and needs, develop and implement nursing care plans, and maintain medical records. Administer nursing care to ill, injured, convalescent, or disabled patients. May advise patients on health maintenance and disease prevention or provide case management. Licensing or registration required. Includes advance practice nurses such as: nurse practitioners, clinical nurse specialists, certified nurse midwives, and certified registered nurse anesthetists. Advanced practice nursing is practiced by RNs who have specialized formal, post-basic education and who function in highly autonomous and specialized roles.

29-1121.00 Audiologists

Assess and treat persons with hearing and related disorders. May fit hearing aids and provide auditory training. May perform research related to hearing problems.

29-1122.00 Occupational Therapists

Assess, plan, organize, and participate in rehabilitative programs that help restore vocational, homemaking, and daily living skills, as well as general independence, to disabled persons.

29-1123.00 Physical Therapists

Assess, plan, organize, and participate in rehabilitative programs that improve mobility, relieve pain, increase strength, and decrease or prevent deformity of patients suffering from disease or injury.

29-1124.00 Radiation Therapists

Provide radiation therapy to patients as prescribed by a radiologist according to established practices and standards. Duties may include reviewing prescription and diagnosis; acting as liaison with physician and supportive care personnel; preparing equipment, such as immobilization, treatment, and protection devices; and maintaining records, reports, and files. May assist in dosimetry procedures and tumor localization.

29-1125.00 Recreational Therapists

Plan, direct, or coordinate medically-approved recreation programs for patients in hospitals, nursing homes, or other institutions. Activities include sports, trips, dramatics, social activities, and arts and crafts. May assess a patient condition and recommend appropriate recreational activity.

29-1126.00 Respiratory Therapists

Assess, treat, and care for patients with breathing disorders. Assume primary responsibility for all respiratory care modalities, including the supervision of respiratory therapy technicians. Initiate and conduct therapeutic procedures; maintain patient records; and select, assemble, check, and operate equipment.

29-1127.00 Speech-Language Pathologists

Assess and treat persons with speech, language, voice, and fluency disorders. May select alternative communication systems and teach their use. May perform research related to speech and language problems.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

29-1129.99 Therapists, All Other²

All therapists not listed separately.

29-1131.00 Veterinarians

Diagnose and treat diseases and dysfunctions of animals. May engage in a particular function, such as research and development, consultation, administration, technical writing, sale or production of commercial products, or rendering of technical services to commercial firms or other organizations. Includes veterinarians who inspect livestock.

29-1199.99 Health Diagnosing and Treating Practitioners, All Other²

All health diagnosing and treating practitioners not listed separately.

29-2011.00 Medical and Clinical Laboratory Technologists

Perform complex medical laboratory tests for diagnosis, treatment, and prevention of disease. May train or supervise staff.

29-2012.00 Medical and Clinical Laboratory Technicians

Perform routine medical laboratory tests for the diagnosis, treatment, and prevention of disease. May work under the supervision of a medical technologist.

29-2021.00 Dental Hygienists

Clean teeth and examine oral areas, head, and neck for signs of oral disease. May educate patients on oral hygiene, take and develop X-rays, or apply fluoride or sealants.

29-2031.00 Cardiovascular Technologists and Technicians

Conduct tests on pulmonary or cardiovascular systems of patients for diagnostic purposes. May conduct or assist in electrocardiograms, cardiac catheterizations, pulmonary-functions, lung capacity, and similar tests.

29-2032.00 Diagnostic Medical Sonographers

Produce ultrasonic recordings of internal organs for use by physicians.

29-2033.00 Nuclear Medicine Technologists

Prepare, administer, and measure radioactive isotopes in therapeutic, diagnostic, and tracer studies utilizing a variety of radioisotope equipment. Prepare stock solutions of radioactive materials and calculate doses to be administered by radiologists. Subject patients to radiation. Execute blood volume, red cell survival, and fat absorption studies following standard laboratory techniques.

29-2034.00 Radiologic Technologists and Technicians¹

Take X-rays and CAT scans or administer nonradioactive materials into patient's blood stream for diagnostic purposes. Includes technologists who specialize in other modalities, such as computed tomography and magnetic resonance. Includes workers whose primary duties are to demonstrate portions of the human body on X-ray film or fluoroscopic screen.

29-2034.01 Radiologic Technologists

Take X-rays and CAT scans or administer nonradioactive materials into patient's blood stream for diagnostic purposes. Includes technologists who specialize in other modalities, such as computed tomography, ultrasound, and magnetic resonance.

29-2034.02 Radiologic Technicians

Maintain and use equipment and supplies necessary to demonstrate portions the human body on X-ray film or fluoroscopic screen for diagnostic purposes.

29-2041.00 Emergency Medical Technicians and Paramedics

Assess injuries, administer emergency medical care, and extricate trapped individuals. Transport injured or sick persons to medical facilities.

29-2051.00 Dietetic Technicians

Assist dietitians in the provision of food service and nutritional programs. Under the supervision of dietitians, may plan and produce meals based on established guidelines, teach principles of food and nutrition, or counsel individuals.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

29-2052.00 Pharmacy Technicians

Prepare medications under the direction of a pharmacist. May measure, mix, count out, label, and record amounts and dosages of medications.

29-2053.00 Psychiatric Technicians

Care for mentally impaired or emotionally disturbed individuals, following physician instructions and hospital procedures. Monitor patients' physical and emotional well-being and report to medical staff. May participate in rehabilitation and treatment programs, help with personal hygiene, and administer oral medications and hypodermic injections.

29-2054.00 Respiratory Therapy Technicians

Provide specific, well defined respiratory care procedures under the direction of respiratory therapists and physicians.

29-2055.00 Surgical Technologists

Assist in operations, under the supervision of surgeons, registered nurses, or other surgical personnel. May help set up operating room, prepare and transport patients for surgery, adjust lights and equipment, pass instruments and other supplies to surgeons and surgeon's assistants, hold retractors, cut sutures, and help count sponges, needles, supplies, and instruments.

29-2056.00 Veterinary Technologists and Technicians

Perform medical tests in a laboratory environment for use in the treatment and diagnosis of diseases in animals. Prepare vaccines and serums for prevention of diseases. Prepare tissue samples, take blood samples, and execute laboratory tests, such as urinalysis and blood counts. Clean and sterilize instruments and materials and maintain equipment and machines.

29-2061.00 Licensed Practical and Licensed Vocational Nurses

Care for ill, injured, convalescent, or disabled persons in hospitals, nursing homes, clinics, private homes, group homes, and similar institutions. May work under the supervision of a registered nurse. Licensing required.

29-2071.00 Medical Records and Health Information Technicians

Compile, process, and maintain medical records of hospital and clinic patients in a manner consistent with medical, administrative, ethical, legal, and regulatory requirements of the health care system. Process, maintain, compile, and report patient information for health requirements and standards.

29-2081.00 Opticians, Dispensing

Design, measure, fit, and adapt lenses and frames for client according to written optical prescription or specification. Assist client with selecting frames. Measure customer for size of eyeglasses and coordinate frames with facial and eye measurements and optical prescription. Prepare work order for optical laboratory containing instructions for grinding and mounting lenses in frames. Verify exactness of finished lens spectacles. Adjust frame and lens position to fit client. May shape or reshape frames.

29-2091.00 Orthotists and Prosthetists

Assist patients with disabling conditions of limbs and spine or with partial or total absence of limb by fitting and preparing orthopedic braces or prostheses.

29-2099.99 Health Technologists and Technicians, All Other ²

All health technologists and technicians not listed separately.

29-9011.00 Occupational Health and Safety Specialists

Review, evaluate, and analyze work environments and design programs and procedures to control, eliminate, and prevent disease or injury caused by chemical, physical, and biological agents or ergonomic factors. May conduct inspections and enforce adherence to laws and regulations governing the health and safety of individuals. May be employed in the public or private sector.

29-9012.00 Occupational Health and Safety Technicians

Collect data on work environments for analysis by occupational health and safety specialists. Implement and conduct evaluation of programs designed to limit chemical, physical, biological, and ergonomic risks to workers.

29-9091.00 Athletic Trainers

Evaluate, advise, and treat athletes to assist recovery from injury, avoid injury, or maintain peak physical fitness.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

29-9099.99 Healthcare Practitioners and Technical Workers, All Other²

All healthcare practitioners and technical workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

31-0000 HEALTHCARE SUPPORT OCCUPATIONS

31-1011.00 Home Health Aides

Provide routine, personal healthcare, such as bathing, dressing, or grooming, to elderly, convalescent, or disabled persons in the home of patients or in a residential care facility.

31-1012.00 Nursing Aides, Orderlies, and Attendants

Provide basic patient care under direction of nursing staff. Perform duties, such as feed, bathe, dress, groom, or move patients, or change linens.

31-1013.00 Psychiatric Aides

Assist mentally impaired or emotionally disturbed patients, working under direction of nursing and medical staff.

31-2011.00 Occupational Therapist Assistants

Assist occupational therapists in providing occupational therapy treatments and procedures. May, in accordance with State laws, assist in development of treatment plans, carry out routine functions, direct activity programs, and document the progress of treatments. Generally requires formal training.

31-2012.00 Occupational Therapist Aides

Under close supervision of an occupational therapist or occupational therapy assistant, perform only delegated, selected, or routine tasks in specific situations. These duties include preparing patient and treatment room.

31-2021.00 Physical Therapist Assistants

Assist physical therapists in providing physical therapy treatments and procedures. May, in accordance with State laws, assist in the development of treatment plans, carry out routine functions, document the progress of treatment, and modify specific treatments in accordance with patient status and within the scope of treatment plans established by a physical therapist. Generally requires formal training.

31-2022.00 Physical Therapist Aides

Under close supervision of a physical therapist or physical therapy assistant, perform only delegated, selected, or routine tasks in specific situations. These duties include preparing the patient and the treatment area.

31-9011.00 Massage Therapists

Massage customers for hygienic or remedial purposes.

31-9091.00 Dental Assistants

Assist dentist, set up patient and equipment, and keep records.

31-9092.00 Medical Assistants

Perform administrative and certain clinical duties under the direction of physician. Administrative duties may include scheduling appointments, maintaining medical records, billing, and coding for insurance purposes. Clinical duties may include taking and recording vital signs and medical histories, preparing patients for examination, drawing blood, and administering medications as directed by physician.

31-9093.00 Medical Equipment Preparers

Prepare, sterilize, install, or clean laboratory or healthcare equipment. May perform routine laboratory tasks and operate or inspect equipment.

31-9094.00 Medical Transcriptionists

Use transcribing machines with headset and foot pedal to listen to recordings by physicians and other healthcare professionals dictating a variety of medical reports, such as emergency room visits, diagnostic imaging studies, operations, chart reviews, and final summaries. Transcribe dictated reports and translate medical jargon and abbreviations into their expanded forms. Edit as necessary and return reports in either printed or electronic form to the dictator for review and signature, or correction.

31-9095.00 Pharmacy Aides

Record drugs delivered to the pharmacy, store incoming merchandise, and inform the supervisor of stock needs. May operate cash register and accept prescriptions for filling.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

31-9096.00 Veterinary Assistants and Laboratory Animal Caretakers

Feed, water, and examine pets and other nonfarm animals for signs of illness, disease, or injury in laboratories and animal hospitals and clinics. Clean and disinfect cages and work areas, and sterilize laboratory and surgical equipment. May provide routine post-operative care, administer medication orally or topically, or prepare samples for laboratory examination under the supervision of veterinary or laboratory animal technologists or technicians, veterinarians, or scientists.

31-9099.99 Healthcare Support Workers, All Other ²

All healthcare support workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

33-0000 PROTECTIVE SERVICE OCCUPATIONS

33-1011.00 First-Line Supervisors/Managers of Correctional Officers

Supervise and coordinate activities of correctional officers and jailers.

33-1012.00 First-Line Supervisors/Managers of Police and Detectives

Supervise and coordinate activities of members of police force.

33-1021.00 First-Line Supervisors/Managers of Fire Fighting and Prevention Workers ¹

Supervise and coordinate activities of workers engaged in fire fighting and fire prevention and control.

33-1021.01 Municipal Fire Fighting and Prevention Supervisors

Supervise fire fighters who control and extinguish municipal fires, protect life and property, and conduct rescue efforts.

33-1021.02 Forest Fire Fighting and Prevention Supervisors

Supervise fire fighters who control and suppress fires in forests or vacant public land.

33-1099.99 First-Line Supervisors/Managers, Protective Service Workers, All Other ²

All protective service supervisors not listed separately above.

33-2011.00 Fire Fighters ¹

Control and extinguish fires or respond to emergency situations where life, property, or the environment is at risk. Duties may include fire prevention, emergency medical service, hazardous material response, search and rescue, and disaster management.

33-2011.01 Municipal Fire Fighters

Control and extinguish municipal fires, protect life and property and conduct rescue efforts.

33-2011.02 Forest Fire Fighters

Control and suppress fires in forests or vacant public land.

33-2021.00 Fire Inspectors and Investigators ¹

Inspect buildings to detect fire hazards and enforce local ordinances and State laws. Investigate and gather facts to determine cause of fires and explosions.

33-2021.01 Fire Inspectors

Inspect buildings and equipment to detect fire hazards and enforce state and local regulations.

33-2021.02 Fire Investigators

Conduct investigations to determine causes of fires and explosions.

33-2022.00 Forest Fire Inspectors and Prevention Specialists

Enforce fire regulations and inspect for forest fire hazards. Report forest fires and weather conditions.

33-3011.00 Bailiffs

Maintain order in courts of law.

33-3012.00 Correctional Officers and Jailers

Guard inmates in penal or rehabilitative institution in accordance with established regulations and procedures. May guard prisoners in transit between jail, courtroom, prison, or other point. Includes deputy sheriffs and police who spend the majority of their time guarding prisoners in correctional institutions.

33-3021.00 Detectives and Criminal Investigators ¹

Conduct investigations related to suspected violations of Federal, State, or local laws to prevent or solve crimes.

33-3021.01 Police Detectives

Conduct investigations to prevent crimes or solve criminal cases.

33-3021.02 Police Identification and Records Officers

Collect evidence at crime scene, classify and identify fingerprints, and photograph evidence for use in criminal and civil cases.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

33-3021.03 Criminal Investigators and Special Agents

Investigate alleged or suspected criminal violations of Federal, state, or local laws to determine if evidence is sufficient to recommend prosecution.

33-3021.04 Child Support, Missing Persons, and Unemployment Insurance Fraud Investigators

Conduct investigations to locate, arrest, and return fugitives and persons wanted for non-payment of support payments and unemployment insurance fraud, and to locate missing persons.

33-3021.05 Immigration and Customs Inspectors

Investigate and inspect persons, common carriers, goods, and merchandise, arriving in or departing from the United States or between states to detect violations of immigration and customs laws and regulations.

33-3031.00 Fish and Game Wardens

Patrol assigned area to prevent fish and game law violations. Investigate reports of damage to crops or property by wildlife. Compile biological data.

33-3041.00 Parking Enforcement Workers

Patrol assigned area, such as public parking lot or section of city to issue tickets to overtime parking violators and illegally parked vehicles.

33-3051.00 Police and Sheriff's Patrol Officers¹

Maintain order, enforce laws and ordinances, and protect life and property in an assigned patrol district. Perform combination of following duties: patrol a specific area on foot or in a vehicle; direct traffic; issue traffic summonses; investigate accidents; apprehend and arrest suspects, or serve legal processes of courts.

33-3051.01 Police Patrol Officers

Patrol assigned area to enforce laws and ordinances, regulate traffic, control crowds, prevent crime, and arrest violators.

33-3051.02 Highway Patrol Pilots

Pilot aircraft to patrol highway and enforce traffic laws.

33-3051.03 Sheriffs and Deputy Sheriffs

Enforce law and order in rural or unincorporated districts or serve legal processes of courts. May patrol courthouse, guard court or grand jury, or escort defendants.

33-3052.00 Transit and Railroad Police

Protect and police railroad and transit property, employees, or passengers.

33-9011.00 Animal Control Workers

Handle animals for the purpose of investigations of mistreatment, or control of abandoned, dangerous, or unattended animals.

33-9021.00 Private Detectives and Investigators

Detect occurrences of unlawful acts or infractions of rules in private establishment, or seek, examine, and compile information for client.

33-9031.00 Gaming Surveillance Officers and Gaming Investigators

Act as oversight and security agent for management and customers. Observe casino or casino hotel operation for irregular activities such as cheating or theft by either employees or patrons. May utilize one-way mirrors above the casino floor, cashier's cage, and from desk. Use of audio/video equipment is also common to observe operation of the business. Usually required to provide verbal and written reports of all violations and suspicious behavior to supervisor.

33-9032.00 Security Guards

Guard, patrol, or monitor premises to prevent theft, violence, or infractions of rules.

33-9091.00 Crossing Guards

Guide or control vehicular or pedestrian traffic at such places as streets, schools, railroad crossings or construction sites.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

33-9092.00 Lifeguards, Ski Patrol, and Other Recreational Protective Service Workers

Monitor recreational areas, such as pools, beaches, or ski slopes to provide assistance and protection to participants.

33-9099.99 Protective Service Workers, All Other ²

All protective service workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

35-0000 FOOD PREPARATION AND SERVING RELATED OCCUPATIONS

35-1011.00 Chefs and Head Cooks

Direct the preparation, seasoning, and cooking of salads, soups, fish, meats, vegetables, desserts, or other foods. May plan and price menu items, order supplies, and keep records and accounts. May participate in cooking.

35-1012.00 First-Line Supervisors/Managers of Food Preparation and Serving Workers

Supervise workers engaged in preparing and serving food.

35-2011.00 Cooks, Fast Food

Prepare and cook food in a fast food restaurant with a limited menu. Duties of the cooks are limited to preparation of a few basic items and normally involve operating large-volume single-purpose cooking equipment.

35-2012.00 Cooks, Institution and Cafeteria

Prepare and cook large quantities of food for institutions, such as schools, hospitals, or cafeterias.

35-2013.00 Cooks, Private Household

Prepare meals in private homes.

35-2014.00 Cooks, Restaurant

Prepare, season, and cook soups, meats, vegetables, desserts, or other foodstuffs in restaurants. May order supplies, keep records and accounts, price items on menu, or plan menu.

35-2015.00 Cooks, Short Order

Prepare and cook to order a variety of foods that require only a short preparation time. May take orders from customers and serve patrons at counters or tables.

35-2019.99 Cooks, All Other ²

All cooks not listed separately.

35-2021.00 Food Preparation Workers

Perform a variety of food preparation duties other than cooking, such as preparing cold foods and shellfish, slicing meat, and brewing coffee or tea.

35-3011.00 Bartenders

Mix and serve drinks to patrons, directly or through waitstaff.

35-3021.00 Combined Food Preparation and Serving Workers, Including Fast Food

Perform duties which combine both food preparation and food service.

35-3022.00 Counter Attendants, Cafeteria, Food Concession, and Coffee Shop

Serve food to diners at counter or from a steam table. Includes counter attendants who also wait tables.

35-3031.00 Waiters and Waitresses

Take orders and serve food and beverages to patrons at tables in dining establishment.

35-3041.00 Food Servers, Nonrestaurant

Serve food to patrons outside of a restaurant environment, such as in hotels, hospital rooms, or cars.

35-9011.00 Dining Room and Cafeteria Attendants and Bartender Helpers

Facilitate food service. Clean tables, carry dirty dishes, replace soiled table linens; set tables; replenish supply of clean linens, silverware, glassware, and dishes; supply service bar with food, and serve water, butter, and coffee to patrons.

35-9021.00 Dishwashers

Clean dishes, kitchen, food preparation equipment, or utensils.

35-9031.00 Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop

Welcome patrons, seat them at tables or in lounge, and help ensure quality of facilities and service.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

35-9099.99 Food Preparation and Serving Related Workers, All Other ²

All food preparation and serving related workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

37-0000 BUILDING AND GROUNDS CLEANING AND MAINTENANCE OCCUPATIONS

37-1011.00 First-Line Supervisors/Managers of Housekeeping and Janitorial Workers ¹

Supervise work activities of cleaning personnel in hotels, hospitals, offices, and other establishments.

37-1011.01 Housekeeping Supervisors

Supervise work activities of cleaning personnel to ensure clean, orderly, and attractive rooms in hotels, hospitals, educational institutions, and similar establishments. Assign duties, inspect work, and investigate complaints regarding housekeeping service and equipment and take corrective action. May purchase housekeeping supplies and equipment, take periodic inventories, screen applicants, train new employees, and recommend dismissals.

37-1011.02 Janitorial Supervisors

Supervise work activities of janitorial personnel in commercial and industrial establishments. Assign duties, inspect work, and investigate complaints regarding janitorial services and take corrective action. May purchase janitorial supplies and equipment, take periodic inventories, screen applicants, train new employees, and recommend dismissals.

37-1012.00 First-Line Supervisors/Managers of Landscaping, Lawn Service, and Groundskeeping Workers ¹

Plan, organize, direct, or coordinate activities of workers engaged in landscaping or groundskeeping activities, such as planting and maintaining ornamental trees, shrubs, flowers, and lawns, and applying fertilizers, pesticides, and other chemicals, according to contract specifications. May also coordinate activities of workers engaged in terracing hillsides, building retaining walls, constructing pathways, installing patios, and similar activities in following a landscape design plan. Work may involve reviewing contracts to ascertain service, machine, and work force requirements; answering inquiries from potential customers regarding methods, material, and price ranges; and preparing estimates according to labor, material, and machine costs.

37-1012.01 Lawn Service Managers

Plan, direct, and coordinate activities of workers engaged in pruning trees and shrubs, cultivating lawns, and applying pesticides and other chemicals according to service contract specifications.

37-1012.02 First-Line Supervisors and Manager/Supervisors - Landscaping Workers

Directly supervise and coordinate activities of landscaping workers. Manager/Supervisors are generally found in smaller establishment where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same landscaping work as the workers they supervise.

37-2011.00 Janitors and Cleaners, Except Maids and Housekeeping Cleaners

Keep buildings in clean and orderly condition. Perform heavy cleaning duties, such as cleaning floors, shampooing rugs, washing walls and glass, and removing rubbish. Duties may include tending furnace and boiler, performing routine maintenance activities, notifying management of need for repairs, and cleaning snow or debris from sidewalk.

37-2012.00 Maids and Housekeeping Cleaners

Perform any combination of light cleaning duties to maintain private households or commercial establishments, such as hotels, restaurants, and hospitals, in a clean and orderly manner. Duties include making beds, replenishing linens, cleaning rooms and halls, and vacuuming.

37-2019.99 Building Cleaning Workers, All Other ²

All building cleaning workers not listed separately.

37-2021.00 Pest Control Workers

Spray or release chemical solutions or toxic gases and set traps to kill pests and vermin, such as mice, termites, and roaches, that infest buildings and surrounding areas.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

37-3011.00 Landscaping and Groundskeeping Workers

Landscape or maintain grounds of property using hand or power tools or equipment. Workers typically perform a variety of tasks, which may include any combination of the following: sod laying, mowing, trimming, planting, watering, fertilizing, digging, raking, sprinkler installation, and installation of mortarless segmental concrete masonry wall units.

37-3012.00 Pesticide Handlers, Sprayers, and Applicators, Vegetation

Mix or apply pesticides, herbicides, fungicides, or insecticides through sprays, dusts, vapors, soil incorporation or chemical application on trees, shrubs, lawns, or botanical crops. Usually requires specific training and State or Federal certification.

37-3013.00 Tree Trimmers and Pruners

Cut away dead or excess branches from trees or shrubs to maintain right-of-way for roads, sidewalks, or utilities, or to improve appearance, health, and value of tree. Prune or treat trees or shrubs using handsaws, pruning hooks, sheers, and clippers. May use truck-mounted lifts and power pruners. May fill cavities in trees to promote healing and prevent deterioration.

37-3019.99 Grounds Maintenance Workers, All Other ²

All grounds maintenance workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

39-0000 PERSONAL CARE AND SERVICE OCCUPATIONS

39-1011.00 Gaming Supervisors

Supervise gaming operations and personnel in an assigned area. Circulate among tables and observe operations. Ensure that stations and games are covered for each shift. May explain and interpret operating rules of house to patrons. May plan and organize activities and create friendly atmosphere for guests in hotels/casinos. May adjust service complaints.

39-1012.00 Slot Key Persons

Coordinate/supervise functions of slot department workers to provide service to patrons. Handle and settle complaints of players. Verify and payoff jackpots. Reset slot machines after payoffs. Make minor repairs or adjustments to slot machines. Recommend removal of slot machines for repair. Report hazards and enforces safety rules.

39-1021.00 First-Line Supervisors/Managers of Personal Service Workers

Supervise and coordinate activities of personal service workers, such as supervisors of flight attendants, hairdressers, or caddies.

39-2011.00 Animal Trainers

Train animals for riding, harness, security, performance, or obedience, or assisting persons with disabilities. Accustom animals to human voice and contact; and condition animals to respond to commands. Train animals according to prescribed standards for show or competition. May train animals to carry pack loads or work as part of pack team.

39-2021.00 Nonfarm Animal Caretakers

Feed, water, groom, bathe, exercise, or otherwise care for pets and other nonfarm animals, such as dogs, cats, ornamental fish or birds, zoo animals, and mice. Work in settings such as kennels, animal shelters, zoos, circuses, and aquariums. May keep records of feedings, treatments, and animals received or discharged. May clean, disinfect, and repair cages, pens, or fish tanks.

39-3011.00 Gaming Dealers

Operate table games. Stand or sit behind table and operate games of chance by dispensing the appropriate number of cards or blocks to players, or operating other gaming equipment. Compare the house's hand against players' hands and payoff or collect players' money or chips.

39-3012.00 Gaming and Sports Book Writers and Runners

Assist in the operation of games such as keno and bingo. Scan winning tickets presented by patrons, calculate amount of winnings and pay patrons. May operate keno and bingo equipment. May start gaming equipment that randomly selects numbers. May announce number selected until total numbers specified for each game are selected. May pick up tickets from players, collect bets, receive, verify and record patrons' cash wages.

39-3019.99 Gaming Service Workers, All Other ²

All Gaming Service Workers not listed separately.

39-3021.00 Motion Picture Projectionists

Set up and operate motion picture projection and related sound reproduction equipment.

39-3031.00 Ushers, Lobby Attendants, and Ticket Takers

Assist patrons at entertainment events by performing duties, such as collecting admission tickets and passes from patrons, assisting in finding seats, searching for lost articles, and locating such facilities as rest rooms and telephones.

39-3091.00 Amusement and Recreation Attendants

Perform variety of attending duties at amusement or recreation facility. May schedule use of recreation facilities, maintain and provide equipment to participants of sporting events or recreational pursuits, or operate amusement concessions and rides.

39-3092.00 Costume Attendants

Select, fit, and take care of costumes for cast members, and aid entertainers.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

39-3093.00 Locker Room, Coatroom, and Dressing Room Attendants

Provide personal items to patrons or customers in locker rooms, dressing rooms, or coatrooms.

39-3099.99 Entertainment Attendants and Related Workers, All Other ²

All entertainment attendants and related workers not listed separately.

39-4011.00 Embalmers

Prepare bodies for interment in conformity with legal requirements.

39-4021.00 Funeral Attendants

Perform variety of tasks during funeral, such as placing casket in parlor or chapel prior to service; arranging floral offerings or lights around casket; directing or escorting mourners; closing casket; and issuing and storing funeral equipment.

39-5011.00 Barbers

Provide barbering services, such as cutting, trimming, shampooing, and styling hair, trimming beards, or giving shaves.

39-5012.00 Hairdressers, Hairstylists, and Cosmetologists

Provide beauty services, such as shampooing, cutting, coloring, and styling hair, and massaging and treating scalp. May also apply makeup, dress wigs, perform hair removal, and provide nail and skin care services.

39-5091.00 Makeup Artists, Theatrical and Performance

Apply makeup to performers to reflect period, setting, and situation of their role.

39-5092.00 Manicurists and Pedicurists

Clean and shape customers' fingernails and toenails. May polish or decorate nails.

39-5093.00 Shampooers

Shampoo and rinse customers' hair.

39-5094.00 Skin Care Specialists

Provide skin care treatments to face and body to enhance an individual's appearance.

39-6011.00 Baggage Porters and Bellhops

Handle baggage for travelers at transportation terminals or for guests at hotels or similar establishments.

39-6012.00 Concierges

Assist patrons at hotel, apartment or office building with personal services. May take messages, arrange or give advice on transportation, business services or entertainment, or monitor guest requests for housekeeping and maintenance.

39-6021.00 Tour Guides and Escorts

Escort individuals or groups on sightseeing tours or through places of interest, such as industrial establishments, public buildings, and art galleries.

39-6022.00 Travel Guides

Plan, organize, and conduct long distance cruises, tours, and expeditions for individuals and groups.

39-6031.00 Flight Attendants

Provide personal services to ensure the safety and comfort of airline passengers during flight. Greet passengers, verify tickets, explain use of safety equipment, and serve food or beverages.

39-6032.00 Transportation Attendants, Except Flight Attendants and Baggage Porters

Provide services to ensure the safety and comfort of passengers aboard ships, buses, trains, or within the station or terminal. Perform duties, such as greeting passengers, explaining the use of safety equipment, serving meals or beverages, or answering questions related to travel.

39-9011.00 Child Care Workers

Attend to children at schools, businesses, private households, and child care institutions. Perform a variety of tasks, such as dressing, feeding, bathing, and overseeing play.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

39-9021.00 Personal and Home Care Aides

Assist elderly or disabled adults with daily living activities at the person's home or in a daytime non-residential facility. Duties performed at a place of residence may include keeping house (making beds, doing laundry, washing dishes) and preparing meals. May provide meals and supervised activities at non-residential care facilities. May advise families, the elderly, and disabled on such things as nutrition, cleanliness, and household utilities.

39-9031.00 Fitness Trainers and Aerobics Instructors

Instruct or coach groups or individuals in exercise activities and the fundamentals of sports. Demonstrate techniques and methods of participation. Observe participants and inform them of corrective measures necessary to improve their skills. Those required to hold teaching degrees should be reported in the appropriate teaching category.

39-9032.00 Recreation Workers

Conduct recreation activities with groups in public, private, or volunteer agencies or recreation facilities. Organize and promote activities, such as arts and crafts, sports, games, music, dramatics, social recreation, camping, and hobbies, taking into account the needs and interests of individual members.

39-9041.00 Residential Advisors

Coordinate activities for residents of boarding schools, college fraternities or sororities, college dormitories, or similar establishments. Order supplies and determine need for maintenance, repairs, and furnishings. May maintain household records and assign rooms. May refer residents to counseling resources if needed.

39-9099.99 Personal Care and Service Workers, All Other ²

All personal care and service workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

41-0000 SALES AND RELATED OCCUPATIONS

41-1011.00 First-Line Supervisors/Managers of Retail Sales Workers

Directly supervise sales workers in a retail establishment or department. Duties may include management functions, such as purchasing, budgeting, accounting, and personnel work, in addition to supervisory duties.

41-1012.00 First-Line Supervisors/Managers of Non-Retail Sales Workers

Directly supervise and coordinate activities of sales workers other than retail sales workers. May perform duties, such as budgeting, accounting, and personnel work, in addition to supervisory duties.

41-2011.00 Cashiers

Receive and disburse money in establishments other than financial institutions. Usually involves use of electronic scanners, cash registers, or related equipment. Often involved in processing credit or debit card transactions and validating checks.

41-2012.00 Gaming Change Persons and Booth Cashiers

Exchange coins and tokens for patrons' money. May issue payoffs and obtain customer's signature on receipt when winnings exceed the amount held in the slot machine. May operate a booth in the slot machine area and furnish change persons with money bank at the start of the shift, or count and audit money in drawers.

41-2021.00 Counter and Rental Clerks

Receive orders for repairs, rentals, and services. May describe available options, compute cost, and accept payment.

41-2022.00 Parts Salespersons

Sell spare and replacement parts and equipment in repair shop or parts store.

41-2031.00 Retail Salespersons

Sell merchandise, such as furniture, motor vehicles, appliances, or apparel in a retail establishment.

41-3011.00 Advertising Sales Agents

Sell or solicit advertising, including graphic art, advertising space in publications, custom made signs, or TV and radio advertising time. May obtain leases for outdoor advertising sites or persuade retailer to use sales promotion display items.

41-3021.00 Insurance Sales Agents

Sell life, property, casualty, health, automotive, or other types of insurance. May refer clients to independent brokers, work as independent broker, or be employed by an insurance company.

41-3031.00 Securities, Commodities, and Financial Services Sales Agents ¹

Buy and sell securities in investment and trading firms, or call upon businesses and individuals to sell financial services. Provide financial services, such as loan, tax, and securities counseling. May advise securities customers about such things as stocks, bonds, and market conditions.

41-3031.01 Sales Agents, Securities and Commodities

Buy and sell securities in investment and trading firms and develop and implement financial plans for individuals, businesses, and organizations.

41-3031.02 Sales Agents, Financial Services

Sell financial services, such as loan, tax, and securities counseling to customers of financial institutions and business establishments.

41-3041.00 Travel Agents

Plan and sell transportation and accommodations for travel agency customers. Determine destination, modes of transportation, travel dates, costs, and accommodations required.

41-3099.99 Sales Representatives, Services, All Other ²

All services sales representatives not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

41-4011.00 Sales Representatives, Wholesale and Manufacturing, Technical and Scientific Products ¹

Sell goods for wholesalers or manufacturers where technical or scientific knowledge is required in such areas as biology, engineering, chemistry, and electronics, normally obtained from at least 2 years of post-secondary education.

41-4011.01 Sales Representatives, Agricultural

Sell agricultural products and services, such as animal feeds, farm and garden equipment, and dairy, poultry, and veterinarian supplies.

41-4011.02 Sales Representatives, Chemical and Pharmaceutical

Sell chemical or pharmaceutical products or services, such as acids, industrial chemicals, agricultural chemicals, medicines, drugs, and water treatment supplies.

41-4011.03 Sales Representatives, Electrical/Electronic

Sell electrical, electronic, or related products or services, such as communication equipment, radiographic-inspection equipment and services, ultrasonic equipment, electronics parts, computers, and EDP systems.

41-4011.04 Sales Representatives, Mechanical Equipment and Supplies

Sell mechanical equipment, machinery, materials, and supplies, such as aircraft and railroad equipment and parts, construction machinery, material-handling equipment, industrial machinery, and welding equipment.

41-4011.05 Sales Representatives, Medical

Sell medical equipment, products, and services. Does not include pharmaceutical sales representatives.

41-4011.06 Sales Representatives, Instruments

Sell precision instruments, such as dynamometers and spring scales, and laboratory, navigation, and surveying instruments.

41-4012.00 Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products

Sell goods for wholesalers or manufacturers to businesses or groups of individuals. Work requires substantial knowledge of items sold.

41-9011.00 Demonstrators and Product Promoters

Demonstrate merchandise and answer questions for the purpose of creating public interest in buying the product. May sell demonstrated merchandise.

41-9012.00 Models

Model garments and other apparel to display clothing before prospective buyers at fashion shows, private showings, retail establishments, or photographer. May pose for photos to be used for advertising purposes. May pose as subject for paintings, sculptures, and other types of artistic expression.

41-9021.00 Real Estate Brokers

Operate real estate office, or work for commercial real estate firm, overseeing real estate transactions. Other duties usually include selling real estate or renting properties and arranging loans.

41-9022.00 Real Estate Sales Agents

Rent, buy, or sell property for clients. Perform duties, such as study property listings, interview prospective clients, accompany clients to property site, discuss conditions of sale, and draw up real estate contracts. Includes agents who represent buyer.

41-9031.00 Sales Engineers

Sell business goods or services, the selling of which requires a technical background equivalent to a baccalaureate degree in engineering.

41-9041.00 Telemarketers

Solicit orders for goods or services over the telephone.

41-9091.00 Door-To-Door Sales Workers, News and Street Vendors, and Related Workers

Sell goods or services door-to-door or on the street.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

41-9099.99 Sales and Related Workers, All Other²

All sales and related workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-0000 OFFICE AND ADMINISTRATIVE SUPPORT OCCUPATIONS

43-1011.00 First-Line Supervisors/Managers of Office and Administrative Support Workers ¹

Supervise and coordinate the activities of clerical and administrative support workers.

43-1011.01 First-Line Supervisors, Customer Service

Supervise and coordinate activities of workers involved in providing customer service.

43-1011.02 First-Line Supervisors, Administrative Support

Supervise and coordinate activities of workers involved in providing administrative support.

43-2011.00 Switchboard Operators, Including Answering Service

Operate telephone business systems equipment or switchboards to relay incoming, outgoing, and interoffice calls. May supply information to callers and record messages.

43-2021.00 Telephone Operators ¹

Provide information by accessing alphabetical and geographical directories. Assist customers with special billing requests, such as charges to a third party and credits or refunds for incorrectly dialed numbers or bad connections. May handle emergency calls and assist children or people with physical disabilities to make telephone calls.

43-2021.01 Directory Assistance Operators

Provide telephone information from central office switchboard. Refer to alphabetical or geographical reels or directories to answer questions or suggest answer sources.

43-2021.02 Central Office Operators

Operate telephone switchboard to establish or assist customers in establishing local or long-distance telephone connections.

43-2099.99 Communications Equipment Operators, All Other ²

All communications equipment operators not listed separately.

43-3011.00 Bill and Account Collectors

Locate and notify customers of delinquent accounts by mail, telephone, or personal visit to solicit payment. Duties include receiving payment and posting amount to customer's account; preparing statements to credit department if customer fails to respond; initiating repossession proceedings or service disconnection; keeping records of collection and status of accounts.

43-3021.00 Billing and Posting Clerks and Machine Operators ¹

Compile, compute, and record billing, accounting, statistical, and other numerical data for billing purposes. Prepare billing invoices for services rendered or for delivery or shipment of goods.

43-3021.01 Statement Clerks

Prepare and distribute bank statements to customers, answer inquiries, and reconcile discrepancies in records and accounts.

43-3021.02 Billing, Cost, and Rate Clerks

Compile data, compute fees and charges, and prepare invoices for billing purposes. Duties include computing costs and calculating rates for goods, services, and shipment of goods; posting data; and keeping other relevant records. May involve use of computer or typewriter, calculator, and adding and bookkeeping machines.

43-3021.03 Billing, Posting, and Calculating Machine Operators

Operate machines that automatically perform mathematical processes, such as addition, subtraction, multiplication, and division, to calculate and record billing, accounting, statistical, and other numerical data. Duties include operating special billing machines to prepare statements, bills, and invoices, and operating bookkeeping machines to copy and post data, make computations, and compile records of transactions.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-3031.00 Bookkeeping, Accounting, and Auditing Clerks

Compute, classify, and record numerical data to keep financial records complete. Perform any combination of routine calculating, posting, and verifying duties to obtain primary financial data for use in maintaining accounting records. May also check the accuracy of figures, calculations, and postings pertaining to business transactions recorded by other workers.

43-3041.00 Gaming Cage Workers

In a gaming establishment, conduct financial transactions for patrons. May reconcile daily summaries of transactions to balance books. Accept patron's credit application and verify credit references to provide check-cashing authorization or to establish house credit accounts. May sell gambling chips, tokens, or tickets to patrons, or to other workers for resale to patrons. May convert gaming chips, tokens, or tickets to currency upon patron's request. May use a cash register or computer to record transaction.

43-3051.00 Payroll and Timekeeping Clerks

Compile and post employee time and payroll data. May compute employees' time worked, production, and commission. May compute and post wages and deductions. May prepare paychecks.

43-3061.00 Procurement Clerks

Compile information and records to draw up purchase orders for procurement of materials and services.

43-3071.00 Tellers

Receive and pay out money. Keep records of money and negotiable instruments involved in a financial institution's various transactions.

43-4011.00 Brokerage Clerks

Perform clerical duties involving the purchase or sale of securities. Duties include writing orders for stock purchases and sales, computing transfer taxes, verifying stock transactions, accepting and delivering securities, tracking stock price fluctuations, computing equity, distributing dividends, and keeping records of daily transactions and holdings.

43-4021.00 Correspondence Clerks

Compose letters in reply to requests for merchandise, damage claims, credit and other information, delinquent accounts, incorrect billings, or unsatisfactory services. Duties may include gathering data to formulate reply and typing correspondence.

43-4031.00 Court, Municipal, and License Clerks ¹

Perform clerical duties in courts of law, municipalities, and governmental licensing agencies and bureaus. May prepare docket of cases to be called; secure information for judges and court; prepare draft agendas or bylaws for town or city council; answer official correspondence; keep fiscal records and accounts; issue licenses or permits; record data, administer tests, or collect fees.

43-4031.01 Court Clerks

Perform clerical duties in court of law; prepare docket of cases to be called; secure information for judges; and contact witnesses, attorneys, and litigants to obtain information for court.

43-4031.02 Municipal Clerks

Draft agendas and bylaws for town or city council; record minutes of council meetings; answer official correspondence; keep fiscal records and accounts; and prepare reports on civic needs.

43-4031.03 License Clerks

Issue licenses or permits to qualified applicants. Obtain necessary information; record data; advise applicants on requirements; collect fees; and issue licenses. May conduct oral, written, visual, or performance testing.

43-4041.00 Credit Authorizers, Checkers, and Clerks ¹

Authorize credit charges against customers' accounts. Investigate history and credit standing of individuals or business establishments applying for credit. May interview applicants to obtain personal and financial data; determine credit worthiness; process applications; and notify customers of acceptance or rejection of credit.

43-4041.01 Credit Authorizers

Authorize credit charges against customers' accounts.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-4041.02 Credit Checkers

Investigate history and credit standing of individuals or business establishments applying for credit. Telephone or write to credit departments of business and service establishments to obtain information about applicant's credit standing.

43-4051.00 Customer Service Representatives ¹

Interact with customers to provide information in response to inquiries about products and services and to handle and resolve complaints.

43-4051.01 Adjustment Clerks

Investigate and resolve customers' inquiries concerning merchandise, service, billing, or credit rating. Examine pertinent information to determine accuracy of customers' complaints and responsibility for errors. Notify customers and appropriate personnel of findings, adjustments, and recommendations, such as exchange of merchandise, refund of money, credit to customers' accounts, or adjustment to customers' bills.

43-4051.02 Customer Service Representatives, Utilities

Interview applicants for water, gas, electric, or telephone service. Talk with customer by phone or in person and receive orders for installation, turn-on, discontinuance, or change in services.

43-4061.00 Eligibility Interviewers, Government Programs ¹

Determine eligibility of persons applying to receive assistance from government programs and agency resources, such as welfare, unemployment benefits, social security, and public housing.

43-4061.01 Claims Takers, Unemployment Benefits

Interview unemployed workers and compile data to determine eligibility for unemployment benefits.

43-4061.02 Welfare Eligibility Workers and Interviewers

Interview and investigate applicants and recipients to determine eligibility for use of social programs and agency resources. Duties include recording and evaluating personal and financial data obtained from individuals; initiating procedures to grant, modify, deny, or terminate eligibility for various aid programs; authorizing grant amounts; and preparing reports. These workers generally receive specialized training and assist social service caseworkers.

43-4071.00 File Clerks

File correspondence, cards, invoices, receipts, and other records in alphabetical or numerical order or according to the filing system used. Locate and remove material from file when requested.

43-4081.00 Hotel, Motel, and Resort Desk Clerks

Accommodate hotel, motel, and resort patrons by registering and assigning rooms to guests, issuing room keys, transmitting and receiving messages, keeping records of occupied rooms and guests' accounts, making and confirming reservations, and presenting statements to and collecting payments from departing guests.

43-4111.00 Interviewers, Except Eligibility and Loan

Interview persons by telephone, mail, in person, or by other means for the purpose of completing forms, applications, or questionnaires. Ask specific questions, record answers, and assist persons with completing form. May sort, classify, and file forms.

43-4121.00 Library Assistants, Clerical

Compile records, sort and shelve books, and issue and receive library materials such as pictures, cards, slides and microfilm. Locate library materials for loan and replace material in shelving area, stacks, or files according to identification number and title. Register patrons to permit them to borrow books, periodicals, and other library materials.

43-4131.00 Loan Interviewers and Clerks

Interview loan applicants to elicit information; investigate applicants' backgrounds and verify references; prepare loan request papers; and forward findings, reports, and documents to appraisal department. Review loan papers to ensure completeness, and complete transactions between loan establishment, borrowers, and sellers upon approval of loan.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-4141.00 New Accounts Clerks

Interview persons desiring to open bank accounts. Explain banking services available to prospective customers and assist them in preparing application form.

43-4151.00 Order Clerks

Receive and process incoming orders for materials, merchandise, classified ads, or services such as repairs, installations, or rental of facilities. Duties include informing customers of receipt, prices, shipping dates, and delays; preparing contracts; and handling complaints.

43-4161.00 Human Resources Assistants, Except Payroll and Timekeeping

Compile and keep personnel records. Record data for each employee, such as address, weekly earnings, absences, amount of sales or production, supervisory reports on ability, and date of and reason for termination. Compile and type reports from employment records. File employment records. Search employee files and furnish information to authorized persons.

43-4171.00 Receptionists and Information Clerks

Answer inquiries and obtain information for general public, customers, visitors, and other interested parties. Provide information regarding activities conducted at establishment; location of departments, offices, and employees within organization.

43-4181.00 Reservation and Transportation Ticket Agents and Travel Clerks ¹

Make and confirm reservations and sell tickets to passengers for large hotel or motel chains. May check baggage and direct passengers to designated concourse, pier, or track; make reservations, deliver tickets, arrange for visas, contact individuals and groups to inform them of package tours, or provide tourists with travel information, such as points of interest, restaurants, rates, and emergency service.

43-4181.01 Travel Clerks

Provide tourists with travel information, such as points of interest, restaurants, rates, and emergency service. Duties include answering inquiries, offering suggestions, and providing literature pertaining to trips, excursions, sporting events, concerts and plays. May make reservations, deliver tickets, arrange for visas, or contact individuals and groups to inform them of package tours.

43-4181.02 Reservation and Transportation Ticket Agents

Make and confirm reservations for passengers and sell tickets for transportation agencies such as airlines, bus companies, railroads, and steamship lines. May check baggage and direct passengers to designated concourse, pier, or track.

43-4199.99 Information and Record Clerks, All Other ²

All information and record clerks not listed separately.

43-5011.00 Cargo and Freight Agents

Expedite and route movement of incoming and outgoing cargo and freight shipments in airline, train, and trucking terminals, and shipping docks. Take orders from customers and arrange pickup of freight and cargo for delivery to loading platform. Prepare and examine bills of lading to determine shipping charges and tariffs.

43-5021.00 Couriers and Messengers

Pick up and carry messages, documents, packages, and other items between offices or departments within an establishment or to other business concerns, traveling by foot, bicycle, motorcycle, automobile, or public conveyance.

43-5031.00 Police, Fire, and Ambulance Dispatchers

Receive complaints from public concerning crimes and police emergencies. Broadcast orders to police patrol units in vicinity of complaint to investigate. Operate radio, telephone, or computer equipment to receive reports of fires and medical emergencies and relay information or orders to proper officials.

43-5032.00 Dispatchers, Except Police, Fire, and Ambulance

Schedule and dispatch workers, work crews, equipment, or service vehicles for conveyance of materials, freight, or passengers, or for normal installation, service, or emergency repairs rendered outside the place of business. Duties may include using radio, telephone, or computer to transmit assignments and compiling statistics and reports on work progress.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-5041.00 Meter Readers, Utilities

Read meter and record consumption of electricity, gas, water, or steam.

43-5051.00 Postal Service Clerks

Perform any combination of tasks in a post office, such as receive letters and parcels; sell postage and revenue stamps, postal cards, and stamped envelopes; fill out and sell money orders; place mail in pigeon holes of mail rack or in bags according to State, address, or other scheme; and examine mail for correct postage.

43-5052.00 Postal Service Mail Carriers

Sort mail for delivery. Deliver mail on established route by vehicle or on foot.

43-5053.00 Postal Service Mail Sorters, Processors, and Processing Machine Operators

Prepare incoming and outgoing mail for distribution. Examine, sort, and route mail by State, type of mail, or other scheme. Load, operate, and occasionally adjust and repair mail processing, sorting, and canceling machinery. Keep records of shipments, pouches, and sacks and other duties related to mail handling within the postal service. Must complete a competitive exam.

43-5061.00 Production, Planning, and Expediting Clerks

Coordinate and expedite the flow of work and materials within or between departments of an establishment according to production schedule. Duties include reviewing and distributing production, work, and shipment schedules; conferring with department supervisors to determine progress of work and completion dates; and compiling reports on progress of work, inventory levels, costs, and production problems.

43-5071.00 Shipping, Receiving, and Traffic Clerks

Verify and keep records on incoming and outgoing shipments. Prepare items for shipment. Duties include assembling, addressing, stamping, and shipping merchandise or material; receiving, unpacking, verifying and recording incoming merchandise or material; and arranging for the transportation of products.

43-5081.00 Stock Clerks and Order Fillers ¹

Receive, store, and issue sales floor merchandise, materials, equipment, and other items from stockroom, warehouse, or storage yard to fill shelves, racks, tables, or customers' orders. May mark prices on merchandise and set up sales displays.

43-5081.01 Stock Clerks, Sales Floor

Receive, store, and issue sales floor merchandise. Stock shelves, racks, cases, bins, and tables with merchandise and arrange merchandise displays to attract customers. May periodically take physical count of stock or check and mark merchandise.

43-5081.02 Marking Clerks

Print and attach price tickets to articles of merchandise using one or several methods, such as marking price on tickets by hand or using ticket-printing machine.

43-5081.03 Stock Clerks- Stockroom, Warehouse, or Storage Yard

Receive, store, and issue materials, equipment, and other items from stockroom, warehouse, or storage yard. Keep records and compile stock reports.

43-5081.04 Order Fillers, Wholesale and Retail Sales

Fill customers' mail and telephone orders from stored merchandise in accordance with specifications on sales slips or order forms. Duties include computing prices of items, completing order receipts, keeping records of out-going orders, and requisitioning additional materials, supplies, and equipment.

43-5111.00 Weighers, Measurers, Checkers, and Samplers, Recordkeeping

Weigh, measure, and check materials, supplies, and equipment for the purpose of keeping relevant records. Duties are primarily clerical by nature.

43-6011.00 Executive Secretaries and Administrative Assistants

Provide high-level administrative support by conducting research, preparing statistical reports, handling information requests, and performing clerical functions such as preparing correspondence, receiving visitors, arranging conference calls, and scheduling meetings. May also train and supervise lower-level clerical staff.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-6012.00 Legal Secretaries

Perform secretarial duties utilizing legal terminology, procedures, and documents. Prepare legal papers and correspondence, such as summonses, complaints, motions, and subpoenas. May also assist with legal research.

43-6013.00 Medical Secretaries

Perform secretarial duties utilizing specific knowledge of medical terminology and hospital, clinic, or laboratory procedures. Duties include scheduling appointments, billing patients, and compiling and recording medical charts, reports, and correspondence.

43-6014.00 Secretaries, Except Legal, Medical, and Executive

Perform routine clerical and administrative functions such as drafting correspondence, scheduling appointments, organizing and maintaining paper and electronic files, or providing information to callers.

43-9011.00 Computer Operators

Monitor and control electronic computer and peripheral electronic data processing equipment to process business, scientific, engineering, and other data according to operating instructions. May enter commands at a computer terminal and set controls on computer and peripheral devices. Monitor and respond to operating and error messages.

43-9021.00 Data Entry Keyers

Operate data entry device, such as keyboard or photo composing perforator. Duties may include verifying data and preparing materials for printing.

43-9022.00 Word Processors and Typists

Use word processor/computer or typewriter to type letters, reports, forms, or other material from rough draft, corrected copy, or voice recording. May perform other clerical duties as assigned.

43-9031.00 Desktop Publishers

Format typescript and graphic elements using computer software to produce publication-ready material.

43-9041.00 Insurance Claims and Policy Processing Clerks ¹

Process new insurance policies, modifications to existing policies, and claims forms. Obtain information from policyholders to verify the accuracy and completeness of information on claims forms, applications and related documents, and company records. Update existing policies and company records to reflect changes requested by policyholders and insurance company representatives.

43-9041.01 Insurance Claims Clerks

Obtain information from insured or designated persons for purpose of settling claim with insurance carrier.

43-9041.02 Insurance Policy Processing Clerks

Process applications for, changes to, reinstatement of, and cancellation of insurance policies. Duties include reviewing insurance applications to ensure that all questions have been answered, compiling data on insurance policy changes, changing policy records to conform to insured party's specifications, compiling data on lapsed insurance policies to determine automatic reinstatement according to company policies, canceling insurance policies as requested by agents, and verifying the accuracy of insurance company records.

43-9051.00 Mail Clerks and Mail Machine Operators, Except Postal Service ¹

Prepare incoming and outgoing mail for distribution. Use hand or mail handling machines to time stamp, open, read, sort, and route incoming mail; and address, seal, stamp, fold, stuff, and affix postage to outgoing mail or packages. Duties may also include keeping necessary records and completed forms.

43-9051.01 Mail Machine Operators, Preparation and Handling

Operate machines that: Emboss names, addresses, and other matter onto metal plates for use in addressing machines; print names, addresses, and similar information onto items such as envelopes, accounting forms, and advertising literature; address, fold, stuff, seal, and stamp mail; and open envelopes.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

43-9051.02 Mail Clerks, Except Mail Machine Operators and Postal Service

Prepare incoming and outgoing mail for distribution. Duties include time stamping, opening, reading, sorting, and routing incoming mail; sealing, stamping, and affixing postage to outgoing mail or packages; and keeping necessary records and completed forms.

43-9061.00 Office Clerks, General

Perform duties too varied and diverse to be classified in any specific office clerical occupation, requiring limited knowledge of office management systems and procedures. Clerical duties may be assigned in accordance with the office procedures of individual establishments and may include a combination of answering telephones, bookkeeping, typing or word processing, stenography, office machine operation, and filing.

43-9071.00 Office Machine Operators, Except Computer³

Operate one or more of a variety of office machines, such as photocopying, photographic, and duplicating machines, or other office machines.

43-9071.01 Duplicating Machine Operators

Operate one of a variety of office machines such as photocopying, photographic, mimeograph, and duplicating machines to make copies.

43-9081.00 Proofreaders and Copy Markers

Read transcript or proof type setup to detect and mark for correction any grammatical, typographical, or compositional errors.

43-9111.00 Statistical Assistants

Compile and compute data according to statistical formulas for use in statistical studies. May perform actuarial computations and compile charts and graphs for use by actuaries. Includes actuarial clerks.

43-9199.99 Office and Administrative Support Workers, All Other²

All office and administrative support workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

45-0000 FARMING, FISHING, AND FORESTRY OCCUPATIONS

45-1011.00 First-Line Supervisors/Managers of Farming, Fishing, and Forestry Workers¹

Directly supervise and coordinate the activities of agricultural, forestry, aquacultural, and related workers.

45-1011.01 First-Line Supervisors and Manager/Supervisors - Agricultural Crop Workers

Directly supervise and coordinate activities of agricultural crop workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same agricultural work as the workers they supervise.

45-1011.02 First-Line Supervisors and Manager/Supervisors - Animal Husbandry Workers

Directly supervise and coordinate activities of animal husbandry workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same animal husbandry work as the workers they supervise.

45-1011.03 First-Line Supervisors and Manager/Supervisors - Animal Care Workers, Except Livestock

Directly supervise and coordinate activities of animal care workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same animal care work as the workers they supervise.

45-1011.04 First-Line Supervisors and Manager/Supervisors - Horticultural Workers

Directly supervise and coordinate activities of horticultural workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same horticultural work as the workers they supervise.

45-1011.05 First-Line Supervisors and Manager/Supervisors - Logging Workers

Directly supervise and coordinate activities of logging workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same logging work as the workers they supervise.

45-1011.06 First-Line Supervisors and Manager/Supervisors - Fishery Workers

Directly supervise and coordinate activities of fishery workers. Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same fishery work as the workers they supervise.

45-2011.00 Agricultural Inspectors

Inspect agricultural commodities, processing equipment, and facilities, and fish and logging operations, to ensure compliance with regulations and laws governing health, quality, and safety.

45-2021.00 Animal Breeders

Breed animals, including cattle, goats, horses, sheep, swine, poultry, dogs, cats, or pet birds. Select and breed animals according to their genealogy, characteristics, and offspring. May require a knowledge of artificial insemination techniques and equipment use. May involve keeping records on heats, birth intervals, or pedigree.

45-2031.00 Farm Labor Contractors

Recruit, hire, furnish, and supervise seasonal or temporary agricultural laborers for a fee. May transport, house, and provide meals for workers.

45-2041.00 Graders and Sorters, Agricultural Products

Grade, sort, or classify unprocessed food and other agricultural products by size, weight, color, or condition.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

45-2091.00 Agricultural Equipment Operators

Drive and control farm equipment to till soil and to plant, cultivate, and harvest crops. May perform tasks, such as crop baling or hay bucking. May operate stationary equipment to perform post-harvest tasks, such as husking, shelling, threshing, and ginning.

45-2092.00 Farmworkers and Laborers, Crop, Nursery, and Greenhouse ¹

Manually plant, cultivate, and harvest vegetables, fruits, nuts, horticultural specialties, and field crops. Use hand tools, such as shovels, trowels, hoes, tampers, pruning hooks, shears, and knives. Duties may include tilling soil and applying fertilizers transplanting, weeding, thinning, or pruning crops applying pesticides cleaning, grading, sorting, packing and loading harvested products. May construct trellises, repair fences and farm buildings, or participate in irrigation activities.

45-2092.01 Nursery Workers

Work in nursery facilities or at customer location planting, cultivating, harvesting, and transplanting trees, shrubs, or plants.

45-2092.02 General Farmworkers

Apply pesticides, herbicides, and fertilizer to crops and livestock; plant, maintain, and harvest food crops; and tend livestock and poultry. Repair farm buildings and fences. Duties may include: operating milking machines and other dairy processing equipment; supervising seasonal help; irrigating crops; and hauling livestock products to market.

45-2093.00 Farmworkers, Farm and Ranch Animals

Attend to live farm, ranch, or aquacultural animals that may include cattle, sheep, swine, goats, horses and other equines, poultry, finfish, shellfish, and bees. Attend to animals produced for animal products, such as meat, fur, skins, feathers, eggs, milk, and honey. Duties may include feeding, watering, herding, grazing, castrating, branding, de-beaking, weighing, catching, and loading animals. May maintain records on animals examine animals to detect diseases and injuries assist in birth deliveries and administer medications, vaccinations, or insecticides as appropriate. May clean and maintain animal housing areas.

45-2099.99 Agricultural Workers, All Other ²

All agricultural workers not listed separately.

45-3011.00 Fishers and Related Fishing Workers

Use nets, fishing rods, traps, or other equipment to catch and gather fish or other aquatic animals from rivers, lakes, or oceans, for human consumption or other uses. May haul game onto ship.

45-3021.00 Hunters and Trappers

Hunt and trap wild animals for human consumption, fur, feed, bait, or other purposes.

45-4011.00 Forest and Conservation Workers

Under supervision, perform manual labor necessary to develop, maintain, or protect forest, forested areas, and woodlands through such activities as raising and transporting tree seedlings; combating insects, pests, and diseases harmful to trees; and building erosion and water control structures and leaching of forest soil. Includes forester aides, seedling pullers, and tree planters.

45-4021.00 Fallers

Use axes or chainsaws to fell trees using knowledge of tree characteristics and cutting techniques to control direction of fall and minimize tree damage.

45-4022.00 Logging Equipment Operators ³

Drive logging tractor or wheeled vehicle equipped with one or more accessories, such as bulldozer blade, frontal shear, grapple, logging arch, cable winches, hoisting rack, or crane boom, to fell tree; to skid, load, unload, or stack logs; or to pull stumps or clear brush.

45-4022.01 Logging Tractor Operators

Drive tractor equipped with one or more accessories, such as bulldozer blade, frontal hydraulic shear, grapple, logging arch, cable winches, hoisting rack, or crane boom, to fell tree, to skid, load and unload, or stack logs, or to pull stumps or clear brush.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

45-4023.00 Log Graders and Scalers

Grade logs or estimate the marketable content or value of logs or pulpwood in sorting yards, millpond, log deck, or similar locations. Inspect logs for defects or measure logs to determine volume.

45-4029.99 Logging Workers, All Other ²

All logging workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-0000 CONSTRUCTION AND EXTRACTION OCCUPATIONS

47-1011.00 First-Line Supervisors/Managers of Construction Trades and Extraction Workers¹

Directly supervise and coordinate activities of construction or extraction workers.

47-1011.01 First-Line Supervisors and Manager/Supervisors- Construction Trades Workers

Directly supervise and coordinate activities of construction trades workers and their helpers.

Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work and may also engage in the same construction trades work as the workers they supervise.

47-1011.02 First-Line Supervisors and Manager/Supervisors- Extractive Workers

Directly supervise and coordinate activities of extractive workers and their helpers.

Manager/Supervisors are generally found in smaller establishments where they perform both supervisory and management functions, such as accounting, marketing, and personnel work, and may also engage in the same extractive work as the workers they supervise.

47-2011.00 Boilermakers

Construct, assemble, maintain, and repair stationary steam boilers and boiler house auxiliaries. Align structures or plate sections to assemble boiler frame tanks or vats, following blueprints. Work involves use of hand and power tools, plumb bobs, levels, wedges, dogs, or turnbuckles. Assist in testing assembled vessels. Direct cleaning of boilers and boiler furnaces. Inspect and repair boiler fittings, such as safety valves, regulators, automatic-control mechanisms, water columns, and auxiliary machines.

47-2021.00 Brickmasons and Blockmasons

Lay and bind building materials, such as brick, structural tile, concrete block, cinder block, glass block, and terra-cotta block, with mortar and other substances to construct or repair walls, partitions, arches, sewers, and other structures.

47-2022.00 Stonemasons

Build stone structures, such as piers, walls, and abutments. Lay walks, curbstones, or special types of masonry for vats, tanks, and floors.

47-2031.00 Carpenters¹

Construct, erect, install, or repair structures and fixtures made of wood, such as concrete forms; building frameworks, including partitions, joists, studding, and rafters; wood stairways, window and door frames, and hardwood floors. May also install cabinets, siding, drywall and batt or roll insulation. Includes brattice builders who build doors or brattices (ventilation walls or partitions) in underground passageways to control the proper circulation of air through the passageways and to the working places.

47-2031.01 Construction Carpenters

Construct, erect, install, and repair structures and fixtures of wood, plywood, and wallboard, using carpenter's hand tools and power tools.

47-2031.02 Rough Carpenters

Build rough wooden structures, such as concrete forms, scaffolds, tunnel, bridge, or sewer supports, billboard signs, and temporary frame shelters, according to sketches, blueprints, or oral instructions.

47-2031.03 Carpenter Assemblers and Repairers

Perform a variety of tasks requiring a limited knowledge of carpentry, such as applying siding and weatherboard to building exteriors or assembling and erecting prefabricated buildings.

47-2031.04 Ship Carpenters and Joiners

Fabricate, assemble, install, or repair wooden furnishings in ships or boats.

47-2031.05 Boat Builders and Shipwrights

Construct and repair ships or boats, according to blueprints.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-2031.06 Brattice Builders

Build doors or brattices (ventilation walls or partitions) in underground passageways to control the proper circulation of air through the passageways and to the working places.

47-2041.00 Carpet Installers

Lay and install carpet from rolls or blocks on floors. Install padding and trim flooring materials.

47-2042.00 Floor Layers, Except Carpet, Wood, and Hard Tiles

Apply blocks, strips, or sheets of shock-absorbing, sound-deadening, or decorative coverings to floors.

47-2043.00 Floor Sanders and Finishers

Scrape and sand wooden floors to smooth surfaces using floor scraper and floor sanding machine, and apply coats of finish.

47-2044.00 Tile and Marble Setters

Apply hard tile, marble, and wood tile to walls, floors, ceilings, and roof decks.

47-2051.00 Cement Masons and Concrete Finishers

Smooth and finish surfaces of poured concrete, such as floors, walks, sidewalks, roads, or curbs using a variety of hand and power tools. Align forms for sidewalks, curbs, or gutters; patch voids; use saws to cut expansion joints.

47-2053.00 Terrazzo Workers and Finishers

Apply a mixture of cement, sand, pigment, or marble chips to floors, stairways, and cabinet fixtures to fashion durable and decorative surfaces.

47-2061.00 Construction Laborers

Perform tasks involving physical labor at building, highway, and heavy construction projects, tunnel and shaft excavations, and demolition sites. May operate hand and power tools of all types: air hammers, earth tampers, cement mixers, small mechanical hoists, surveying and measuring equipment, and a variety of other equipment and instruments. May clean and prepare sites, dig trenches, set braces to support the sides of excavations, erect scaffolding, clean up rubble and debris, and remove asbestos, lead, and other hazardous waste materials. May assist other craft workers.

47-2071.00 Paving, Surfacing, and Tamping Equipment Operators

Operate equipment used for applying concrete, asphalt, or other materials to road beds, parking lots, or airport runways and taxiways, or equipment used for tamping gravel, dirt, or other materials. Includes concrete and asphalt paving machine operators, form tampers, tamping machine operators, and stone spreader operators.

47-2072.00 Pile-Driver Operators

Operate pile drivers mounted on skids, barges, crawler treads, or locomotive cranes to drive pilings for retaining walls, bulkheads, and foundations of structures, such as buildings, bridges, and piers.

47-2073.00 Operating Engineers and Other Construction Equipment Operators¹

Operate one or several types of power construction equipment, such as motor graders, bulldozers, scrapers, compressors, pumps, derricks, shovels, tractors, or front-end loaders to excavate, move, and grade earth, erect structures, or pour concrete or other hard surface pavement. May repair and maintain equipment in addition to other duties.

47-2073.01 Grader, Bulldozer, and Scraper Operators

Operate machines or vehicles equipped with blades to remove, distribute, level, or grade earth.

47-2073.02 Operating Engineers

Operate several types of power construction equipment, such as compressors, pumps, hoists, derricks, cranes, shovels, tractors, scrapers, or motor graders to excavate, move and grade earth, erect structures, or pour concrete or other hard surface pavement. May repair and maintain equipment in addition to other duties.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-2081.00 Drywall and Ceiling Tile Installers¹

Apply plasterboard or other wallboard to ceilings or interior walls of buildings. Apply or mount acoustical tiles or blocks, strips, or sheets of shock-absorbing materials to ceilings and walls of buildings to reduce or reflect sound. Materials may be of decorative quality. Includes lathers who fasten wooden, metal, or rockboard lath to walls, ceilings or partitions of buildings to provide support base for plaster, fire-proofing, or acoustical material.

47-2081.01 Ceiling Tile Installers

Apply plasterboard or other wallboard to ceilings or interior walls of buildings. Apply or mount acoustical tiles or blocks, strips, or sheets of shock-absorbing materials to ceilings and walls of buildings to reduce or reflect sound. Materials may be of decorative quality. Includes lathers who fasten wooden, metal, or rockboard lath to walls, ceilings or partitions of buildings to provide support base for plaster, fire-proofing, or acoustical material.

47-2081.02 Drywall Installers

Apply plasterboard or other wallboard to ceilings and interior walls of buildings.

47-2082.00 Tapers

Seal joints between plasterboard or other wallboard to prepare wall surface for painting or papering.

47-2111.00 Electricians

Install, maintain, and repair electrical wiring, equipment, and fixtures. Ensure that work is in accordance with relevant codes. May install or service street lights, intercom systems, or electrical control systems.

47-2121.00 Glaziers

Install glass in windows, skylights, store fronts, and display cases, or on surfaces, such as building fronts, interior walls, ceilings, and tabletops.

47-2131.00 Insulation Workers, Floor, Ceiling, and Wall

Line and cover structures with insulating materials. May work with batt, roll, or blown insulation materials.

47-2132.00 Insulation Workers, Mechanical

Apply insulating materials to pipes or ductwork, or other mechanical systems in order to help control and maintain temperature.

47-2141.00 Painters, Construction and Maintenance

Paint walls, equipment, buildings, bridges, and other structural surfaces, using brushes, rollers, and spray guns. May remove old paint to prepare surface prior to painting. May mix colors or oils to obtain desired color or consistency.

47-2142.00 Paperhangers

Cover interior walls and ceilings of rooms with decorative wallpaper or fabric, or attach advertising posters on surfaces, such as walls and billboards. Duties include removing old materials from surface to be papered.

47-2151.00 Pipelayers

Lay pipe for storm or sanitation sewers, drains, and water mains. Perform any combination of the following tasks: grade trenches or culverts, position pipe, or seal joints.

47-2152.00 Plumbers, Pipefitters, and Steamfitters¹

Assemble, install, alter, and repair pipelines or pipe systems that carry water, steam, air, or other liquids or gases. May install heating and cooling equipment and mechanical control systems.

47-2152.01 Pipe Fitters

Lay out, assemble, install, and maintain pipe systems, pipe supports, and related hydraulic and pneumatic equipment for steam, hot water, heating, cooling, lubricating, sprinkling, and industrial production and processing systems.

47-2152.02 Plumbers

Assemble, install, and repair pipes, fittings, and fixtures of heating, water, and drainage systems, according to specifications and plumbing codes.

47-2152.03 Pipelaying Fitters

Align pipeline section in preparation of welding. Signal tractor driver for placement of pipeline sections in proper alignment. Insert steel spacer.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-2161.00 Plasterers and Stucco Masons

Apply interior or exterior plaster, cement, stucco, or similar materials. May also set ornamental plaster.

47-2171.00 Reinforcing Iron and Rebar Workers

Position and secure steel bars or mesh in concrete forms in order to reinforce concrete. Use a variety of fasteners, rod-bending machines, blowtorches, and hand tools.

47-2181.00 Roofers

Cover roofs of structures with shingles, slate, asphalt, aluminum, wood, and related materials. May spray roofs, sidings, and walls with material to bind, seal, insulate, or soundproof sections of structures.

47-2211.00 Sheet Metal Workers

Fabricate, assemble, install, and repair sheet metal products and equipment, such as ducts, control boxes, drainpipes, and furnace casings. Work may involve any of the following: setting up and operating fabricating machines to cut, bend, and straighten sheet metal shaping metal over anvils, blocks, or forms using hammer operating soldering and welding equipment to join sheet metal parts inspecting, assembling, and smoothing seams and joints of burred surfaces.

47-2221.00 Structural Iron and Steel Workers

Raise, place, and unite iron or steel girders, columns, and other structural members to form completed structures or structural frameworks. May erect metal storage tanks and assemble prefabricated metal buildings.

47-3011.00 Helpers--Brickmasons, Blockmasons, Stonemasons, and Tile and Marble Setters

Help brickmasons, blockmasons, stonemasons, or tile and marble setters by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3012.00 Helpers--Carpenters

Help carpenters by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3013.00 Helpers--Electricians

Help electricians by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3014.00 Helpers--Painters, Paperhangers, Plasterers, and Stucco Masons

Help painters, paperhangers, plasterers, or stucco masons by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3015.00 Helpers--Pipefitters, Plumbers, Pipefitters, and Steamfitters

Help plumbers, pipefitters, steamfitters, or pipelayers by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3016.00 Helpers--Roofers

Help roofers by performing duties of lesser skill. Duties include using, supplying or holding materials or tools, and cleaning work area and equipment.

47-3019.99 Helpers, Construction Trades, All Other ²

All construction trades helpers not listed separately.

47-4011.00 Construction and Building Inspectors

Inspect structures using engineering skills to determine structural soundness and compliance with specifications, building codes, and other regulations. Inspections may be general in nature or may be limited to a specific area, such as electrical systems or plumbing.

47-4021.00 Elevator Installers and Repairers

Assemble, install, repair, or maintain electric or hydraulic freight or passenger elevators, escalators, or dumbwaiters.

47-4031.00 Fence Erectors

Erect and repair metal and wooden fences and fence gates around highways, industrial establishments, residences, or farms, using hand and power tools.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-4041.00 Hazardous Materials Removal Workers³

Identify, remove, pack, transport, or dispose of hazardous materials, including asbestos, lead-based paint, waste oil, fuel, transmission fluid, radioactive materials, contaminated soil, etc. Specialized training and certification in hazardous materials handling or a confined entry permit are generally required. May operate earth-moving equipment or trucks.

47-4041.01 Irradiated-Fuel Handlers

Package, store, and convey irradiated fuels and wastes, using hoists, mechanical arms, shovels, and industrial truck.

47-4051.00 Highway Maintenance Workers

Maintain highways, municipal and rural roads, airport runways, and rights-of-way. Duties include patching broken or eroded pavement, repairing guard rails, highway markers, and snow fences. May also mow or clear brush from along road or plow snow from roadway.

47-4061.00 Rail-Track Laying and Maintenance Equipment Operators

Lay, repair, and maintain track for standard or narrow-gauge railroad equipment used in regular railroad service or in plant yards, quarries, sand and gravel pits, and mines. Includes ballast cleaning machine operators and road bed tamping machine operators.

47-4071.00 Septic Tank Servicers and Sewer Pipe Cleaners

Clean and repair septic tanks, sewer lines, or drains. May patch walls and partitions of tank, replace damaged drain tile, or repair breaks in underground piping.

47-4091.00 Segmental Pavers

Lay out, cut, and paste segmental paving units. Includes installers of bedding and restraining materials for the paving units.

47-4099.99 Construction and Related Workers, All Other²

All construction and related workers not listed separately.

47-5011.00 Derrick Operators, Oil and Gas

Rig derrick equipment and operate pumps to circulate mud through drill hole.

47-5012.00 Rotary Drill Operators, Oil and Gas

Set up or operate a variety of drills to remove petroleum products from the earth and to find and remove core samples for testing during oil and gas exploration.

47-5013.00 Service Unit Operators, Oil, Gas, and Mining

Operate equipment to increase oil flow from producing wells or to remove stuck pipe, casing, tools, or other obstructions from drilling wells. May also perform similar services in mining exploration operations.

47-5021.00 Earth Drillers, Except Oil and Gas¹

Operate a variety of drills--such as rotary, churn, and pneumatic--to tap sub-surface water and salt deposits, to remove core samples during mineral exploration or soil testing, and to facilitate the use of explosives in mining or construction. May use explosives. Includes horizontal and earth boring machine operators.

47-5021.01 Construction Drillers

Operate machine to drill or bore through earth or rock.

47-5021.02 Well and Core Drill Operators

Operate machine to drill wells and take samples or cores for analysis of strata.

47-5031.00 Explosives Workers, Ordnance Handling Experts, and Blasters

Place and detonate explosives to demolish structures or to loosen, remove, or displace earth, rock, or other materials. May perform specialized handling, storage, and accounting procedures. Includes seismograph shooters.

47-5041.00 Continuous Mining Machine Operators

Operate self-propelled mining machines that rip coal, metal and nonmetal ores, rock, stone, or sand from the face and load it onto conveyors or into shuttle cars in a continuous operation.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

47-5042.00 Mine Cutting and Channeling Machine Operators

Operate machinery--such as longwall shears, plows, and cutting machines--to cut or channel along the face or seams of coal mines, stone quarries, or other mining surfaces to facilitate blasting, separating, or removing minerals or materials from mines or from the earth's surface.

47-5049.99 Mining Machine Operators, All Other ²

All mining machine operators not listed separately.

47-5051.00 Rock Splitters, Quarry

Separate blocks of rough dimension stone from quarry mass using jackhammer and wedges.

47-5061.00 Roof Bolters, Mining

Operate machinery to install roof support bolts in underground mine.

47-5071.00 Roustabouts, Oil and Gas

Assemble or repair oil field equipment using hand and power tools. Perform other tasks as needed.

47-5081.00 Helpers--Extraction Workers

Help extraction craft workers, such as earth drillers, blasters and explosives workers, derrick operators, and mining machine operators, by performing duties of lesser skill. Duties include supplying equipment or cleaning work area.

47-5099.99 Extraction Workers, All Other ²

All extraction workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

49-0000 INSTALLATION, MAINTENANCE, AND REPAIR OCCUPATIONS

49-1011.00 First-Line Supervisors/Managers of Mechanics, Installers, and Repairers

Supervise and coordinate the activities of mechanics, installers, and repairers.

49-2011.00 Computer, Automated Teller, and Office Machine Repairers¹

Repair, maintain, or install computers, word processing systems, automated teller machines, and electronic office machines, such as duplicating and fax machines.

49-2011.01 Automatic Teller Machine Servicers

Collect deposits and replenish automatic teller machines with cash and supplies.

49-2011.02 Data Processing Equipment Repairers

Repair, maintain, and install computer hardware such as peripheral equipment and word processing systems.

49-2011.03 Office Machine and Cash Register Servicers

Repair and service office machines, such as adding, accounting, calculating, duplicating, and typewriting machines. Includes the repair of manual, electrical, and electronic office machines.

49-2021.00 Radio Mechanics

Test or repair mobile or stationary radio transmitting and receiving equipment and two-way radio communications systems used in ship-to-shore communications and found in service and emergency vehicles.

49-2022.00 Telecommunications Equipment Installers and Repairers, Except Line Installers¹

Set-up, rearrange, or remove switching and dialing equipment used in central offices. Service or repair telephones and other communication equipment on customers' property. May install equipment in new locations or install wiring and telephone jacks in buildings under construction.

49-2022.01 Central Office and PBX Installers and Repairers

Test, analyze, and repair telephone or telegraph circuits and equipment at a central office location using test meters and hand tools. Analyze and repair defects in communications equipment on customers' premises using circuit diagrams, polarity probes, meters, and a telephone test set. May install equipment.

49-2022.02 Frame Wires, Central Office

Connect wires from telephone lines and cables to distributing frames in telephone company central office, using soldering iron and other hand tools.

49-2022.03 Communication Equipment Mechanics, Installers, and Repairers

Install, maintain, test, and repair communication cables and equipment.

49-2022.04 Telecommunications Facility Examiners

Examine telephone transmission facilities to determine equipment requirements for providing subscribers with new or additional telephone services.

49-2022.05 Station Installers and Repairers, Telephone

Install and repair telephone station equipment, such as telephones, coin collectors, telephone booths, and switching-key equipment.

49-2091.00 Avionics Technicians

Install, inspect, test, adjust, or repair avionics equipment, such as radar, radio, navigation, and missile control systems in aircraft or space vehicles.

49-2092.00 Electric Motor, Power Tool, and Related Repairers¹

Repair, maintain, or install electric motors, wiring, or switches.

49-2092.01 Electric Home Appliance and Power Tool Repairers

Repair, adjust, and install all types of electric household appliances.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

49-2092.02 Electric Motor and Switch Assemblers and Repairers

Test, repair, rebuild, and assemble electric motors, generators, and equipment.

49-2092.03 Battery Repairers

Inspect, repair, recharge, and replace batteries.

49-2092.04 Transformer Repairers

Clean and repair electrical transformers.

49-2092.05 Electrical Parts Reconditioners

Recondition and rebuild salvaged, electrical parts of equipment and wind new coils on armatures of used generators and motors.

49-2092.06 Hand and Portable Power Tool Repairers

Repair and adjust hand and power tools.

49-2093.00 Electrical and Electronics Installers and Repairers, Transportation Equipment

Install, adjust, or maintain mobile electronics communication equipment, including sound, sonar, security, navigation, and surveillance systems on trains, watercraft, or other mobile equipment.

49-2094.00 Electrical and Electronics Repairers, Commercial and Industrial Equipment

Repair, test, adjust, or install electronic equipment, such as industrial controls, transmitters, and antennas.

49-2095.00 Electrical and Electronics Repairers, Powerhouse, Substation, and Relay

Inspect, test, repair, or maintain electrical equipment in generating stations, substations, and in-service relays.

49-2096.00 Electronic Equipment Installers and Repairers, Motor Vehicles

Install, diagnose, or repair communications, sound, security, or navigation equipment in motor vehicles.

49-2097.00 Electronic Home Entertainment Equipment Installers and Repairers

Repair, adjust, or install audio or television receivers, stereo systems, camcorders, video systems, or other electronic home entertainment equipment.

49-2098.00 Security and Fire Alarm Systems Installers

Install, program, maintain, and repair security and fire alarm wiring and equipment. Ensure that work is in accordance with relevant codes.

49-3011.00 Aircraft Mechanics and Service Technicians ¹

Diagnose, adjust, repair, or overhaul aircraft engines and assemblies, such as hydraulic and pneumatic systems.

49-3011.01 Airframe-and-Power-Plant Mechanics

Inspect, test, repair, maintain, and service aircraft.

49-3011.02 Aircraft Engine Specialists

Repair and maintain the operating condition of aircraft engines. Includes helicopter engine mechanics.

49-3011.03 Aircraft Body and Bonded Structure Repairers

Repair body or structure of aircraft according to specifications.

49-3021.00 Automotive Body and Related Repairers

Repair and refinish automotive vehicle bodies and straighten vehicle frames.

49-3022.00 Automotive Glass Installers and Repairers

Replace or repair broken windshields and window glass in motor vehicles.

49-3023.00 Automotive Service Technicians and Mechanics ¹

Diagnose, adjust, repair, or overhaul automotive vehicles.

49-3023.01 Automotive Master Mechanics

Repair automobiles, trucks, buses, and other vehicles. Master mechanics repair virtually any part on the vehicle or specialize in the transmission system.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

49-3023.02 Automotive Specialty Technicians

Repair only one system or component on a vehicle, such as brakes, suspension, or radiator.

49-3031.00 Bus and Truck Mechanics and Diesel Engine Specialists

Inspect, repair, and maintain diesel engines used to power machines.

49-3041.00 Farm Equipment Mechanics

Diagnose, adjust, repair, or overhaul farm machinery and vehicles, such as tractors, harvesters, dairy equipment, and irrigation systems.

49-3042.00 Mobile Heavy Equipment Mechanics, Except Engines

Diagnose, adjust, repair, or overhaul mobile mechanical, hydraulic, and pneumatic equipment, such as cranes, bulldozers, graders, and conveyors, used in construction, logging, and surface mining.

49-3043.00 Rail Car Repairers

Diagnose, adjust, repair, or overhaul railroad rolling stock, mine cars, or mass transit rail cars.

49-3051.00 Motorboat Mechanics

Repairs and adjusts electrical and mechanical equipment of gasoline or diesel powered inboard or inboard-outboard boat engines.

49-3052.00 Motorcycle Mechanics

Diagnose, adjust, repair, or overhaul motorcycles, scooters, mopeds, dirt bikes, or similar motorized vehicles.

49-3053.00 Outdoor Power Equipment and Other Small Engine Mechanics

Diagnose, adjust, repair, or overhaul small engines used to power lawn mowers, chain saws, and related equipment.

49-3091.00 Bicycle Repairers

Repair and service bicycles.

49-3092.00 Recreational Vehicle Service Technicians

Diagnose, inspect, adjust, repair, or overhaul recreational vehicles including travel trailers. May specialize in maintaining gas, electrical, hydraulic, plumbing, or chassis/towing systems as well as repairing generators, appliances, and interior components.

49-3093.00 Tire Repairers and Changers

Repair and replace tires.

49-9011.00 Mechanical Door Repairers

Install, service, or repair opening and closing mechanisms of automatic doors and hydraulic door closers. Includes garage door mechanics.

49-9012.00 Control and Valve Installers and Repairers, Except Mechanical Door ¹

Install, repair, and maintain mechanical regulating and controlling devices, such as electric meters, gas regulators, thermostats, safety and flow valves, and other mechanical governors.

49-9012.01 Electric Meter Installers and Repairers

Install electric meters on customers' premises or on pole. Test meters and perform necessary repairs. Turn current on/off by connecting/disconnecting service drop.

49-9012.02 Valve and Regulator Repairers

Test, repair, and adjust mechanical regulators and valves.

49-9012.03 Meter Mechanics

Test, adjust, and repair gas, water, and oil meters.

49-9021.00 Heating, Air Conditioning, and Refrigeration Mechanics and Installers ¹

Install or repair heating, central air conditioning, or refrigeration systems, including oil burners, hot-air furnaces, and heating stoves.

49-9021.01 Heating and Air Conditioning Mechanics

Install, service, and repair heating and air conditioning systems in residences and commercial establishments.

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² Residual or "All Other" occupation. O*NET data not collected.

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⁴ Military Specific Occupation. O*NET data not collected.

49-9021.02 Refrigeration Mechanics

Install and repair industrial and commercial refrigerating systems.

49-9031.00 Home Appliance Repairers¹

Repair, adjust, or install all types of electric or gas household appliances, such as refrigerators, washers, dryers, and ovens.

49-9031.01 Home Appliance Installers

Install household appliances, such as refrigerators, washing machines, and stoves in mobile homes or customers' homes.

49-9031.02 Gas Appliance Repairers

Repair and install gas appliances and equipment, such as ovens, dryers, and hot water heaters.

49-9041.00 Industrial Machinery Mechanics

Repair, install, adjust, or maintain industrial production and processing machinery or refinery and pipeline distribution systems.

49-9042.00 Maintenance and Repair Workers, General

Perform work involving the skills of two or more maintenance or craft occupations to keep machines, mechanical equipment, or the structure of an establishment in repair. Duties may involve pipe fitting; boiler making; insulating; welding; machining; carpentry; repairing electrical or mechanical equipment; installing, aligning, and balancing new equipment; and repairing buildings, floors, or stairs.

49-9043.00 Maintenance Workers, Machinery

Lubricate machinery, change parts, or perform other routine machinery maintenance.

49-9044.00 Millwrights

Install, dismantle, or move machinery and heavy equipment according to layout plans, blueprints, or other drawings.

49-9045.00 Refractory Materials Repairers, Except Brickmasons

Build or repair furnaces, kilns, cupolas, boilers, converters, ladles, soaking pits, ovens, etc., using refractory materials.

49-9051.00 Electrical Power-Line Installers and Repairers

Install or repair cables or wires used in electrical power or distribution systems. May erect poles and light or heavy duty transmission towers.

49-9052.00 Telecommunications Line Installers and Repairers

String and repair telephone and television cable, including fiber optics and other equipment for transmitting messages or television programming.

49-9061.00 Camera and Photographic Equipment Repairers

Repair and adjust cameras and photographic equipment, including commercial video and motion picture camera equipment.

49-9062.00 Medical Equipment Repairers

Test, adjust, or repair biomedical or electromedical equipment.

49-9063.00 Musical Instrument Repairers and Tuners¹

Repair percussion, stringed, reed, or wind instruments. May specialize in one area, such as piano tuning.

49-9063.01 Keyboard Instrument Repairers and Tuners

Repair, adjust, refinish, and tune musical keyboard instruments.

49-9063.02 Stringed Instrument Repairers and Tuners

Repair, adjust, refinish, and tune musical stringed instruments.

49-9063.03 Reed or Wind Instrument Repairers and Tuners

Repair, adjust, refinish, and tune musical reed and wind instruments.

49-9063.04 Percussion Instrument Repairers and Tuners

Repair and tune musical percussion instruments.

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² Residual or "All Other" occupation. O*NET data not collected.

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⁴ Military Specific Occupation. O*NET data not collected.

49-9064.00 Watch Repairers

Repair, clean, and adjust mechanisms of timing instruments, such as watches and clocks.

49-9069.99 Precision Instrument and Equipment Repairers, All Other ²

All precision instrument and equipment repairers not listed separately.

49-9091.00 Coin, Vending, and Amusement Machine Servicers and Repairers

Install, service, adjust, or repair coin, vending, or amusement machines including video games, juke boxes, pinball machines, or slot machines.

49-9092.00 Commercial Divers

Work below surface of water, using scuba gear to inspect, repair, remove, or install equipment and structures. May use a variety of power and hand tools, such as drills, sledgehammers, torches, and welding equipment. May conduct tests or experiments, rig explosives, or photograph structures or marine life.

49-9093.00 Fabric Menders, Except Garment

Repair tears, holes, and other defects in fabrics, such as draperies, linens, parachutes, and tents.

49-9094.00 Locksmiths and Safe Repairers

Repair and open locks; make keys; change locks and safe combinations; and install and repair safes.

49-9095.00 Manufactured Building and Mobile Home Installers

Move or install mobile homes or prefabricated buildings.

49-9096.00 Riggers

Set up or repair rigging for construction projects, manufacturing plants, logging yards, ships and shipyards, or for the entertainment industry.

49-9097.00 Signal and Track Switch Repairers

Install, inspect, test, maintain, or repair electric gate crossings, signals, signal equipment, track switches, section lines, or intercommunications systems within a railroad system.

49-9098.00 Helpers--Installation, Maintenance, and Repair Workers

Help installation, maintenance, and repair workers in maintenance, parts replacement, and repair of vehicles, industrial machinery, and electrical and electronic equipment. Perform duties, such as furnishing tools, materials, and supplies to other workers cleaning work area, machines, and tools and holding materials or tools for other workers.

49-9099.99 Installation, Maintenance, and Repair Workers, All Other ²

All mechanical, installation, and repair workers and helpers not listed separately.

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51-0000 PRODUCTION OCCUPATIONS

51-1011.00 First-Line Supervisors/Managers of Production and Operating Workers

Supervise and coordinate the activities of production and operating workers, such as inspectors, precision workers, machine setters and operators, assemblers, fabricators, and plant and system operators.

51-2011.00 Aircraft Structure, Surfaces, Rigging, and Systems Assemblers ¹

Assemble, fit, fasten, and install parts of airplanes, space vehicles, or missiles, such as tails, wings, fuselage, bulkheads, stabilizers, landing gear, rigging and control equipment, or heating and ventilating systems.

51-2011.01 Aircraft Structure Assemblers, Precision

Assemble tail, wing, fuselage, or other structural section of aircraft, space vehicles, and missiles from parts, subassemblies, and components and install functional units, parts, or equipment, such as landing gear, control surfaces, doors, and floorboards.

51-2011.02 Aircraft Systems Assemblers, Precision

Lay out, assemble, install, and test aircraft systems, such as armament, environmental control, plumbing and hydraulic.

51-2011.03 Aircraft Rigging Assemblers

Fabricate and assemble aircraft tubing or cable components or assemblies.

51-2021.00 Coil Winders, Tapers, and Finishers

Wind wire coils used in electrical components, such as resistors and transformers, and in electrical equipment and instruments, such as field cores, bobbins, armature cores, electrical motors, generators, and control equipment.

51-2022.00 Electrical and Electronic Equipment Assemblers

Assemble or modify electrical or electronic equipment, such as computers, test equipment telemetering systems, electric motors, and batteries.

51-2023.00 Electromechanical Equipment Assemblers

Assemble or modify electromechanical equipment or devices, such as servomechanisms, gyros, dynamometers, magnetic drums, tape drives, brakes, control linkage, actuators, and appliances.

51-2031.00 Engine and Other Machine Assemblers

Construct, assemble, or rebuild machines, such as engines, turbines, and similar equipment used in such industries as construction, extraction, textiles, and paper manufacturing.

51-2041.00 Structural Metal Fabricators and Fitters ¹

Fabricate, lay out, position, align, and fit parts of structural metal products.

51-2041.01 Metal Fabricators, Structural Metal Products

Fabricate and assemble structural metal products, such as frameworks or shells for machinery, ovens, tanks, and stacks, and metal parts for buildings and bridges, according to job order or blueprints.

51-2041.02 Fitters, Structural Metal- Precision

Lay out, position, align, and fit together fabricated parts of structural metal products preparatory to welding or riveting.

51-2091.00 Fiberglass Laminators and Fabricators

Laminate layers of fiberglass on molds to form boat decks and hulls, bodies for golf carts, automobiles, or other products.

51-2092.00 Team Assemblers

Work as part of a team having responsibility for assembling an entire product or component of a product. Team assemblers can perform all tasks conducted by the team in the assembly process and rotate through all or most of them rather than being assigned to a specific task on a permanent basis. May participate in making management decisions affecting the work. Team leaders who work as part of the team should be included.

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² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-2093.00 Timing Device Assemblers, Adjusters, and Calibrators

Perform precision assembling or adjusting, within narrow tolerances, of timing devices, such as watches, clocks, or chronometers.

51-2099.99 Assemblers and Fabricators, All Other ²

All assemblers and fabricators not listed separately.

51-3011.00 Bakers ¹

Mix and bake ingredients according to recipes to produce breads, rolls, cookies, cakes, pies, pastries, or other baked goods.

51-3011.01 Bakers, Bread and Pastry

Mix and bake ingredients according to recipes to produce small quantities of breads, pastries, and other baked goods for consumption on premises or for sale as specialty baked goods.

51-3011.02 Bakers, Manufacturing

Mix and bake ingredients according to recipes to produce breads, pastries, and other baked goods. Goods are produced in large quantities for sale through establishments such as grocery stores. Generally, high volume production equipment is used.

51-3021.00 Butchers and Meat Cutters

Cut, trim, or prepare consumer-sized portions of meat for use or sale in retail establishments.

51-3022.00 Meat, Poultry, and Fish Cutters and Trimmers

Use hand tools to perform routine cutting and trimming of meat, poultry, and fish.

51-3023.00 Slaughterers and Meat Packers

Work in slaughtering, meat packing, or wholesale establishments performing precision functions involving the preparation of meat. Work may include specialized slaughtering tasks, cutting standard or premium cuts of meat for marketing, making sausage, or wrapping meats.

51-3091.00 Food and Tobacco Roasting, Baking, and Drying Machine Operators and Tenders

Operate or tend food or tobacco roasting, baking, or drying equipment, including hearth ovens, kiln driers, roasters, char kilns, and vacuum drying equipment.

51-3092.00 Food Batchmakers

Set up and operate equipment that mixes or blends ingredients used in the manufacturing of food products. Includes candy makers and cheese makers.

51-3093.00 Food Cooking Machine Operators and Tenders

Operate or tend cooking equipment, such as steam cooking vats, deep fry cookers, pressure cookers, kettles, and boilers, to prepare food products.

51-4011.00 Computer-Controlled Machine Tool Operators, Metal and Plastic ³

Operate computer-controlled machines or robots to perform one or more machine functions on metal or plastic work pieces.

51-4011.01 Numerical Control Machine Tool Operators and Tenders, Metal and Plastic

Set up and operate numerical control (magnetic- or punched-tape-controlled) machine tools that automatically mill, drill, broach, and ream metal and plastic parts. May adjust machine feed and speed, change cutting tools, or adjust machine controls when automatic programming is faulty or if machine malfunctions.

51-4012.00 Numerical Tool and Process Control Programmers

Develop programs to control machining or processing of parts by automatic machine tools, equipment, or systems.

51-4021.00 Extruding and Drawing Machine Setters, Operators, and Tenders, Metal and Plastic

Set up, operate, or tend machines to extrude or draw thermoplastic or metal materials into tubes, rods, hoses, wire, bars, or structural shapes.

51-4022.00 Forging Machine Setters, Operators, and Tenders, Metal and Plastic

Set up, operate, or tend forging machines to taper, shape, or form metal or plastic parts.

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² Residual or "All Other" occupation. O*NET data not collected.

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⁴ Military Specific Occupation. O*NET data not collected.

51-4023.00 Rolling Machine Setters, Operators, and Tenders, Metal and Plastic

Set up, operate, or tend machines to roll steel or plastic forming bends, beads, knurls, rolls, or plate or to flatten, temper, or reduce gauge of material.

51-4031.00 Cutting, Punching, and Press Machine Setters, Operators, and Tenders, Metal and Plastic ¹

Set up, operate, or tend machines to saw, cut, shear, slit, punch, crimp, notch, bend, or straighten metal or plastic material.

51-4031.01 Sawing Machine Tool Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate metal or plastic sawing machines to cut straight, curved, irregular, or internal patterns in metal or plastic stock or to trim edges of metal or plastic objects. Involves the use of such machines as band saws, circular saws, friction saws, hacksawing machines, and jigsaws.

51-4031.02 Punching Machine Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate machines to punch, crimp, cut blanks, or notch metal or plastic workpieces between preset dies, according to specifications.

51-4031.03 Press and Press Brake Machine Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate power-press machines or power-brake machines to bend, form, stretch, notch, punch, or straighten metal or plastic plate and structural shapes, as specified by work order, blueprints, drawing, templates, or layout.

51-4031.04 Shear and Slitter Machine Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate power-shear or slitting machines to cut metal or plastic material, such as plates, sheets, slabs, billets or bars, to specified dimensions and angles.

51-4032.00 Drilling and Boring Machine Tool Setters, Operators, and Tenders, Metal and Plastic

Set up, operate, or tend drilling machines to drill, bore, ream, mill, or countersink metal or plastic work pieces.

51-4033.00 Grinding, Lapping, Polishing, and Buffing Machine Tool Setters, Operators, and Tenders, Metal and Plastic ¹

Set up, operate, or tend grinding and related tools that remove excess material or burrs from surfaces, sharpen edges or corners, or buff, hone, or polish metal or plastic work pieces.

51-4033.01 Grinding, Honing, Lapping, and Deburring Machine Set-Up Operators

Set up and operate grinding, honing, lapping, or deburring machines to remove excess materials or burrs from internal and external surfaces.

51-4033.02 Buffing and Polishing Set-Up Operators

Set up and operate buffing or polishing machine.

51-4034.00 Lathe and Turning Machine Tool Setters, Operators, and Tenders Metal and Plastic

Set up, operate, or tend lathe and turning machines to turn, bore, thread, form, or face metal or plastic materials, such as wire, rod, or bar stock.

51-4035.00 Milling and Planing Machine Setters, Operators, and Tenders, Metal and Plastic

Set up, operate, or tend milling or planing machines to mill, plane, shape, groove, or profile metal or plastic work pieces.

51-4041.00 Machinists

Set up and operate a variety of machine tools to produce precision parts and instruments. Includes precision instrument makers who fabricate, modify, or repair mechanical instruments. May also fabricate and modify parts to make or repair machine tools or maintain industrial machines, applying knowledge of mechanics, shop mathematics, metal properties, layout, and machining procedures.

51-4051.00 Metal-Refining Furnace Operators and Tenders

Operate or tend furnaces, such as gas, oil, coal, electric-arc or electric induction, open-hearth, or oxygen furnaces, to melt and refine metal before casting or to produce specified types of steel.

51-4052.00 Pourers and Casters, Metal

Operate hand-controlled mechanisms to pour and regulate the flow of molten metal into molds to produce castings or ingots.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-4061.00 Model Makers, Metal and Plastic

Set up and operate machines, such as lathes, milling and engraving machines, and jig borers to make working models of metal or plastic objects.

51-4062.00 Patternmakers, Metal and Plastic

Lay out, machine, fit, and assemble castings and parts to metal or plastic foundry patterns, core boxes, or match plates.

51-4071.00 Foundry Mold and Coremakers

Make or form wax or sand cores or molds used in the production of metal castings in foundries.

51-4072.00 Molding, Coremaking, and Casting Machine Setters, Operators, and Tenders, Metal and Plastic¹

Set up, operate, or tend metal or plastic molding, casting, or coremaking machines to mold or cast metal or thermoplastic parts or products.

51-4072.01 Plastic Molding and Casting Machine Setters and Set-Up Operators

Set up or set up and operate plastic molding machines, such as compression or injection molding machines, to mold, form, or cast thermoplastic materials to specified shape.

51-4072.02 Plastic Molding and Casting Machine Operators and Tenders

Operate or tend plastic molding machines, such as compression or injection molding machines, to mold, form, or cast thermoplastic materials to specified shape.

51-4072.03 Metal Molding, Coremaking, and Casting Machine Setters and Set-Up Operators

Set up or set up and operate metal casting, molding, and coremaking machines to mold or cast metal parts and products, such as tubes, rods, automobile trim, carburetor housings, and motor parts. Machines include: die casting and continuous casting machines, and roll-over, squeeze, and shell molding machines.

51-4072.04 Metal Molding, Coremaking, and Casting Machine Operators and Tenders

Operate or tend metal molding, casting, or coremaking machines to mold or cast metal products, such as pipes, brake drums, and rods, and metal parts, such as automobile trim, carburetor housings, and motor parts. Machines include centrifugal casting machines, vacuum casting machines, turnover draw-type coremaking machines, conveyor-screw coremaking machines, and die casting machines.

51-4072.05 Casting Machine Set-Up Operators

Set up and operate machines to cast and assemble printing type.

51-4081.00 Multiple Machine Tool Setters, Operators, and Tenders, Metal and Plastic¹

Set up, operate, or tend more than one type of cutting or forming machine tool or robot.

51-4081.01 Combination Machine Tool Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate more than one type of cutting or forming machine tool, such as gear hobbers, lathes, press brakes, shearing, and boring machines.

51-4081.02 Combination Machine Tool Operators and Tenders, Metal and Plastic

Operate or tend more than one type of cutting or forming machine tool which has been previously set up. Includes such machine tools as band saws, press brakes, slitting machines, drills, lathes, and boring machines.

51-4111.00 Tool and Die Makers

Analyze specifications, lay out metal stock, set up and operate machine tools, and fit and assemble parts to make and repair dies, cutting tools, jigs, fixtures, gauges, and machinists' hand tools.

51-4121.00 Welders, Cutters, Solderers, and Brazers¹

Use hand-welding, flame-cutting, hand soldering, or brazing equipment to weld or join metal components or to fill holes, indentations, or seams of fabricated metal products.

51-4121.01 Welders, Production

Assemble and weld metal parts on production line, using welding equipment, requiring only a limited knowledge of welding techniques.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-4121.02 Welders and Cutters

Use hand welding and flame-cutting equipment to weld together metal components and parts or to cut, trim, or scarf metal objects to dimensions, as specified by layouts, work orders, or blueprints.

51-4121.03 Welder-Fitters

Lay out, fit, and fabricate metal components to assemble structural forms, such as machinery frames, bridge parts, and pressure vessels, using knowledge of welding techniques, metallurgy, and engineering requirements. Includes experimental welders who analyze engineering drawings and specifications to plan welding operations where procedural information is unavailable.

51-4121.04 Solderers

Solder together components to assemble fabricated metal products, using soldering iron.

51-4121.05 Brazers

Braze together components to assemble fabricated metal parts, using torch or welding machine and flux.

51-4122.00 Welding, Soldering, and Brazing Machine Setters, Operators, and Tenders ¹

Set up, operate, or tend welding, soldering, or brazing machines or robots that weld, braze, solder, or heat treat metal products, components, or assemblies.

51-4122.01 Welding Machine Setters and Set-Up Operators

Set up or set up and operate welding machines that join or bond together components to fabricate metal products or assemblies, according to specifications and blueprints.

51-4122.02 Welding Machine Operators and Tenders

Operate or tend welding machines that join or bond together components to fabricate metal products and assemblies, according to specifications and blueprints.

51-4122.03 Soldering and Brazing Machine Setters and Set-Up Operators

Set up or set up and operate soldering or brazing machines to braze, solder, heat treat, or spot weld fabricated metal products or components as specified by work orders, blueprints, and layout specifications.

51-4122.04 Soldering and Brazing Machine Operators and Tenders

Operate or tend soldering and brazing machines that braze, solder, or spot weld fabricated metal products or components as specified by work orders, blueprints, and layout specifications.

51-4191.00 Heat Treating Equipment Setters, Operators, and Tenders, Metal and Plastic ¹

Set up, operate, or tend heating equipment, such as heat-treating furnaces, flame-hardening machines, induction machines, soaking pits, or vacuum equipment to temper, harden, anneal, or heat-treat metal or plastic objects.

51-4191.01 Heating Equipment Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate heating equipment, such as heat-treating furnaces, flame-hardening machines, and induction machines, that anneal or heat-treat metal objects.

51-4191.02 Heat Treating, Annealing, and Tempering Machine Operators and Tenders, Metal and Plastic

Operate or tend machines, such as furnaces, baths, flame-hardening machines, and electronic induction machines, to harden, anneal, and heat-treat metal products or metal parts.

51-4191.03 Heaters, Metal and Plastic

Operate or tend heating equipment, such as soaking pits, reheating furnaces, and heating and vacuum equipment, to heat metal sheets, blooms, billets, bars, plate, and rods to a specified temperature for rolling or processing, or to heat and cure preformed plastic parts.

51-4192.00 Lay-Out Workers, Metal and Plastic

Lay out reference points and dimensions on metal or plastic stock or workpieces, such as sheets, plates, tubes, structural shapes, castings, or machine parts, for further processing. Includes shipfitters.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-4193.00 Plating and Coating Machine Setters, Operators, and Tenders, Metal and Plastic¹

Set up, operate, or tend plating or coating machines to coat metal or plastic products with chromium, zinc, copper, cadmium, nickel, or other metal to protect or decorate surfaces. Includes electrolytic processes.

51-4193.01 Electrolytic Plating and Coating Machine Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate electrolytic plating or coating machines, such as continuous multistrand electrogalvanizing machines, to coat metal or plastic products electrolytically with chromium, copper, cadmium, or other metal to provide protective or decorative surfaces or to build up worn surfaces.

51-4193.02 Electrolytic Plating and Coating Machine Operators and Tenders, Metal and Plastic

Operate or tend electrolytic plating or coating machines, such as zinc-plating machines and anodizing machines, to coat metal or plastic products electrolytically with chromium, zinc, copper, cadmium, or other metal to provide protective or decorative surfaces or to build up worn surfaces.

51-4193.03 Nonelectrolytic Plating and Coating Machine Setters and Set-Up Operators, Metal and Plastic

Set up or set up and operate nonelectrolytic plating or coating machines, such as hot-dip lines and metal-spraying machines, to coat metal or plastic products or parts with metal.

51-4193.04 Nonelectrolytic Plating and Coating Machine Operators and Tenders, Metal and Plastic

Operate or tend nonelectrolytic plating or coating machines, such as metal-spraying machines and vacuum metalizing machines, to coat metal or plastic products or parts with metal.

51-4194.00 Tool Grinders, Filers, and Sharpeners

Perform precision smoothing, sharpening, polishing, or grinding of metal objects.

51-4199.99 Metal Workers and Plastic Workers, All Other²

All metalworkers and plastic workers not listed separately.

51-5011.00 Bindery Workers¹

Set up or operate binding machines that produce books and other printed materials.

51-5011.01 Bindery Machine Setters and Set-Up Operators

Set up or set up and operate machines that perform some or all of the following functions in order to produce books, magazines, pamphlets, catalogs, and other printed materials: gathering, folding, cutting, stitching, rounding and backing, supering, casing-in, lining, pressing, and trimming.

51-5011.02 Bindery Machine Operators and Tenders

Operate or tend binding machines that round, back, case, line stitch, press, fold, trim, or perform other binding operations on books and related articles.

51-5012.00 Bookbinders

Perform highly skilled hand finishing operations, such as grooving and lettering to bind books.

51-5021.00 Job Printers

Set type according to copy; operate press to print job order; and read proof for errors and clarity of impression, and correct imperfections. Job printers are often found in small establishments where work combines several job skills.

51-5022.00 Prepress Technicians and Workers¹

Set up and prepare material for printing presses.

51-5022.01 Hand Compositors and Typesetters

Set up and arrange type by hand. Assemble and lock setup of type, cuts, and headings. Pull proofs.

51-5022.02 Paste-Up Workers

Arrange and mount typeset material and illustrations into pasteup for printing reproduction, based on artist's or editor's layout.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-5022.03 Photoengravers

Photograph copy, develop negatives, and prepare photosensitized metal plates for use in letterpress and gravure printing.

51-5022.04 Camera Operators

Operate process camera and related darkroom equipment to photograph and develop negatives of material to be printed.

51-5022.05 Scanner Operators

Operate electronic or computerized scanning equipment to produce and screen film separations of photographs or art for use in producing lithographic printing plates. Evaluate and correct for deficiencies in the film.

51-5022.06 Strippers

Cut and arrange film into flats (layout sheets resembling a film negative of text in its final form) which are used to make plates. Prepare separate flat for each color.

51-5022.07 Platemakers

Produce printing plates by exposing sensitized metal sheets to special light through a photographic negative. May operate machines that process plates automatically.

51-5022.08 Dot Etchers

Increase or reduce size of photographic dots by chemical or photomechanical methods to make color corrections on halftone negatives or positives to be used in preparation of lithographic printing plates.

51-5022.09 Electronic Masking System Operators

Operate computerized masking system to produce stripping masks used in production of offset lithographic printing plates.

51-5022.10 Electrotypers and Stereotypers

Fabricate and finish electrotypes and stereotype printing plates.

51-5022.11 Plate Finishers

Set up and operate equipment to trim and mount electrotypes or stereotype plates.

51-5022.12 Typesetting and Composing Machine Operators and Tenders

Operate or tend typesetting and composing equipment, such as phototypesetters, linotype or monotype keyboard machines, photocomposers, linocasters, and photoletterers.

51-5022.13 Photoengraving and Lithographing Machine Operators and Tenders

Operate or tend photoengraving and lithographing equipment, such as plate graining, pantograph, roll varnishing, and routing machines.

51-5023.00 Printing Machine Operators¹

Set up or operate various types of printing machines, such as offset, letterpress, intaglio, or gravure presses or screen printers to produce print on paper or other materials.

51-5023.01 Precision Printing Workers

Perform variety of precision printing activities, such as duplication of microfilm and reproduction of graphic arts materials.

51-5023.02 Offset Lithographic Press Setters and Set-Up Operators

Set up or set up and operate offset printing press, either sheet or web fed, to print single and multicolor copy from lithographic plates. Examine job order to determine press operating time, quantity to be printed, and stock specifications.

51-5023.03 Letterpress Setters and Set-Up Operators

Set up or set up and operate direct relief letterpresses, either sheet or roll (web) fed, to produce single or multicolor printed material, such as newspapers, books, and periodicals.

51-5023.04 Design Printing Machine Setters and Set-Up Operators

Set up or set up and operate machines to print designs on materials.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-5023.05 Marking and Identification Printing Machine Setters and Set-Up Operators

Set up or set up and operate machines to print trademarks, labels, or multicolored identification symbols on materials.

51-5023.06 Screen Printing Machine Setters and Set-Up Operators

Set up or set up and operate screen printing machines to print designs onto articles and materials, such as glass or plasticware, cloth, and paper.

51-5023.07 Embossing Machine Set-Up Operators

Set up and operate embossing machines.

51-5023.08 Engraver Set-Up Operators

Set up and operate machines to transfer printing designs.

51-5023.09 Printing Press Machine Operators and Tenders

Operate or tend various types of printing machines, such as offset lithographic presses, letter or letterset presses, flexographic or gravure presses, to produce print on paper or other materials such as plastic, cloth, or rubber.

51-6011.00 Laundry and Dry-Cleaning Workers¹

Operate or tend washing or dry-cleaning machines to wash or dry-clean industrial or household articles, such as cloth garments, suede, leather, furs, blankets, draperies, fine linens, rugs, and carpets.

51-6011.01 Spotters, Dry Cleaning

Identify stains in wool, synthetic, and silk garments and household fabrics and apply chemical solutions to remove stain. Determine spotting procedures on basis of type of fabric and nature of stain.

51-6011.02 Precision Dyers

Change or restore the color of articles, such as garments, drapes, and slipcovers, by means of dyes. Work requires knowledge of the composition of the textiles being dyed or restored; the chemical properties of bleaches and dyes; and their effects upon such textiles.

51-6011.03 Laundry and Drycleaning Machine Operators and Tenders, Except Pressing

Operate or tend washing or dry-cleaning machines to wash or dry-clean commercial, industrial, or household articles, such as cloth garments, suede, leather, furs, blankets, draperies, fine linens, rugs, and carpets.

51-6021.00 Pressers, Textile, Garment, and Related Materials¹

Press or shape articles by hand or machine.

51-6021.01 Pressers, Delicate Fabrics

Press dry-cleaned and wet-cleaned silk and synthetic fiber garments by hand or machine, applying knowledge of fabrics and heat to produce high quality finish. Finish pleated or fancy garments, normally by hand.

51-6021.02 Pressing Machine Operators and Tenders- Textile, Garment, and Related Materials

Operate or tend pressing machines, such as hot-head pressing, steam pressing, automatic pressing, ironing, plunger pressing, and hydraulic pressing machines, to press and shape articles such as leather, fur, and cloth garments, drapes, slipcovers, handkerchiefs, and millinery.

51-6021.03 Pressers, Hand

Press articles to remove wrinkles, flatten seams, and give shape by using hand iron. Articles pressed include: drapes, knit goods, millinery parts, parachutes, garments, slip covers, and textiles such as lace, rayon, and silk. May block (shape) knitted garments after cleaning. May press leather goods.

51-6031.00 Sewing Machine Operators¹

Operate or tend sewing machines to join, reinforce, decorate, or perform related sewing operations in the manufacture of garment or nongarment products.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-6031.01 Sewing Machine Operators, Garment

Operate or tend sewing machines to perform garment sewing operations, such as joining, reinforcing, or decorating garments or garment parts.

51-6031.02 Sewing Machine Operators, Non-Garment

Operate or tend sewing machines to join together, reinforce, decorate, or perform related sewing operations in the manufacture of nongarment products, such as upholstery, draperies, linens, carpets, and mattresses.

51-6041.00 Shoe and Leather Workers and Repairers

Construct, decorate, or repair leather and leather-like products, such as luggage, shoes, and saddles.

51-6042.00 Shoe Machine Operators and Tenders

Operate or tend a variety of machines to join, decorate, reinforce, or finish shoes and shoe parts.

51-6051.00 Sewers, Hand

Sew, join, reinforce, or finish, usually with needle and thread, a variety of manufactured items. Includes weavers and stitchers.

51-6052.00 Tailors, Dressmakers, and Custom Sewers ¹

Design, make, alter, repair, or fit garments.

51-6052.01 Shop and Alteration Tailors

Make tailored garments from existing patterns. Alter, repair, or fit made-to-measure or ready-to-wear garments.

51-6052.02 Custom Tailors

Design/make tailored garments, applying knowledge of garment design, construction, styling, and fabrics.

51-6061.00 Textile Bleaching and Dyeing Machine Operators and Tenders

Operate or tend machines to bleach, shrink, wash, dye, or finish textiles or synthetic or glass fibers.

51-6062.00 Textile Cutting Machine Setters, Operators, and Tenders

Set up, operate, or tend machines that cut textiles.

51-6063.00 Textile Knitting and Weaving Machine Setters, Operators, and Tenders

Set up, operate, or tend machines that knit, loop, weave, or draw in textiles.

51-6064.00 Textile Winding, Twisting, and Drawing Out Machine Setters, Operators, and Tenders

Set up, operate, or tend machines that wind or twist textiles; or draw out and combine sliver, such as wool, hemp, or synthetic fibers. Includes slubber machine and drawing frame operators.

51-6091.00 Extruding and Forming Machine Setters, Operators, and Tenders, Synthetic and Glass Fibers ³

Set up, operate, or tend machines that extrude and form continuous filaments from synthetic materials, such as liquid polymer, rayon, and fiberglass.

51-6091.01 Extruding and Forming Machine Operators and Tenders, Synthetic or Glass Fibers

Operate or tend machines that extrude and form continuous filaments from synthetic materials, such as liquid polymer, rayon, and fiberglass, preparatory to further processing.

51-6092.00 Fabric and Apparel Patternmakers

Draw and construct sets of precision master fabric patterns or layouts. May also mark and cut fabrics and apparel.

51-6093.00 Upholsterers

Make, repair, or replace upholstery for household furniture or transportation vehicles.

51-6099.99 Textile, Apparel, and Furnishings Workers, All Other ²

All textile, apparel, and furnishings workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-7011.00 Cabinetmakers and Bench Carpenters

Cut, shape, and assemble wooden articles or set up and operate a variety of woodworking machines, such as power saws, jointers, and mortisers to surface, cut, or shape lumber or to fabricate parts for wood products.

51-7021.00 Furniture Finishers

Shape, finish, and refinish damaged, worn, or used furniture or new high-grade furniture to specified color or finish.

51-7031.00 Model Makers, Wood

Construct full-size and scale wooden precision models of products. Includes wood jig builders and loft workers.

51-7032.00 Patternmakers, Wood

Plan, lay out, and construct wooden unit or sectional patterns used in forming sand molds for castings.

51-7041.00 Sawing Machine Setters, Operators, and Tenders, Wood ¹

Set up, operate, or tend wood sawing machines. Includes head sawyers.

51-7041.01 Sawing Machine Setters and Set-Up Operators

Set up or set up and operate wood-sawing machines. Examine blueprints, drawings, work orders, and patterns to determine size and shape of items to be sawed, sawing machines to set up, and sequence of sawing operations.

51-7041.02 Sawing Machine Operators and Tenders

Operate or tend wood-sawing machines, such as circular saws, band saws, multiple blade sawing machines, scroll saws, rip saws, equalizer saws, power saws, and crozer machines. Duties include: sawing logs to specifications; cutting lumber to specified dimensions; sawing curved or irregular designs; trimming edges and removing defects from lumber; or cutting grooves, bevel, and miter according to specifications or work orders.

51-7042.00 Woodworking Machine Setters, Operators, and Tenders, Except Sawing ¹

Set up, operate, or tend woodworking machines, such as drill presses, lathes, shapers, routers, sanders, planers, and wood nailing machines.

51-7042.01 Woodworking Machine Setters and Set-Up Operators, Except Sawing

Set up or set up and operate woodworking machines, such as lathes, drill presses, sanders, shapers, and planing machines, to perform woodworking operations.

51-7042.02 Woodworking Machine Operators and Tenders, Except Sawing

Operate or tend woodworking machines, such as drill presses, lathes, shapers, routers, sanders, planers, and wood-nailing machines to perform woodworking operations.

51-7099.99 Woodworkers, All Other ²

All woodworkers not listed separately.

51-8011.00 Nuclear Power Reactor Operators

Control nuclear reactors.

51-8012.00 Power Distributors and Dispatchers

Coordinate, regulate, or distribute electricity or steam.

51-8013.00 Power Plant Operators ¹

Control, operate, or maintain machinery to generate electric power. Includes auxiliary equipment operators.

51-8013.01 Power Generating Plant Operators, Except Auxiliary Equipment Operators

Control or operate machinery, such as steam-driven turbogenerators, to generate electric power, often through the use of panelboards, control boards, or semi-automatic equipment.

51-8013.02 Auxiliary Equipment Operators, Power

Control and maintain auxiliary equipment, such as pumps, fans, compressors, condensers, feedwater heaters, filters, and chlorinators, that supply water, fuel, lubricants, air, and auxiliary power for turbines, generators, boilers, and other power-generating plant facilities.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-8021.00 Stationary Engineers and Boiler Operators ¹

Operate or maintain stationary engines, boilers, or other mechanical equipment to provide utilities for buildings or industrial processes. Operate equipment, such as steam engines, generators, motors, turbines, and steam boilers.

51-8021.01 Boiler Operators and Tenders, Low Pressure

Operate or tend low pressure stationary steam boilers and auxiliary steam equipment, such as pumps, compressors and air conditioning equipment, to supply steam heat for office buildings, apartment houses, or industrial establishments; to maintain steam at specified pressure aboard marine vessels; or to generate and supply compressed air for operation of pneumatic tools, hoists, and air lances.

51-8021.02 Stationary Engineers

Operate and maintain stationary engines and mechanical equipment to provide utilities for buildings or industrial processes. Operate equipment such as steam engines, generators, motors, turbines, and steam boilers.

51-8031.00 Water and Liquid Waste Treatment Plant and System Operators

Operate or control an entire process or system of machines, often through the use of control boards, to transfer or treat water or liquid waste.

51-8091.00 Chemical Plant and System Operators

Control or operate an entire chemical process or system of machines.

51-8092.00 Gas Plant Operators ¹

Distribute or process gas for utility companies and others by controlling compressors to maintain specified pressures on main pipelines.

51-8092.01 Gas Processing Plant Operators

Control equipment, such as compressors, evaporators, heat exchangers, and refrigeration equipment to process gas for utility companies and for industrial use.

51-8092.02 Gas Distribution Plant Operators

Control equipment to regulate flow and pressure of gas for utility companies and industrial use. May control distribution of gas for a municipal or industrial plant or a single process in an industrial plant.

51-8093.00 Petroleum Pump System Operators, Refinery Operators, and Gaugers ¹

Control the operation of petroleum refining or processing units. May specialize in controlling manifold and pumping systems, gauging or testing oil in storage tanks, or regulating the flow of oil into pipelines.

51-8093.01 Petroleum Pump System Operators

Control or operate manifold and pumping systems to circulate liquids through a petroleum refinery.

51-8093.02 Petroleum Refinery and Control Panel Operators

Analyze specifications and control continuous operation of petroleum refining and processing units. Operate control panel to regulate temperature, pressure, rate of flow, and tank level in petroleum refining unit, according to process schedules.

51-8093.03 Gaugers

Gauge and test oil in storage tanks. Regulate flow of oil into pipelines at wells, tank farms, refineries, and marine and rail terminals, following prescribed standards and regulations.

51-8099.99 Plant and System Operators, All Other ²

All plant and system operators not listed separately.

51-9011.00 Chemical Equipment Operators and Tenders ¹

Operate or tend equipment to control chemical changes or reactions in the processing of industrial or consumer products. Equipment used includes devulcanizers, steam-jacketed kettles, and reactor vessels.

51-9011.01 Chemical Equipment Controllers and Operators

Control or operate equipment to control chemical changes or reactions in the processing of industrial or consumer products. Typical equipment used are: reaction kettles, catalytic converters, continuous or batch treating equipment, saturator tanks, electrolytic cells, reactor vessels, recovery units, and fermentation chambers.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-9011.02 Chemical Equipment Tenders

Tend equipment in which a chemical change or reaction takes place in the processing of industrial or consumer products. Typical equipment used are: devulcanizers, batch stills, fermenting tanks, steam-jacketed kettles, and reactor vessels.

51-9012.00 Separating, Filtering, Clarifying, Precipitating, and Still Machine Setters, Operators, and Tenders

Set up, operate, or tend continuous flow or vat-type equipment; filter presses; shaker screens; centrifuges; condenser tubes; precipitating, fermenting, or evaporating tanks; scrubbing towers; or batch stills. These machines extract, sort, or separate liquids, gases, or solids from other materials to recover a refined product. Includes dairy processing equipment operators.

51-9021.00 Crushing, Grinding, and Polishing Machine Setters, Operators, and Tenders

Set up, operate, or tend machines to crush, grind, or polish materials, such as coal, glass, grain, stone, food, or rubber.

51-9022.00 Grinding and Polishing Workers, Hand

Grind, sand, or polish, using hand tools or hand-held power tools, a variety of metal, wood, stone, clay, plastic, or glass objects.

51-9023.00 Mixing and Blending Machine Setters, Operators, and Tenders

Set up, operate, or tend machines to mix or blend materials, such as chemicals, tobacco, liquids, color pigments, or explosive ingredients.

51-9031.00 Cutters and Trimmers, Hand

Use hand tools or hand-held power tools to cut and trim a variety of manufactured items, such as carpet, fabric, stone, glass, or rubber.

51-9032.00 Cutting and Slicing Machine Setters, Operators, and Tenders ¹

Set up, operate, or tend machines that cut or slice materials, such as glass, stone, cork, rubber, tobacco, food, paper, or insulating material.

51-9032.01 Fiber Product Cutting Machine Setters and Set-Up Operators

Set up and operate machine to cut or slice fiber material, such as paper, wallboard, and insulation material.

51-9032.02 Stone Sawyers

Set up and operate gang saws, reciprocating saws, circular saws, or wire saws to cut blocks of stone into specified dimensions.

51-9032.03 Glass Cutting Machine Setters and Set-Up Operators

Set up and operate machines to cut glass.

51-9032.04 Cutting and Slicing Machine Operators and Tenders

Operate or tend machines to cut or slice any of a wide variety of products or materials, such as tobacco, food, paper, roofing slate, glass, stone, rubber, cork, and insulating material.

51-9041.00 Extruding, Forming, Pressing, and Compacting Machine Setters, Operators, and Tenders ¹

Set up, operate, or tend machines, such as glass forming machines, plodder machines, and tuber machines, to shape and form products, such as glassware, food, rubber, soap, brick, tile, clay, wax, tobacco, or cosmetics.

51-9041.01 Extruding, Forming, Pressing, and Compacting Machine Setters and Set-Up Operators

Set up or set up and operate machines, such as glass forming machines, plodder machines, and tuber machines, to manufacture any of a wide variety of products, such as soap bars, formed rubber, glassware, food, brick, and tile, by means of extruding, compressing, or compacting.

51-9041.02 Extruding, Forming, Pressing, and Compacting Machine Operators and Tenders

Operate or tend machines to shape and form any of a wide variety of manufactured products, such as glass bulbs, molded food and candy, rubber goods, clay products, wax products, tobacco plugs, cosmetics, or paper products, by means of extruding, compressing or compacting.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-9051.00 Furnace, Kiln, Oven, Drier, and Kettle Operators and Tenders

Operate or tend heating equipment other than basic metal, plastic, or food processing equipment. Includes activities, such as annealing glass, drying lumber, curing rubber, removing moisture from materials, or boiling soap.

51-9061.00 Inspectors, Testers, Sorters, Samplers, and Weighers ¹

Inspect, test, sort, sample, or weigh nonagricultural raw materials or processed, machined, fabricated, or assembled parts or products for defects, wear, and deviations from specifications. May use precision measuring instruments and complex test equipment.

51-9061.01 Materials Inspectors

Examine and inspect materials and finished parts and products for defects and wear and to ensure conformance with work orders, diagrams, blueprints, and template specifications. Usually specialize in a single phase of inspection.

51-9061.02 Mechanical Inspectors

Inspect and test mechanical assemblies and systems, such as motors, vehicles, and transportation equipment for defects and wear to ensure compliance with specifications.

51-9061.03 Precision Devices Inspectors and Testers

Verify accuracy of and adjust precision devices, such as meters and gauges, testing instruments, clock and watch mechanisms, to ensure operation of device is in accordance with design specifications.

51-9061.04 Electrical and Electronic Inspectors and Testers

Inspect and test electrical and electronic systems, such as radar navigational equipment, computer memory units, television and radio transmitters, using precision measuring instruments.

51-9061.05 Production Inspectors, Testers, Graders, Sorters, Samplers, Weighers

Inspect, test, grade, sort, sample, or weigh nonagricultural raw materials or processed, machined, fabricated, or assembled parts or products. Work may be performed before, during, or after processing.

51-9071.00 Jewelers and Precious Stone and Metal Workers ¹

Design, fabricate, adjust, repair, or appraise jewelry, gold, silver, other precious metals, or gems. Includes diamond polishers and gem cutters and persons who perform precision casting and modeling of molds, casting metal in molds, or setting precious and semi-precious stones for jewelry and related products.

51-9071.01 Jewelers

Fabricate and repair jewelry articles.

51-9071.02 Silversmiths

Anneal, solder, hammer, shape, and glue silver articles.

51-9071.03 Model and Mold Makers, Jewelry

Make models or molds to create jewelry items.

51-9071.04 Bench Workers, Jewelry

Cut, file, form, and solder parts for jewelry.

51-9071.05 Pewter Casters and Finishers

Cast and finish pewter alloy to form parts for goblets, candlesticks, and other pewterware.

51-9071.06 Gem and Diamond Workers

Split, saw, cut, shape, polish, or drill gems and diamonds used in jewelry or industrial tools.

51-9081.00 Dental Laboratory Technicians

Construct and repair full or partial dentures or dental appliances.

51-9082.00 Medical Appliance Technicians

Construct, fit, maintain, or repair medical supportive devices, such as braces, artificial limbs, joints, arch supports, and other surgical and medical appliances.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-9083.00 Ophthalmic Laboratory Technicians¹

Cut, grind, and polish eyeglasses, contact lenses, or other precision optical elements. Assemble and mount lenses into frames or process other optical elements.

51-9083.01 Precision Lens Grinders and Polishers

Set up and operate variety of machines and equipment to grind and polish lens and other optical elements.

51-9083.02 Optical Instrument Assemblers

Assemble optical instruments, such as telescopes, level-transits, and gunsights.

51-9111.00 Packaging and Filling Machine Operators and Tenders

Operate or tend machines to prepare industrial or consumer products for storage or shipment. Includes cannery workers who pack food products.

51-9121.00 Coating, Painting, and Spraying Machine Setters, Operators, and Tenders¹

Set up, operate, or tend machines to coat or paint any of a wide variety of products including food, glassware, cloth, ceramics, metal, plastic, paper, or wood, with lacquer, silver, copper, rubber, varnish, glaze, enamel, oil, or rust-proofing materials.

51-9121.01 Coating, Painting, and Spraying Machine Setters and Set-Up Operators

Set up or set up and operate machines to coat or paint any of a wide variety of products, such as food products, glassware, and cloth, ceramic, metal, plastic, paper, and wood products, with lacquer, silver and copper solution, rubber, paint, varnish, glaze, enamel, oil, or rust-proofing materials.

51-9121.02 Coating, Painting, and Spraying Machine Operators and Tenders

Coating Machine Operators and Tenders: Operate or tend machines to coat any of a wide variety of items: Coating food products with sugar, chocolate, or butter; coat paper and paper products with chemical solutions, wax, or glazes; or coat fabric with rubber or plastic. Painting and Spraying Machine Operators and Tenders: Operate or tend machines to spray or paint decorative, protective, or other coating or finish, such as adhesive, lacquer, paint, stain, latex, preservative, oil, or other solutions. May apply coating or finish to any of a wide variety of items or materials, such as wood and wood products, ceramics, and glass. Includes workers who apply coating or finish to materials preparatory to further processing or to consumer use.

51-9122.00 Painters, Transportation Equipment

Operate or tend painting machines to paint surfaces of transportation equipment, such as automobiles, buses, trucks, trains, boats, and airplanes.

51-9123.00 Painting, Coating, and Decorating Workers

Paint, coat, or decorate articles, such as furniture, glass, plateware, pottery, jewelry, cakes, toys, books, or leather.

51-9131.00 Photographic Process Workers¹

Perform precision work involved in photographic processing, such as editing photographic negatives and prints, using photo-mechanical, chemical, or computerized methods.

51-9131.01 Photographic Retouchers and Restorers

Retouch or restore photographic negatives and prints to accentuate desirable features of subject, using pencils, watercolors, or airbrushes.

51-9131.02 Photographic Reproduction Technicians

Duplicate materials to produce prints on sensitized paper, cloth, or film, using photographic equipment.

51-9131.03 Photographic Hand Developers

Develop exposed photographic film or sensitized paper in series of chemical and water baths to produce negative or positive prints.

51-9131.04 Film Laboratory Technicians

Evaluate motion picture film to determine characteristics, such as sensitivity to light, density, and exposure time required for printing.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-9132.00 Photographic Processing Machine Operators

Operate photographic processing machines, such as photographic printing machines, film developing machines, and mounting presses.

51-9141.00 Semiconductor Processors

Perform any or all of the following functions in the manufacture of electronic semiconductors: load semiconductor material into furnace; saw formed ingots into segments; load individual segment into crystal growing chamber and monitor controls; locate crystal axis in ingot using x-ray equipment and saw ingots into wafers; clean, polish, and load wafers into series of special purpose furnaces, chemical baths, and equipment used to form circuitry and change conductive properties.

51-9191.00 Cementing and Gluing Machine Operators and Tenders

Operate or tend cementing and gluing machines to join items for further processing or to form a completed product. Processes include joining veneer sheets into plywood; gluing paper; joining rubber and rubberized fabric parts, plastic, simulated leather, or other materials.

51-9192.00 Cleaning, Washing, and Metal Pickling Equipment Operators and Tenders

Operate or tend machines to wash or clean products, such as barrels or kegs, glass items, tin plate, food, pulp, coal, plastic, or rubber, to remove impurities.

51-9193.00 Cooling and Freezing Equipment Operators and Tenders

Operate or tend equipment, such as cooling and freezing units, refrigerators, batch freezers, and freezing tunnels, to cool or freeze products, food, blood plasma, and chemicals.

51-9194.00 Etchers and Engravers¹

Engrave or etch metal, wood, rubber, or other materials for identification or decorative purposes. Includes such workers as etcher-circuit processors, pantograph engravers, and silk screen etchers.

51-9194.01 Precision Etchers and Engravers, Hand or Machine

Engrave or etch flat or curved metal, wood, rubber, or other materials by hand or machine for printing, identification, or decorative purposes. Includes etchers and engravers of both hard and soft metals or materials, and jewelry and seal engravers.

51-9194.02 Engravers/Carvers

Engrave or carve designs or lettering onto objects, using hand-held power tools.

51-9194.03 Etchers

Etch or cut artistic designs in glass articles, using acid solutions, sandblasting equipment, and design patterns.

51-9194.04 Pantograph Engravers

Affix identifying information onto a variety of materials and products, using engraving machines or equipment.

51-9194.05 Etchers, Hand

Etch patterns, designs, lettering, or figures onto a variety of materials and products.

51-9194.06 Engravers, Hand

Engrave designs and identifying information onto rollers or plates used in printing.

51-9195.00 Molders, Shapers, and Casters, Except Metal and Plastic¹

Mold, shape, form, cast, or carve products such as food products, figurines, tile, pipes, and candles consisting of clay, glass, plaster, concrete, stone, or combinations of materials.

51-9195.01 Precision Mold and Pattern Casters, except Nonferrous Metals

Cast molds and patterns from a variety of materials except nonferrous metals, according to blueprints and specifications.

51-9195.02 Precision Pattern and Die Casters, Nonferrous Metals

Cast metal patterns and dies, according to specifications, from a variety of nonferrous metals, such as aluminum or bronze.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

51-9195.03 Stone Cutters and Carvers

Cut or carve stone according to diagrams and patterns.

51-9195.04 Glass Blowers, Molders, Benders, and Finishers

Shape molten glass according to patterns.

51-9195.05 Potters

Mold clay into ware as clay revolves on potter's wheel.

51-9195.06 Mold Makers, Hand

Construct or form molds from existing forms for use in casting objects.

51-9195.07 Molding and Casting Workers

Perform a variety of duties such as mixing materials, assembling mold parts, filling molds, and stacking molds to mold and cast a wide range of products.

51-9196.00 Paper Goods Machine Setters, Operators, and Tenders

Set up, operate, or tend paper goods machines that perform a variety of functions, such as converting, sawing, corrugating, banding, wrapping, boxing, stitching, forming, or sealing paper or paperboard sheets into products.

51-9197.00 Tire Builders

Operate machines to build tires from rubber components.

51-9198.00 Helpers--Production Workers ¹

Help production workers by performing duties of lesser skill. Duties include supplying or holding materials or tools, and cleaning work area and equipment.

51-9198.01 Production Laborers

Perform variety of routine tasks to assist in production activities.

51-9198.02 Production Helpers

Perform variety of tasks requiring limited knowledge of production processes in support of skilled production workers.

51-9199.99 Production Workers, All Other ²

All production workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

53-0000 TRANSPORTATION AND MATERIAL MOVING OCCUPATIONS

53-1011.00 Aircraft Cargo Handling Supervisors

Direct ground crew in the loading, unloading, securing, and staging of aircraft cargo or baggage. Determine the quantity and orientation of cargo and compute aircraft center of gravity. May accompany aircraft as member of flight crew and monitor and handle cargo in flight, and assist and brief passengers on safety and emergency procedures.

53-1021.00 First-Line Supervisors/Managers of Helpers, Laborers, and Material Movers, Hand

Supervise and coordinate the activities of helpers, laborers, or material movers.

53-1031.00 First-Line Supervisors/Managers of Transportation and Material-Moving Machine and Vehicle Operators

Directly supervise and coordinate activities of transportation and material-moving machine and vehicle operators and helpers.

53-2011.00 Airline Pilots, Copilots, and Flight Engineers

Pilot and navigate the flight of multi-engine aircraft in regularly scheduled service for the transport of passengers and cargo. Requires Federal Air Transport rating and certification in specific aircraft type used.

53-2012.00 Commercial Pilots

Pilot and navigate the flight of small fixed or rotary winged aircraft, primarily for the transport of cargo and passengers. Requires Commercial Rating.

53-2021.00 Air Traffic Controllers

Control air traffic on and within vicinity of airport and movement of air traffic between altitude sectors and control centers according to established procedures and policies. Authorize, regulate, and control commercial airline flights according to government or company regulations to expedite and ensure flight safety.

53-2022.00 Airfield Operations Specialists

Ensure the safe takeoff and landing of commercial and military aircraft. Duties include coordination between air-traffic control and maintenance personnel; dispatching; using airfield landing and navigational aids; implementing airfield safety procedures; monitoring and maintaining flight records; and applying knowledge of weather information.

53-3011.00 Ambulance Drivers and Attendants, Except Emergency Medical Technicians

Drive ambulance or assist ambulance driver in transporting sick, injured, or convalescent persons. Assist in lifting patients.

53-3021.00 Bus Drivers, Transit and Intercity

Drive bus or motor coach, including regular route operations, charters, and private carriage. May assist passengers with baggage. May collect fares or tickets.

53-3022.00 Bus Drivers, School

Transport students or special clients, such as the elderly or persons with disabilities. Ensure adherence to safety rules. May assist passengers in boarding or exiting.

53-3031.00 Driver/Sales Workers

Drive truck or other vehicle over established routes or within an established territory and sell goods, such as food products, including restaurant take-out items, or pick up and deliver items, such as laundry. May also take orders and collect payments. Includes newspaper delivery drivers.

53-3032.00 Truck Drivers, Heavy and Tractor-Trailer¹

Drive a tractor-trailer combination or a truck with a capacity of at least 26,000 GVW, to transport and deliver goods, livestock, or materials in liquid, loose, or packaged form. May be required to unload truck. May require use of automated routing equipment. Requires commercial drivers' license.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

53-3032.01 Truck Drivers, Heavy

Drive truck with capacity of more than three tons to transport materials to specified destinations.

53-3032.02 Tractor-Trailer Truck Drivers

Drive tractor-trailer truck to transport products, livestock, or materials to specified destinations.

53-3033.00 Truck Drivers, Light or Delivery Services

Drive a truck or van with a capacity of under 26,000 GVW, primarily to deliver or pick up merchandise or to deliver packages within a specified area. May require use of automatic routing or location software. May load and unload truck.

53-3041.00 Taxi Drivers and Chauffeurs

Drive automobiles, vans, or limousines to transport passengers. May occasionally carry cargo.

53-3099.99 Motor Vehicle Operators, All Other ²

All motor vehicle operators not listed separately.

53-4011.00 Locomotive Engineers

Drive electric, diesel-electric, steam, or gas-turbine-electric locomotives to transport passengers or freight. Interpret train orders, electronic or manual signals, and railroad rules and regulations.

53-4012.00 Locomotive Firers

Monitor locomotive instruments and watch for dragging equipment, obstacles on rights-of-way, and train signals during run. Watch for and relay traffic signals from yard workers to yard engineer in railroad yard.

53-4013.00 Rail Yard Engineers, Dinkey Operators, and Hostlers

Drive switching or other locomotive or dinkey engines within railroad yard, industrial plant, quarry, construction project, or similar location.

53-4021.00 Railroad Brake, Signal, and Switch Operators ¹

Operate railroad track switches. Couple or uncouple rolling stock to make up or break up trains. Signal engineers by hand or flagging. May inspect couplings, air hoses, journal boxes, and hand brakes.

53-4021.01 Train Crew Members

Inspect couplings, airhoses, journal boxes, and handbrakes on trains to ensure that they function properly.

53-4021.02 Railroad Yard Workers

Perform a variety of activities such as coupling railcars and operating railroad track switches in railroad yard to facilitate the movement of rail cars within the yard.

53-4031.00 Railroad Conductors and Yardmasters

Conductors coordinate activities of train crew on passenger or freight train. Coordinate activities of switch-engine crew within yard of railroad, industrial plant, or similar location. Yardmasters coordinate activities of workers engaged in railroad traffic operations, such as the makeup or breakup of trains, yard switching, and review train schedules and switching orders.

53-4041.00 Subway and Streetcar Operators

Operate subway or elevated suburban train with no separate locomotive, or electric-powered streetcar to transport passengers. May handle fares.

53-4099.99 Rail Transportation Workers, All Other ²

All rail transportation workers not listed separately.

53-5011.00 Sailors and Marine Oilers ¹

Stand watch to look for obstructions in path of vessel, measure water depth, turn wheel on bridge, or use emergency equipment as directed by captain, mate, or pilot. Break out, rig, overhaul, and store cargo-handling gear, stationary rigging, and running gear. Perform a variety of maintenance tasks to preserve the painted surface of the ship and to maintain line and ship equipment. Must hold government-issued certification and tankerman certification when working aboard liquid-carrying vessels.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

53-5011.01 Able Seamen

Stand watch at bow or on wing of bridge to look for obstructions in path of vessel. Measure water depth. Turn wheel on bridge or use emergency equipment as directed by mate. Break out, rig, overhaul, and store cargo-handling gear, stationary rigging, and running gear. Chip rust from and paint deck or ship's structure. Must hold government-issued certification. Must hold certification when working aboard liquid-carrying vessels.

53-5011.02 Ordinary Seamen and Marine Oilers

Stand deck department watches and perform a variety of tasks to preserve the painted surface of the ship and to maintain lines and ship equipment, such as running and cargo-handling gear. May oil and grease moving parts of engines and auxiliary equipment. Must hold government-issued certification. Must hold certification when working aboard liquid-carrying vessels.

53-5021.00 Captains, Mates, and Pilots of Water Vessels ¹

Command or supervise operations of ships and water vessels, such as tugboats and ferryboats, that travel into and out of harbors, estuaries, straits, and sounds and on rivers, lakes, bays, and oceans. Required to hold license issued by U.S. Coast Guard.

53-5021.01 Ship and Boat Captains

Command vessels in oceans, bays, lakes, rivers, and coastal waters.

53-5021.02 Mates- Ship, Boat, and Barge

Supervise and coordinate activities of crew aboard ships, boats, barges, or dredges.

53-5021.03 Pilots, Ship

Command ships to steer them into and out of harbors, estuaries, straits, and sounds, and on rivers, lakes, and bays. Must be licensed by U.S. Coast Guard with limitations indicating class and tonnage of vessels for which license is valid and route and waters that may be piloted.

53-5022.00 Motorboat Operators

Operate small motor-driven boats to carry passengers and freight between ships, or ship to shore. May patrol harbors and beach areas. May assist in navigational activities.

53-5031.00 Ship Engineers

Supervise and coordinate activities of crew engaged in operating and maintaining engines, boilers, deck machinery, and electrical, sanitary, and refrigeration equipment aboard ship.

53-6011.00 Bridge and Lock Tenders

Operate and tend bridges, canal locks, and lighthouses to permit marine passage on inland waterways, near shores, and at danger points in waterway passages. May supervise such operations. Includes drawbridge operators, lock tenders and operators, and slip bridge operators.

53-6021.00 Parking Lot Attendants

Park automobiles or issue tickets for customers in a parking lot or garage. May collect fee.

53-6031.00 Service Station Attendants

Service automobiles, buses, trucks, boats, and other automotive or marine vehicles with fuel, lubricants, and accessories. Collect payment for services and supplies. May lubricate vehicle, change motor oil, install antifreeze, or replace lights or other accessories, such as windshield wiper blades or fan belts. May repair or replace tires.

53-6041.00 Traffic Technicians

Conduct field studies to determine traffic volume, speed, effectiveness of signals, adequacy of lighting, and other factors influencing traffic conditions, under direction of traffic engineer.

53-6051.00 Transportation Inspectors ¹

Inspect equipment or goods in connection with the safe transport of cargo or people. Includes rail transport inspectors, such as freight inspectors, car inspectors, rail inspectors, and other nonprecision inspectors of other types of transportation vehicles.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

53-6051.01 Aviation Inspectors

Inspect aircraft, maintenance procedures, air navigational aids, air traffic controls, and communications equipment to ensure conformance with Federal safety regulations.

53-6051.02 Public Transportation Inspectors

Monitor operation of public transportation systems to ensure good service and compliance with regulations. Investigate accidents, equipment failures, and complaints.

53-6051.03 Marine Cargo Inspectors

Inspect cargoes of seagoing vessels to certify compliance with health and safety regulations in cargo handling and stowage.

53-6051.04 Railroad Inspectors

Inspect railroad equipment, roadbed, and track to ensure safe transport of people or cargo.

53-6051.05 Motor Vehicle Inspectors

Inspect automotive vehicles to ensure compliance with governmental regulations and safety standards.

53-6051.06 Freight Inspectors

Inspect freight for proper storage according to specifications.

53-6099.99 Transportation Workers, All Other ²

All transportation workers not listed separately.

53-7011.00 Conveyor Operators and Tenders

Control or tend conveyors or conveyor systems that move materials or products to and from stockpiles, processing stations, departments, or vehicles. May control speed and routing of materials or products.

53-7021.00 Crane and Tower Operators

Operate mechanical boom and cable or tower and cable equipment to lift and move materials, machines, or products in many directions.

53-7031.00 Dredge Operators

Operate dredge to remove sand, gravel, or other materials from lakes, rivers, or streams; and to excavate and maintain navigable channels in waterways.

53-7032.00 Excavating and Loading Machine and Dragline Operators ¹

Operate or tend machinery equipped with scoops, shovels, or buckets, to excavate and load loose materials.

53-7032.01 Excavating and Loading Machine Operators

Operate machinery equipped with scoops, shovels, or buckets to excavate and load loose materials.

53-7032.02 Dragline Operators

Operate power-driven crane equipment with dragline bucket to excavate or move sand, gravel, mud, or other materials.

53-7033.00 Loading Machine Operators, Underground Mining

Operate underground loading machine to load coal, ore, or rock into shuttle or mine car or onto conveyors. Loading equipment may include power shovels, hoisting engines equipped with cable-drawn scraper or scoop, or machines equipped with gathering arms and conveyor.

53-7041.00 Hoist and Winch Operators

Operate or tend hoists or winches to lift and pull loads using power-operated cable equipment.

53-7051.00 Industrial Truck and Tractor Operators

Operate industrial trucks or tractors equipped to move materials around a warehouse, storage yard, factory, construction site, or similar location.

53-7061.00 Cleaners of Vehicles and Equipment

Wash or otherwise clean vehicles, machinery, and other equipment. Use such materials as water, cleaning agents, brushes, cloths, and hoses.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

53-7062.00 Laborers and Freight, Stock, and Material Movers, Hand ¹

Manually move freight, stock, or other materials or perform other unskilled general labor. Includes all unskilled manual laborers not elsewhere classified.

53-7062.01 Stevedores, Except Equipment Operators

Manually load and unload ship cargo. Stack cargo in transit shed or in hold of ship using pallet or cargo board. Attach and move slings to lift cargo. Guide load lift.

53-7062.02 Grips and Set-Up Workers, Motion Picture Sets, Studios, and Stages

Arrange equipment; raise and lower scenery; move dollies, cranes, and booms; and perform other duties for motion-picture, recording, or television industry.

53-7062.03 Freight, Stock, and Material Movers, Hand

Load, unload and move materials at plant, yard, or other work site.

53-7063.00 Machine Feeders and Offbearers

Feed materials into or remove materials from machines or equipment that is automatic or tended by other workers.

53-7064.00 Packers and Packagers, Hand

Pack or package by hand a wide variety of products and materials.

53-7071.00 Gas Compressor and Gas Pumping Station Operators ¹

Operate steam, gas, electric motor, or internal combustion engine driven compressors. Transmit, compress, or recover gases, such as butane, nitrogen, hydrogen, and natural gas.

53-7071.01 Gas Pumping Station Operators

Control the operation of steam, gas, or electric-motor-driven compressor to maintain specified pressures on high and low-pressure mains dispensing gas from gasholders.

53-7071.02 Gas Compressor Operators

Operate steam or internal combustion engines to transmit, compress, or recover gases, such as butane, nitrogen, hydrogen, and natural gas, in various production processes.

53-7072.00 Pump Operators, Except Wellhead Pumpers

Tend, control, or operate power-driven, stationary, or portable pumps and manifold systems to transfer gases, oil, other liquids, slurries, or powdered materials to and from various vessels and processes.

53-7073.00 Wellhead Pumpers

Operate power pumps and auxiliary equipment to produce flow of oil or gas from wells in oil field.

53-7081.00 Refuse and Recyclable Material Collectors

Collect and dump refuse or recyclable materials from containers into truck. May drive truck.

53-7111.00 Shuttle Car Operators

Operate diesel or electric-powered shuttle car in underground mine to transport materials from working face to mine cars or conveyor.

53-7121.00 Tank Car, Truck, and Ship Loaders

Load and unload chemicals and bulk solids, such as coal, sand, and grain into or from tank cars, trucks, or ships using material moving equipment. May perform a variety of other tasks relating to shipment of products. May gauge or sample shipping tanks and test them for leaks.

53-7199.99 Material Moving Workers, All Other ²

All material moving workers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

55-0000 MILITARY SPECIFIC OCCUPATIONS

55-1011.00 Air Crew Officers ⁴

Perform and direct in-flight duties to ensure the successful completion of combat, reconnaissance, transport, and search and rescue missions. Duties include operating aircraft communications and radar equipment, such as establishing satellite linkages and jamming enemy communications capabilities operating aircraft weapons and defensive systems conducting pre-flight, in-flight, and post-flight inspections of onboard equipment and directing cargo and personnel drops.

55-1012.00 Aircraft Launch and Recovery Officers ⁴

Plan and direct the operation and maintenance of catapults, arresting gear, and associated mechanical, hydraulic, and control systems involved primarily in aircraft carrier takeoff and landing operations. Duties include supervision of readiness and safety of arresting gear, launching equipment, barricades, and visual landing aid systems planning and coordinating the design, development, and testing of launch and recovery systems preparing specifications for catapult and arresting gear installations evaluating design proposals determining handling equipment needed for new aircraft preparing technical data and instructions for operation of landing aids and training personnel in carrier takeoff and landing procedures.

55-1013.00 Armored Assault Vehicle Officers ⁴

Direct the operation of tanks, light armor, and amphibious assault vehicle units during combat situations on land or in aquatic environments. Duties include directing crew members in the operation of targeting and firing systems; coordinating the operation of advanced onboard communications and navigation equipment; directing the transport of personnel and equipment during combat; formulating and implementing battle plans, including the tactical employment of armored vehicle units; and coordinating with infantry, artillery, and air support units.

55-1014.00 Artillery and Missile Officers ⁴

Manage personnel and weapons operations to destroy enemy positions, aircraft, and vessels. Duties include planning, targeting, and coordinating the tactical deployment of field artillery and air defense artillery missile systems units; directing the establishment and operation of fire control communications systems; targeting and launching intercontinental ballistic missiles; directing the storage and handling of nuclear munitions and components; overseeing security of weapons storage and launch facilities; and managing maintenance of weapons systems.

55-1015.00 Command and Control Center Officers ⁴

Manage the operation of communications, detection, and weapons systems essential for controlling air, ground, and naval operations. Duties include managing critical communication links between air, naval, and ground forces; formulating and implementing emergency plans for natural and wartime disasters; coordinating emergency response teams and agencies; evaluating command center information and need for high-level military and government reporting; managing the operation of surveillance and detection systems; providing technical information and advice on capabilities and operational readiness; and directing operation of weapons targeting, firing, and launch computer systems.

55-1016.00 Infantry Officers ⁴

Direct, train, and lead infantry units in ground combat operations. Duties include directing deployment of infantry weapons, vehicles, and equipment; directing location, construction, and camouflage of infantry positions and equipment; managing field communications operations; coordinating with armor, artillery, and air support units; performing strategic and tactical planning, including battle plan development; and leading basic reconnaissance operations.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

55-1017.00 Special Forces Officers⁴

Lead elite teams that implement unconventional operations by air, land, or sea during combat or peacetime. These activities include offensive raids, demolitions, reconnaissance, search and rescue, and counterterrorism. In addition to their combat training, special forces officers often have specialized training in swimming, diving, parachuting, survival, emergency medicine, and foreign languages. Duties include directing advanced reconnaissance operations and evaluating intelligence information recruiting, training, and equipping friendly forces leading raids and invasions on enemy territories training personnel to implement individual missions and contingency plans performing strategic and tactical planning for politically sensitive missions and operating sophisticated communications equipment.

55-1019.99 Military Officer Special and Tactical Operations Leaders/Managers, All Other^{2,4}

All military officer special and tactical operations leaders/managers not listed separately.

55-2011.00 First-Line Supervisors/Managers of Air Crew Members⁴

Supervise and coordinate the activities of air crew members. Supervisors may also perform the same activities as the workers they supervise.

55-2012.00 First-Line Supervisors/Managers of Weapons Specialists/Crew Members⁴

Supervise and coordinate the activities of weapons specialists/crew members. Supervisors may also perform the same activities as the workers they supervise.

55-2013.00 First-Line Supervisors/Managers of All Other Tactical Operations Specialists⁴

Supervise and coordinate the activities of all other tactical operations specialists not classified separately above. Supervisors may also perform the same activities as the workers they supervise.

55-3011.00 Air Crew Members⁴

Perform in-flight duties to ensure the successful completion of combat, reconnaissance, transport, and search and rescue missions. Duties include operating aircraft communications and detection equipment, including establishing satellite linkages and jamming enemy communications capabilities; conducting pre-flight, in-flight, and post-flight inspections of onboard equipment; operating and maintaining aircraft weapons and defensive systems; operating and maintaining aircraft in-flight refueling systems; executing aircraft safety and emergency procedures; computing and verifying passenger, cargo, fuel, and emergency and special equipment weight and balance data; and conducting cargo and personnel drops.

55-3012.00 Aircraft Launch and Recovery Specialists⁴

Operate and maintain catapults, arresting gear, and associated mechanical, hydraulic, and control systems involved primarily in aircraft carrier takeoff and landing operations. Duties include installing and maintaining visual landing aids; testing and maintaining launch and recovery equipment using electric and mechanical test equipment and hand tools; activating airfield arresting systems, such as crash barriers and cables, during emergency landing situations; directing aircraft launch and recovery operations using hand or light signals; and maintaining logs of airplane launches, recoveries, and equipment maintenance.

55-3013.00 Armored Assault Vehicle Crew Members⁴

Operate tanks, light armor, and amphibious assault vehicles during combat situations on land or in aquatic environments. Duties include driving armored vehicles which require specialized training; operating and maintaining targeting and firing systems; operating and maintaining advanced onboard communications and navigation equipment; transporting personnel and equipment in a combat environment; and operating and maintaining auxiliary weapons, including machine guns and grenade launchers.

55-3014.00 Artillery and Missile Crew Members⁴

Target, fire, and maintain weapons used to destroy enemy positions, aircraft, and vessels. Field artillery crew members predominantly use guns, cannons, and howitzers in ground combat operations, while air defense artillery crew members predominantly use missiles and rockets. Naval artillery crew members predominantly use torpedoes and missiles launched from a ship or submarine. Duties include testing, inspecting, and storing ammunition, missiles, and torpedoes conducting preventive and routine maintenance on weapons and related equipment establishing and maintaining radio and wire communications and operating weapons targeting, firing, and launch computer systems.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.

55-3015.00 Command and Control Center Specialists⁴

Operate and monitor communications, detection, and weapons systems essential for controlling air, ground, and naval operations. Duties include maintaining and relaying critical communications between air, naval, and ground forces; implementing emergency plans for natural and wartime disasters; relaying command center information to high-level military and government decision makers; monitoring surveillance and detection systems, such as air defense; interpreting and evaluating tactical situations and making recommendations to superiors; and operating weapons targeting, firing, and launch computer systems.

55-3016.00 Infantry⁴

Operate weapons and equipment in ground combat operations. Duties include operating and maintaining weapons, such as rifles, machine guns, mortars, and hand grenades; locating, constructing, and camouflaging infantry positions and equipment; evaluating terrain and recording topographical information; operating and maintaining field communications equipment; assessing need for and directing supporting fire; placing explosives and performing minesweeping activities on land; and participating in basic reconnaissance operations.

55-3017.00 Radar and Sonar Technicians⁴

Operate equipment using radio or sound wave technology to identify, track, and analyze objects or natural phenomena of military interest. Includes airborne, shipboard, and terrestrial positions. May perform minor maintenance.

55-3018.00 Special Forces⁴

Implement unconventional operations by air, land, or sea during combat or peacetime as members of elite teams. These activities include offensive raids, demolitions, reconnaissance, search and rescue, and counterterrorism. In addition to their combat training, Special Forces members often have specialized training in swimming, diving, parachuting, survival, emergency medicine, and foreign languages. Duties include conducting advanced reconnaissance operations and collecting intelligence information recruiting, training, and equipping friendly forces conducting raids and invasions on enemy territories laying and detonating explosives for demolition targets locating, identifying, defusing, and disposing of ordnance and operating and maintaining sophisticated communications equipment.

55-3019.99 Military Enlisted Tactical Operations and Air/Weapons Specialists and Crew Members, All Other^{2,4}

All military enlisted tactical operations and air/weapons specialists and crewmembers not listed separately.

¹ O*NET data not collected at this level. See more detailed O*NET 3.1 occupation(s) below.

² Residual or "All Other" occupation. O*NET data not collected.

³ O*NET data from incumbents will be collected for this occupation and not for the detailed O*NET 3.1 occupation below.

⁴ Military Specific Occupation. O*NET data not collected.